

BAYESIAN ESTIMATION OF RANDOM PARAMETERS



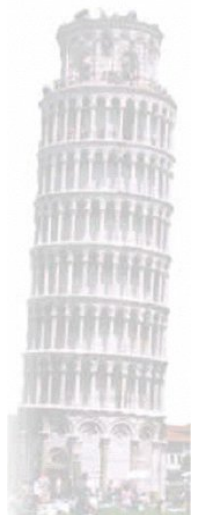
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Random parameters:

- parameters of a signal of which we know the a-priori pdf
- random signals (values of a realization of a random process)

Random parameters:

- Bayesian estimators (MMSE, LMMSE, MAP)
- LMMSE (for filtering/smoothing/prediction of random signals)

MMSE = Minimum Mean Square Error

LMMSE = Linear Minimum Mean Square Error

MAP=Maximum A Posteriori

■ **Problem:** Derive an estimate of the value assumed by the random parameter Y , which is not directly observable, given the observed data $\{X_{ij}\}$ that are related to Y .

■ From a mathematical point of view, the estimator is a function of the observed data vector $\mathbf{X}$ and it should provide a value that is the closest possible to the true value assumed by the random parameter Y in that specific trial of the experiment under observation:

$$\hat{Y} = \hat{Y}(\mathbf{X}) = \Phi(X_1, X_2, X_3, \dots, X_N)$$

■ Clearly, in order the observed data vector $\mathbf{X}$ to be useful for the estimation of the value assumed by Y , there must be a relationship between them and this relationship should be known.

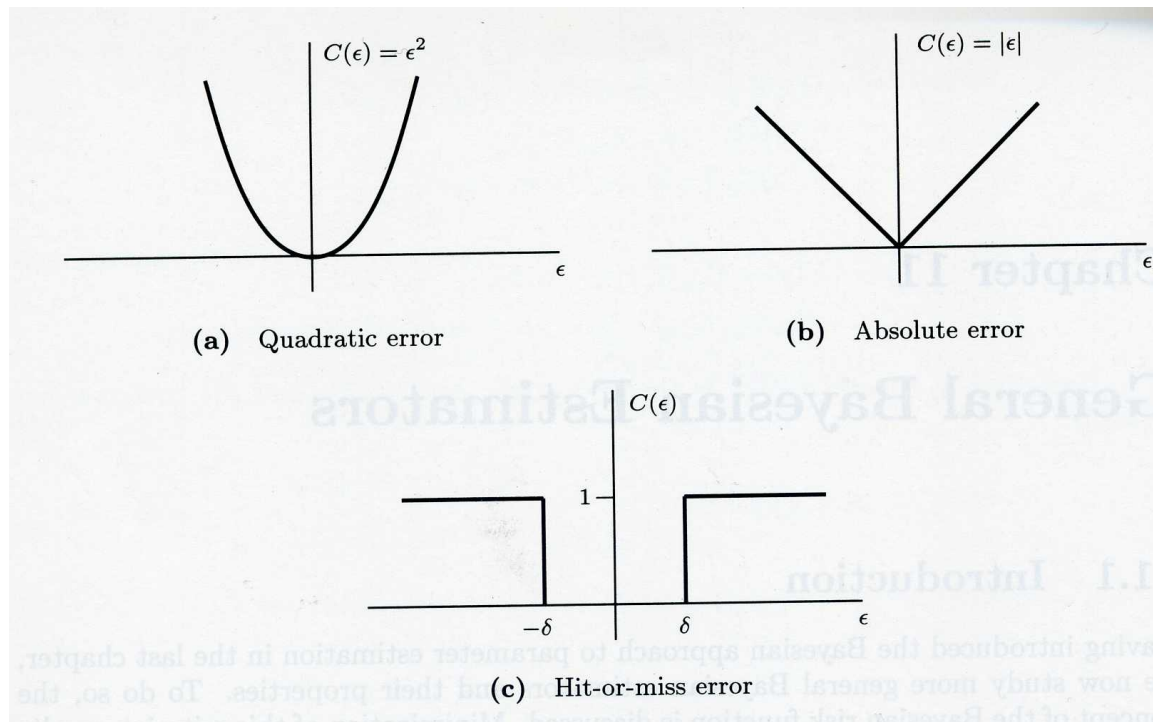
■ In our problems this relationship is mathematically described by pdf of Y conditioned to $\mathbf{X}$ (the *a posteriori* pdf), or equivalently by the joint pdf of Y and $\mathbf{X}$.

■ **Bayesian approach:** the Bayes method to derive the estimator $\Phi(\mathbf{x})$ consists on the definition of a cost function C , which is a function of the estimation error ε and measures the cost of making an error, and then attempts to minimize the average cost R , which is called **Bayes' risk**:

$$\begin{aligned} R &\triangleq E_{Y,\mathbf{X}} \{C(\varepsilon)\} = E_{Y,\mathbf{X}} \left\{ C(Y - \hat{Y}(\mathbf{X})) \right\} \\ &= \iiint C(y - \hat{y}(\mathbf{x})) f_{Y,\mathbf{X}}(y, \mathbf{x}) dy d\mathbf{x} \\ &= \iint \left[\int_{-\infty}^{+\infty} C(y - \hat{y}(\mathbf{x})) f_{Y|\mathbf{X}}(y|\mathbf{x}) dy \right] f_{\mathbf{X}}(\mathbf{x}) d\mathbf{x} \\ &= E_{\mathbf{X}} \left\{ E_{Y|\mathbf{X}} \left\{ C(Y - \hat{Y}(\mathbf{X})) \right\} \right\} \end{aligned}$$

Since $f_{\mathbf{X}}(\mathbf{x}) \geq 0 \Rightarrow \hat{Y} = \arg \min_{\hat{y}} R = \arg \min_{\hat{y}} E_{Y|\mathbf{X}} \left\{ C(Y - \hat{Y}(\mathbf{X})) \right\}$

- It is possible to define various cost functions; the most frequently used are the **quadratic cost function** and the **uniform cost function**, which produce the **Minimum Mean Square Error (MMSE)** estimator and the **Maximum A Posteriori (MAP)** estimator, respectively.
- The **absolute value cost function** produces the **Median estimator**.



■ **Quadratic cost function:** $C(\varepsilon) = |\varepsilon|^2 \Rightarrow R = E\{|\varepsilon|^2\} = MSE$

$$\hat{Y} = \arg \min_{\hat{Y}} E_{Y|\mathbf{X}} \left\{ C(Y - \hat{Y}(\mathbf{X})) \right\} = \arg \min_{\hat{Y}} E_{Y|\mathbf{X}} \left\{ |Y - \hat{Y}(\mathbf{X})|^2 \right\}$$

■ i.e.

$$\hat{Y} = \arg \min_{\hat{Y}} E_{Y|\mathbf{X}} \left\{ |Y - \hat{Y}(\mathbf{X})|^2 \right\} = \arg \min_{\hat{Y}} \left[\int_{-\infty}^{+\infty} |y - \hat{Y}(\mathbf{x})|^2 f_{Y|\mathbf{X}}(y|\mathbf{x}) dy \right]$$


$$\frac{\partial}{\partial \hat{Y}(\mathbf{X})} \left[\int_{-\infty}^{+\infty} |y - \hat{Y}(\mathbf{x})|^2 f_{Y|\mathbf{X}}(y|\mathbf{x}) dy \right] = -2 \int_{-\infty}^{+\infty} (y - \hat{Y}(\mathbf{x})) f_{Y|\mathbf{X}}(y|\mathbf{x}) dy = 0$$

$$\int_{-\infty}^{+\infty} (y - \hat{Y}(\mathbf{x})) f_{Y|\mathbf{X}}(y|\mathbf{x}) dy = 0 \quad \Rightarrow \quad \int_{-\infty}^{+\infty} y f_{Y|\mathbf{X}}(y|\mathbf{x}) dy = \int_{-\infty}^{+\infty} \hat{Y}(\mathbf{x}) f_{Y|\mathbf{X}}(y|\mathbf{x}) dy$$

$$\Rightarrow \hat{Y}(\mathbf{x}) \int_{-\infty}^{+\infty} f_{Y|\mathbf{X}}(y|\mathbf{x}) dy = \int_{-\infty}^{+\infty} y f_{Y|\mathbf{X}}(y|\mathbf{x}) dy$$

$$\Rightarrow \hat{Y}(\mathbf{x}) = \int_{-\infty}^{+\infty} y f_{Y|\mathbf{X}}(y|\mathbf{x}) dy \quad \Rightarrow \quad \hat{Y}_{MMSE} = E\{Y|\mathbf{X}\}$$


$$\begin{aligned} R &= E_{\mathbf{X}} \left\{ E_{Y|\mathbf{X}} \left\{ |Y - \hat{Y}(\mathbf{X})|^2 \right\} \right\} = E_{\mathbf{X}} \left\{ E_{Y|\mathbf{X}} \left\{ |Y - E\{Y|\mathbf{X}\}|^2 \right\} \right\} \\ &= E_{\mathbf{X}} \left\{ \text{var}\{Y|\mathbf{X}\} \right\} = \text{MSE}\{\hat{Y}_{MMSE}\} \end{aligned}$$

- The MMSE estimator of Y given the data vector $\mathbf{X}$ is an **unbiased** estimator:

$$\hat{Y}_{MMSE} = E\{Y|\mathbf{X}\} = E_{Y|\mathbf{X}}\{Y\} = \eta_{Y|\mathbf{X}} \quad \text{mean value of the a-posteriori pdf of } Y$$

$$\begin{aligned} E\{Y - \hat{Y}_{MMSE}\} &= E_{Y,\mathbf{X}}\{Y - \hat{Y}_{MMSE}\} = E_{Y,\mathbf{X}}\{Y\} - E_{Y,\mathbf{X}}\{\hat{Y}_{MMSE}\} \\ &= E_Y\{Y\} - E_{\mathbf{X}}\{\hat{Y}_{MMSE}\} = \eta_Y - E_{\mathbf{X}}\{E_{Y|\mathbf{X}}\{Y\}\} \\ &= \eta_Y - E_{Y,\mathbf{X}}\{Y\} = \eta_Y - \eta_Y = 0 \quad \rightarrow \quad \text{unbiased} \end{aligned}$$

$$\begin{aligned} MSE\{\hat{Y}_{MMSE}\} &= E\left\{\left(Y - \hat{Y}_{MMSE}\right)^2\right\} = E\left\{\left(Y - E_{Y|\mathbf{X}}\{Y\}\right)^2\right\} \\ &= E_{Y,\mathbf{X}}\left\{\left(Y - \eta_{Y|\mathbf{X}}\right)^2\right\} = E_{\mathbf{X}}\left\{E_{Y|\mathbf{X}}\left\{\left(Y - \eta_{Y|\mathbf{X}}\right)^2\right\}\right\} \\ &= E_{\mathbf{X}}\left\{\sigma_{Y|\mathbf{X}}^2\right\} \quad \rightarrow \quad \text{mean value of the conditional variance} \end{aligned}$$

- In summary, for the quadratic cost function we get: $R=\text{MSE}$ and

$$\hat{Y}_{MMSE} = E\{Y|\mathbf{X}\} = \int_{-\infty}^{+\infty} y f_{Y|\mathbf{X}}(y|\mathbf{x}) dy \quad \text{mean value of the a-posteriori pdf}$$

$$MSE\{\hat{Y}_{MMSE}\} = E_{\mathbf{X}}\{\text{var}\{Y|\mathbf{X}\}\} \quad \text{mean value w.r.t. } \mathbf{x} \text{ of the conditional variance}$$

$\uparrow$
 per costruzione non polarizzati

- Hence, to derive the **MMSE estimator** of the random parameter Y given the observed data vector $\mathbf{X}$, we should derive the **a-posteriori pdf**, i.e. the pdf of Y given $\mathbf{X}$, and then derive its (conditional) mean and variance. These quantities provide the MMSE estimator and its MSE, respectively (after averaging w.r.t. $\mathbf{X}$).
- The **a-posteriori pdf** describes the statistical behavior of Y after we have observed $\mathbf{X}$. Before the observation of $\mathbf{X}$, the statistical model of Y is provided by its a-priori pdf $f_Y(y)$.

- **Uniform cost function ("hit-or-miss"):** *(l'effetto non dipende dall'entità dell'errore,*

$$C(\varepsilon) = 1 - \text{rect}\left(\frac{\varepsilon}{2\delta}\right) = \begin{cases} 0 & |\varepsilon| \leq \delta \\ 1 & |\varepsilon| > \delta \end{cases}$$

- We should solve the minimization problem: $\hat{Y} = \arg \min_{\hat{y}} E_{Y|\mathbf{X}} \left\{ C(Y - \hat{Y}(\mathbf{X})) \right\}$ *valor medio condizionato*

■ where:

$$\begin{aligned} E_{Y|\mathbf{X}} \left\{ C(Y - \hat{Y}(\mathbf{X})) \right\} &= \int_{-\infty}^{+\infty} C(y - \hat{y}(\mathbf{x})) f_{Y|\mathbf{X}}(y|\mathbf{x}) dy \\ &= \int_{-\infty}^{+\infty} \left(1 - \text{rect}\left(\frac{y - \hat{y}(\mathbf{x})}{2\delta}\right) \right) f_{Y|\mathbf{X}}(y|\mathbf{x}) dy \\ &= \int_{-\infty}^{+\infty} f_{Y|\mathbf{X}}(y|\mathbf{x}) dy - \int_{-\infty}^{+\infty} \text{rect}\left(\frac{y - \hat{y}(\mathbf{x})}{2\delta}\right) f_{Y|\mathbf{X}}(y|\mathbf{x}) dy \\ &= 1 - \int_{\hat{y}-\delta}^{\hat{y}+\delta} f_{Y|\mathbf{X}}(y|\mathbf{x}) dy \end{aligned}$$

Soddisfare la quantità di normalizzazione

$$\begin{aligned}\hat{Y} &= \arg \min_{\hat{y}} E_{Y|\mathbf{X}} \left\{ C(Y - \hat{Y}(\mathbf{X})) \right\} \\ &= \arg \min_{\hat{y}} \left[1 - \int_{\hat{y}-\delta}^{\hat{y}+\delta} f_{Y|\mathbf{X}}(y|\mathbf{x}) dy \right] \\ &= \arg \max_{\hat{y}} \left[\int_{\hat{y}-\delta}^{\hat{y}+\delta} f_{Y|\mathbf{X}}(y|\mathbf{x}) dy \right]\end{aligned}$$

■ If δ is small, the a-posteriori pdf can be assumed constant over the interval of size 2δ , so that:

cerco dove centrare l'intervallo affinché
l'intervallo sia massimo.

$$\hat{Y} = \arg \max_{\hat{y}} \left[\int_{\hat{y}-\delta}^{\hat{y}+\delta} f_{Y|\mathbf{X}}(y|\mathbf{x}) dy \right] = \arg \max_{\hat{y}} \left[\underbrace{f_{Y|\mathbf{X}}(\hat{y}|\mathbf{x}) \cdot 2\delta}_{\text{moda}} \right] = \arg \max_y f_{Y|\mathbf{X}}(y|\mathbf{x})$$

$$\text{NMSE} = \text{CM}$$

individuale la moda rispetto valore medio
può essere meglio

Bayesian Estimation of Random Parameters

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■ In summary, if we adopt the "hit-or-miss" cost function, the estimator is given by the **mode** of the a-posteriori pdf and it is called the **Maximum A Posteriori (MAP)** estimator:

$$\hat{Y}_{MAP} = \arg \max_y f_{Y|X}(y|\mathbf{x})$$

Se scelgo MAP ho due $\int$ in meno ma non minimizza NMSE

■ Note that to maximize the a-posteriori pdf is equivalent to maximize the joint pdf of Y and $\mathbf{X}$, since the denominator $f_{\mathbf{X}}(\mathbf{x})$ does not depend on y :

$$\begin{aligned} \hat{Y}_{MAP} &= \arg \max_y f_{Y|X}(y|\mathbf{x}) = \arg \max_y \frac{f_{Y,\mathbf{X}}(y,\mathbf{x})}{f_{\mathbf{X}}(\mathbf{x})} = \arg \max_y \frac{f_{\mathbf{X}|Y}(\mathbf{x}|y) f_Y(y)}{f_{\mathbf{X}}(\mathbf{x})} \quad \xrightarrow{\text{then bayes}} \\ &= \arg \max_y f_{\mathbf{X}|Y}(\mathbf{x}|y) f_Y(y) = \arg \max_y \underbrace{LF(y) f_Y(y)}_{\substack{\text{Data } \mathbf{x} \text{ deterministic} \\ \downarrow \\ \text{MI} \rightarrow \text{param. deterministic e max LF}}} \end{aligned}$$

se sono interessato alla moda me ne freghe

where $f_{\mathbf{X}}(\mathbf{x}) = \int f_{Y,\mathbf{X}}(y,\mathbf{x}) dy = \int f_{\mathbf{X}|Y}(\mathbf{x}|y) f_Y(y) dy$

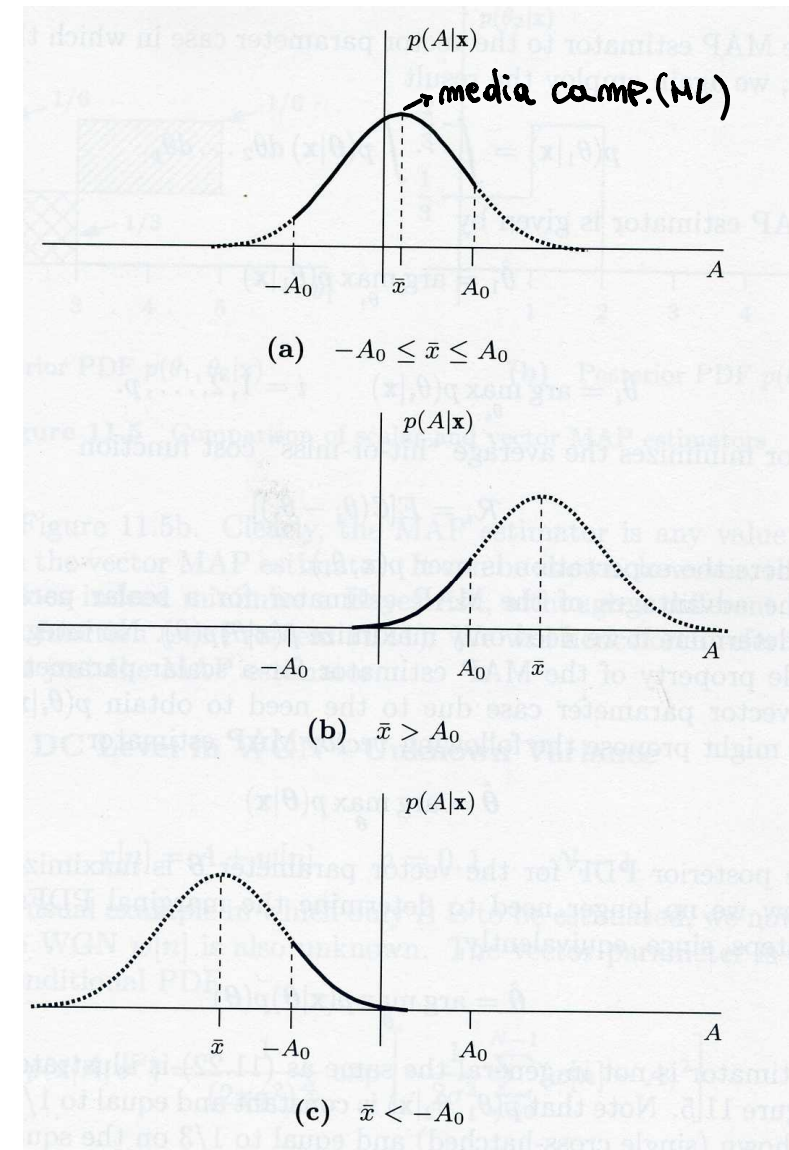
MAP $\rightarrow$ polarizza verso i valori più probabili a priori

MAP $\rightarrow$ distance LF moltiplicando per f_Y (data a priori) 12 che aumenta i valori più probabili.

■ **Example:** derive the MAP estimator of a constant random signal A based on the observation of the N samples $X_i = A + W_i$, where the noise samples W_i are IID, zero mean and Gaussian, and knowing that A is uniformly distributed between $-A_0$ and A_0 :

$$\hat{A}_{MAP} = \arg \max_a f_{\mathbf{X}|A}(\mathbf{x}|a) f_A(a)$$

■ As we already know, the maximum of the pdf of $\mathbf{X}$ conditioned to A is given by the sample mean of the data (which is also the ML estimate of A). The a-priori pdf of A relocates the position of the maximum, favouring those values of A that are more likely to be observed.



- Moreover, to maximize the joint pdf is equivalent to maximize its logarithm (or any monotonic increasing function of it):

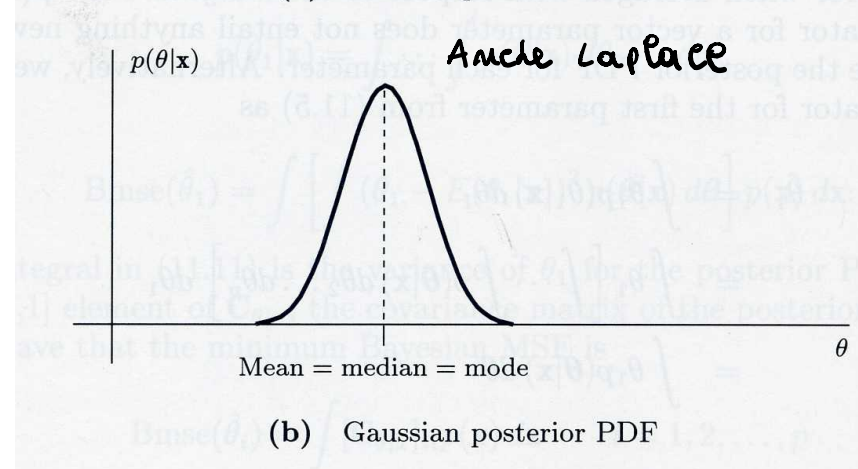
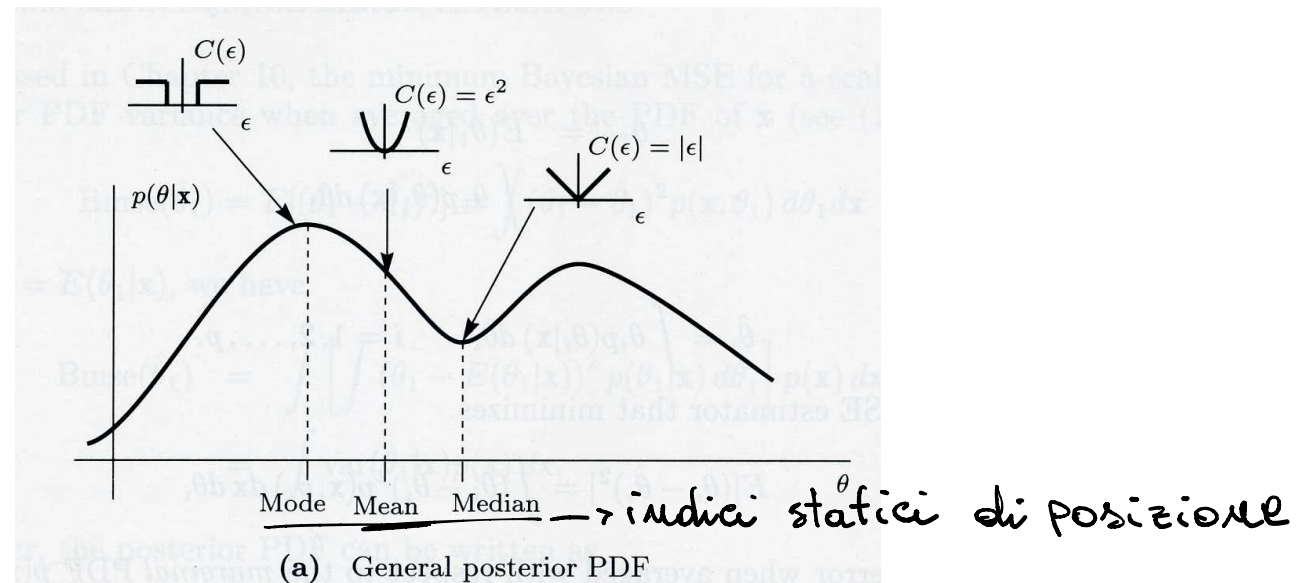
$$\begin{aligned}\hat{Y}_{MAP} &= \arg \max_y \ln f_{Y,\mathbf{X}}(y, \mathbf{x}) = \arg \max_y \ln \left[f_{\mathbf{X}|Y}(\mathbf{x}|y) f_Y(y) \right] \\ &= \arg \max_y \left[\ln f_{\mathbf{X}|Y}(\mathbf{x}|y) + \ln f_Y(y) \right]\end{aligned}$$

- Hence, so as for the ML estimate, also the MAP estimate can be derived by differentiating (if the function is differentiable) the log of the joint pdf and solving the equation that we get by setting the derivative equal to zero (it is the so-called **MAP equation**):

$$\frac{\partial \left[f_{\mathbf{X}|Y}(\mathbf{x}|y) f_Y(y) \right]}{\partial y} = 0, \quad \text{or} \quad \frac{\partial \left[\ln f_{\mathbf{X}|Y}(\mathbf{x}|y) + \ln f_Y(y) \right]}{\partial y} = 0$$

→ sarebbe ML
no info a priori
→ ML

- In the case we adopt the **absolute cost function**, $C(\epsilon)=|\epsilon|$, we obtain that the optimal estimate is given by the **median** of the a-posteriori pdf:



■ The concept of Cramér-Rao lower bound (CRB) introduced for the estimation of deterministic parameters has been also extended to the case of random parameters. It is usually called the **Bayesian CRB (BCRB)**:

$$MSE\{\hat{Y}\} = E_{Y,\mathbf{X}}\left\{\left(Y - \hat{Y}(\mathbf{X})\right)^2\right\} \geq BCRB(Y) = -\frac{1}{E_{Y,\mathbf{X}}\left\{\frac{\partial^2 \ln f_{Y,\mathbf{X}}(y, \mathbf{x})}{\partial y^2}\right\}}$$

■ The **Bayesian CRB** is usually below the **CRB** for deterministic parameters, since the knowledge of the a-priori pdf brings additional information that allows to improve the estimation performance (if properly exploited, e.g. by deriving the MMSE estimator).

$$\begin{aligned}
 -E_{Y,\mathbf{X}} \left\{ \frac{\partial^2 \ln f_{Y,\mathbf{X}}(y,\mathbf{x})}{\partial y^2} \right\} &= -E_{Y,\mathbf{X}} \left\{ \frac{\partial^2 \ln f_{\mathbf{X}|Y}(\mathbf{x}|y)}{\partial y^2} + \frac{\partial^2 \ln f_Y(y)}{\partial y^2} \right\} \\
 &= -E_{Y,\mathbf{X}} \left\{ \frac{\partial^2 \ln f_{\mathbf{X}|Y}(\mathbf{x}|y)}{\partial y^2} \right\} - E_Y \left\{ \frac{\partial^2 \ln f_Y(y)}{\partial y^2} \right\} \\
 &= E_Y \left\{ -E_{\mathbf{X}|Y} \left\{ \frac{\partial^2 \ln f_{\mathbf{X}|Y}(\mathbf{x}|y)}{\partial y^2} \right\} \right\} - E_Y \left\{ \frac{\partial^2 \ln f_Y(y)}{\partial y^2} \right\} \\
 &= E_Y \{ I(y) \} - E_Y \left\{ \frac{\partial^2 \ln f_Y(y)}{\partial y^2} \right\} \geq E_Y \{ I(y) \} \\
 &\rightarrow \text{since } -E_Y \left\{ \frac{\partial^2 \ln f_Y(y)}{\partial y^2} \right\} \geq 0
 \end{aligned}$$

where $I(y) \triangleq -E_{\mathbf{X}|Y} \left\{ \frac{\partial^2 \ln f_{\mathbf{X}|Y}(\mathbf{x}|y)}{\partial y^2} \right\}$ is the Fisher information.

MMSE estimate by using only the a priori information 18

■ Which is the MMSE estimate of a random parameter θ if only the *a-priori* pdf of θ is available?

■ If no data are available, we have to estimate θ with a constant $c \rightarrow$ Which is the value of c which minimizes the MSE?

$$\begin{aligned}\hat{\theta}_{ap} = c &\Rightarrow \varepsilon = \theta - \hat{\theta}_{ap} = \theta - c \Rightarrow \hat{\theta}_{ap} = \arg \min_c E\left\{(\theta - c)^2\right\} \\ E\left\{(\theta - c)^2\right\} &= E\left\{(\theta - \eta_\theta + \eta_\theta - c)^2\right\} = E\left\{((\theta - \eta_\theta) - (c - \eta_\theta))^2\right\} \\ &= E\left\{(\theta - \eta_\theta)^2 - 2(\theta - \eta_\theta)(c - \eta_\theta) + (c - \eta_\theta)^2\right\} = \sigma_\theta^2 + (c - \eta_\theta)^2 \geq \sigma_\theta^2\end{aligned}$$

Errore quadratico medio che dipende dalla varianza.

$$\hat{\theta}_{ap} = \arg \min_c E\left\{(\theta - c)^2\right\} = \eta_\theta \Rightarrow \begin{cases} E\left\{\varepsilon_{ap}\right\} = E\left\{\theta - \hat{\theta}_{ap}\right\} = E\left\{\theta - \eta_\theta\right\} = 0 \\ MSE\left\{\hat{\theta}_{ap}\right\} = E\left\{\varepsilon_{ap}^2\right\} = E\left\{(\theta - \eta_\theta)^2\right\} = \sigma_\theta^2 \end{cases}$$

- The Bayes estimate of a random parameter θ , when both the *a-priori* pdf of θ and the data vector $\mathbf{X}$ are available, exploits both these two “sources” of information through the *a-posteriori* pdf:

$$\hat{\theta}_{MMSE} = E\{\theta|\mathbf{x}\} = \int_{-\infty}^{+\infty} \theta f_{\theta|\mathbf{x}}(\theta|\mathbf{x}) d\theta = \int_{-\infty}^{+\infty} \theta \frac{f_{\mathbf{x}|\theta}(\mathbf{x}|\theta) f_{\theta}(\theta)}{f_{\mathbf{x}}(\mathbf{x})} d\theta = \frac{\int_{-\infty}^{+\infty} \theta f_{\mathbf{x}|\theta}(\mathbf{x}|\theta) f_{\theta}(\theta) d\theta}{\int_{-\infty}^{+\infty} f_{\mathbf{x}|\theta}(\mathbf{x}|\theta) f_{\theta}(\theta) d\theta}$$

$$\hat{\theta}_{MAP} = \arg \max_{\theta} f_{\theta|\mathbf{x}}(\theta|\mathbf{x}) = \arg \max_{\theta} \frac{f_{\mathbf{x}|\theta}(\mathbf{x}|\theta) f_{\theta}(\theta)}{f_{\mathbf{x}}(\mathbf{x})} = \arg \max_{\theta} LF(\theta) f_{\theta}(\theta)$$

- The ML estimator of θ , modelled as a deterministic unknown parameter, relies only on the data vector $\mathbf{X}$ (but not on the *a-priori* pdf of θ):

$$\hat{\theta}_{ML} = \arg \max_{\theta} f_{\mathbf{x}|\theta}(\mathbf{x}|\theta) = \arg \max_{\theta} LF(\theta)$$

The Gaussian case: single observation

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ipotesi congiuntamente gaussiane.

■ Let us consider a system of 2 r.v.'s, X and Y , jointly Gaussian:

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sqrt{(1-\rho_{XY}^2)\sigma_X^2\sigma_Y^2}} \exp\left[-\frac{1}{2(1-\rho_{XY}^2)}Q(x,y)\right]$$

Attuale situazione

$$y(t) = x(t) + w(t)$$

Gaussiani

quindi gaussiano

$$Q(x,y) = \left(\frac{x-\eta_X}{\sigma_X}\right)^2 - 2\rho_{XY}\left(\frac{x-\eta_X}{\sigma_X}\right)\left(\frac{y-\eta_Y}{\sigma_Y}\right) + \left(\frac{y-\eta_Y}{\sigma_Y}\right)^2$$

where

$$\eta_X = E\{X\}, \quad \eta_Y = E\{Y\}$$

7 sezioni orizzontali vengono ellissi

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

= indipendenti

→ Dipende da 5 parametri

$$\sigma_X^2 = E\{(X-\eta_X)^2\}, \quad \sigma_Y^2 = E\{(Y-\eta_Y)^2\}$$

$$\rho_{XY} = \frac{c_{XY}}{\sigma_X\sigma_Y} = \frac{E\{(X-\eta_X)(Y-\eta_Y)\}}{\sigma_X\sigma_Y}$$

sono linearmente indipendenti

is the correlation coefficient between X and Y

indipendenza $\Rightarrow$ no corr.

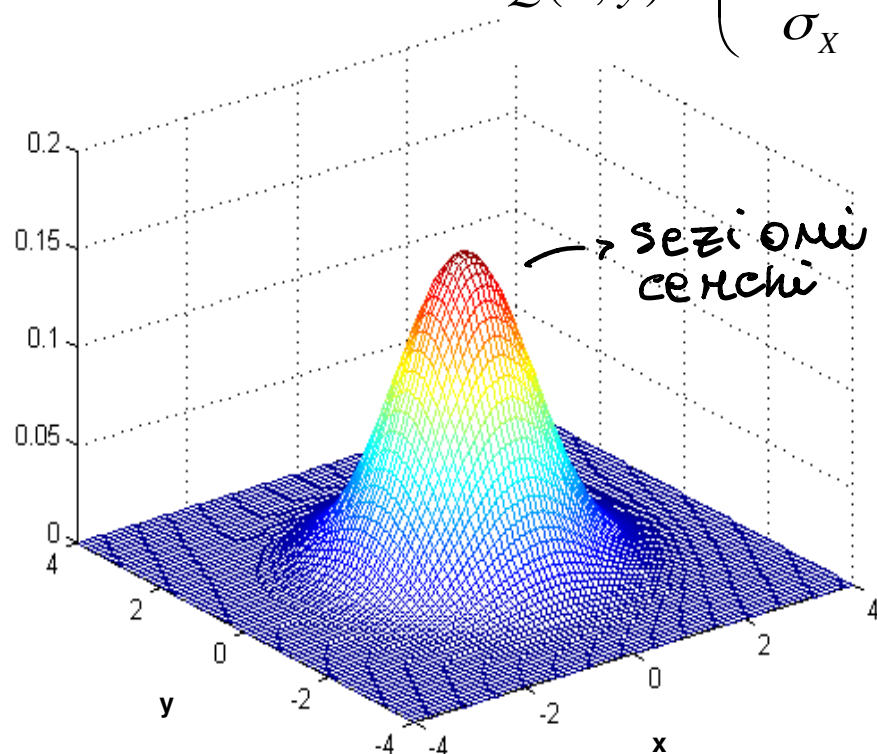
$\Leftarrow$ solo caso gaussiano.

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sqrt{(1-\rho_{XY}^2)}\sigma_X\sigma_Y} \exp\left[-\frac{1}{2(1-\rho_{XY}^2)}Q(x,y)\right]$$

$$Q(x,y) = \left(\frac{x-\eta_X}{\sigma_X}\right)^2 - 2\rho_{XY}\left(\frac{x-\eta_X}{\sigma_X}\right)\left(\frac{y-\eta_Y}{\sigma_Y}\right) + \left(\frac{y-\eta_Y}{\sigma_Y}\right)^2$$

$\neq$ inclinazione rispetto x,y

$\neq$ ellisse



$$\eta_X = \eta_Y = 0$$

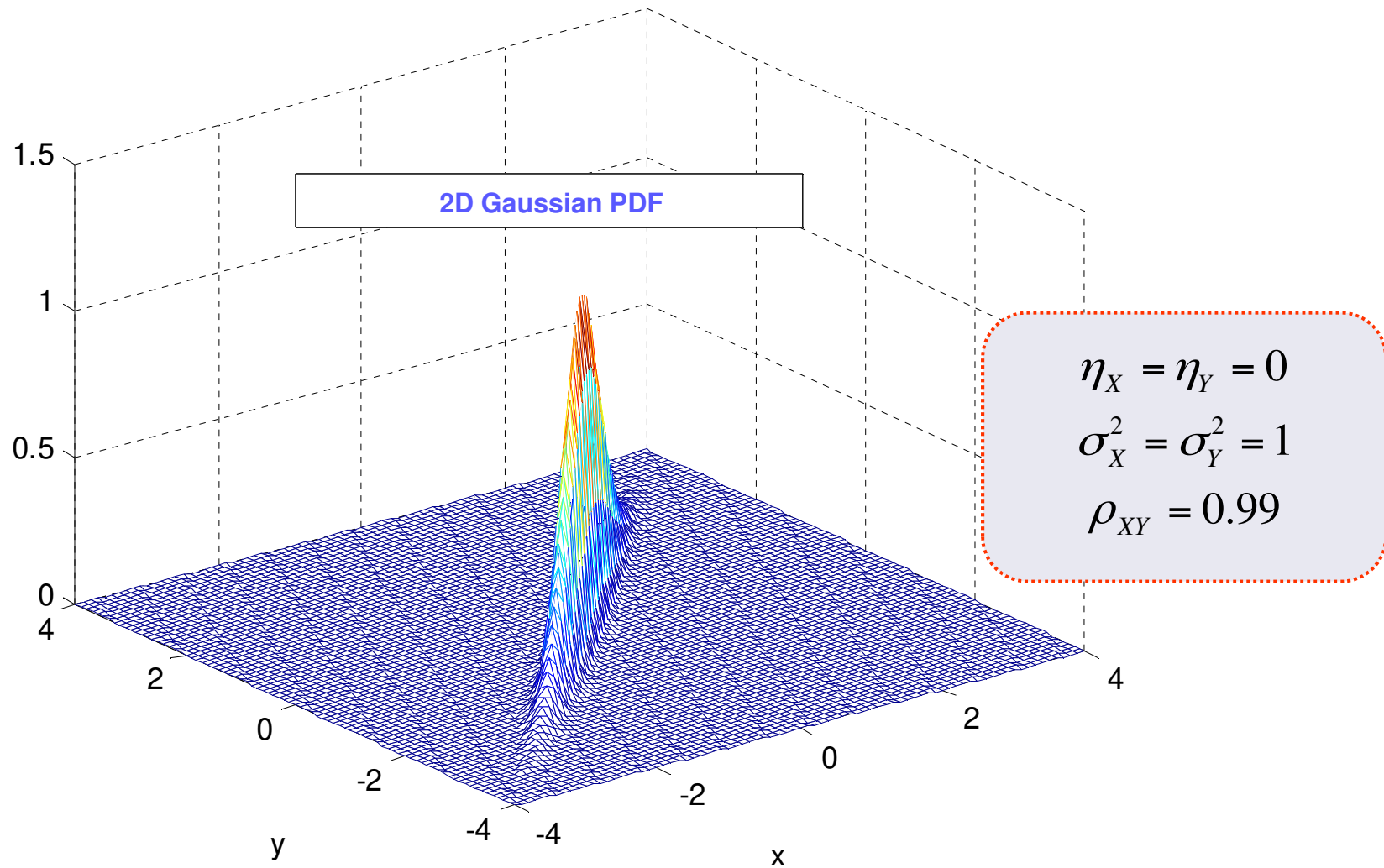
$$\sigma_X^2 = \sigma_Y^2 = 1$$

$$\rho_{XY} = 0$$

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

The Gaussian case: single observation

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■ **Problem:** estimate X (which is not directly observable), given the observation of Y , which is correlated with X (they are jointly Gaussian).

■ The **MMSE** estimator of X given the observed data Y , in the case where X and Y are jointly Gaussian is given by the conditional mean, i.e. the mean of the a-posteriori distribution:

$$\hat{X}_{MMSE} = E\{X|Y\} = \int_{-\infty}^{+\infty} x f_{X|Y}(x|y) dx$$


$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{\frac{1}{2\pi\sqrt{(1-\rho_{XY}^2)\sigma_X^2\sigma_Y^2}} e^{-\frac{1}{2(1-\rho_{XY}^2)}\left[\left(\frac{x-\eta_X}{\sigma_X}\right)^2 - 2\rho_{XY}\left(\frac{x-\eta_X}{\sigma_X}\right)\left(\frac{y-\eta_Y}{\sigma_Y}\right) + \left(\frac{y-\eta_Y}{\sigma_Y}\right)^2\right]}}{\frac{1}{\sqrt{2\pi\sigma_Y^2}} e^{-\frac{1}{2}\left(\frac{y-\eta_Y}{\sigma_Y}\right)^2}}$$

$$\begin{aligned}
 f_{X|Y}(x|y) &= \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{\frac{1}{2\pi\sqrt{(1-\rho_{XY}^2)}\sigma_X^2\sigma_Y^2} e^{-\frac{1}{2(1-\rho_{XY}^2)}\left[\left(\frac{x-\eta_X}{\sigma_X}\right)^2 - 2\rho_{XY}\left(\frac{x-\eta_X}{\sigma_X}\right)\left(\frac{y-\eta_Y}{\sigma_Y}\right) + \left(\frac{y-\eta_Y}{\sigma_Y}\right)^2\right]}}{\frac{1}{\sqrt{2\pi}\sigma_Y^2} e^{-\frac{1}{2}\left(\frac{y-\eta_Y}{\sigma_Y}\right)^2}} \\
 &= \frac{1}{\sqrt{2\pi(1-\rho_{XY}^2)}\sigma_X^2} e^{-\frac{1}{2(1-\rho_{XY}^2)}\left[\left(\frac{x-\eta_X}{\sigma_X}\right)^2 - 2\rho_{XY}\left(\frac{x-\eta_X}{\sigma_X}\right)\left(\frac{y-\eta_Y}{\sigma_Y}\right) + \left(\frac{y-\eta_Y}{\sigma_Y}\right)^2\right] + \frac{1}{2}\left(\frac{y-\eta_Y}{\sigma_Y}\right)^2} \\
 &= \frac{1}{\sqrt{2\pi(1-\rho_{XY}^2)}\sigma_X^2} e^{-\frac{1}{2(1-\rho_{XY}^2)}\left[\left(\frac{x-\eta_X}{\sigma_X}\right)^2 - 2\rho_{XY}\left(\frac{x-\eta_X}{\sigma_X}\right)\left(\frac{y-\eta_Y}{\sigma_Y}\right) + \left(\frac{y-\eta_Y}{\sigma_Y}\right)^2 - (1-\rho_{XY}^2)\left(\frac{y-\eta_Y}{\sigma_Y}\right)^2\right]} \\
 &= \frac{1}{\sqrt{2\pi(1-\rho_{XY}^2)}\sigma_X^2} e^{-\frac{1}{2(1-\rho_{XY}^2)}\left[\left(\frac{x-\eta_X}{\sigma_X}\right)^2 - 2\rho_{XY}\left(\frac{x-\eta_X}{\sigma_X}\right)\left(\frac{y-\eta_Y}{\sigma_Y}\right) + \rho_{XY}^2\left(\frac{y-\eta_Y}{\sigma_Y}\right)^2\right]}
 \end{aligned}$$

$$\begin{aligned}
 f_{X|Y}(x|y) &= \frac{1}{\sqrt{2\pi(1-\rho_{XY}^2)\sigma_X^2}} e^{-\frac{1}{2(1-\rho_{XY}^2)} \left[\left(\frac{x-\eta_X}{\sigma_X} \right)^2 - 2\rho_{XY} \left(\frac{x-\eta_X}{\sigma_X} \right) \left(\frac{y-\eta_Y}{\sigma_Y} \right) + \rho_{XY}^2 \left(\frac{y-\eta_Y}{\sigma_Y} \right)^2 \right]} \\
 &= \frac{1}{\sqrt{2\pi(1-\rho_{XY}^2)\sigma_X^2}} e^{-\frac{1}{2(1-\rho_{XY}^2)} \left(\frac{x-\eta_X}{\sigma_X} - \rho_{XY} \frac{y-\eta_Y}{\sigma_Y} \right)^2} \\
 &= \frac{1}{\sqrt{2\pi(1-\rho_{XY}^2)\sigma_X^2}} e^{-\frac{1}{2(1-\rho_{XY}^2)\sigma_X^2} \left(x - \eta_X - \rho_{XY} \frac{\sigma_X}{\sigma_Y} (y - \eta_Y) \right)^2} = \frac{1}{\sqrt{2\pi\sigma_{X|Y}^2}} e^{-\frac{(x-\eta_{X|Y})^2}{2\sigma_{X|Y}^2}}
 \end{aligned}$$

→ Tra due variabili congiuntamente Gaussianne ci può essere solo legame lineare.

$$\eta_{X|Y} = E\{X|Y\} = \eta_X + \rho_{XY} \frac{\sigma_X}{\sigma_Y} (Y - \eta_Y) \rightarrow \text{Regression line}$$

$$\sigma_{X|Y}^2 = \text{var}\{X|Y\} = E\left\{\left(X - \eta_{X|Y}\right)^2 | Y\right\} = E\{X^2|Y\} - \left(E\{X|Y\}\right)^2 = \sigma_X^2 (1 - \rho_{XY}^2)$$

The Gaussian case: single observation

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$\rho = 1$, legame lineare deterministico

- The **MMSE** estimator of X given the observed data Y , in the case where X and Y are jointly Gaussian r.v.'s is given by:

$$\hat{X}_{MMSE} = E\{X|Y\} = \eta_{X|Y} = \eta_X + \rho_{XY} \frac{\sigma_X}{\sigma_Y} (Y - \eta_Y)$$

→ per il dato in base a quanto è forte il suo legame col parametro da stimare

$$MSE\{\hat{X}_{MMSE}\} = E_{Y,X} \left\{ \left(X - \hat{X}_{MMSE} \right)^2 \right\} = E_Y \left\{ E_{X|Y} \left\{ \left(X - \hat{X}_{MMSE} \right)^2 \right\} \right\}$$

$$\begin{aligned} E_{X|Y} \left\{ \left(X - \hat{X}_{MMSE} \right)^2 \right\} &= E \left\{ \left(X - \hat{X}_{MMSE} \right)^2 \middle| Y \right\} = E \left\{ \left(X - \eta_{X|Y} \right)^2 \middle| Y \right\} \\ &= \text{var}\{X|Y\} = \sigma_{X|Y}^2 = \sigma_X^2 (1 - \rho_{XY}^2) \end{aligned}$$

$$MSE\{\hat{X}_{MMSE}\} = E_Y \left\{ \sigma_X^2 (1 - \rho_{XY}^2) \right\} = \sigma_X^2 (1 - \rho_{XY}^2)$$

→ ottima cosa

■ The conditional mean and conditional variance can also be expressed in terms of the covariance between X and Y :

$$\eta_{X|Y} = \eta_X + \rho_{XY} \frac{\sigma_X}{\sigma_Y} (Y - \eta_Y) = \eta_X + \rho_{XY} \frac{\sigma_X \sigma_Y}{\sigma_Y^2} (Y - \eta_Y) = \eta_X + c_{XY} (\sigma_Y^2)^{-1} (Y - \eta_Y)$$

$$\sigma_{X|Y}^2 = \sigma_X^2 (1 - \rho_{XY}^2) = \sigma_X^2 \left(1 - \frac{c_{XY}^2}{\sigma_X^2 \sigma_Y^2} \right) = \sigma_X^2 - \frac{c_{XY}^2}{\sigma_Y^2} = \sigma_X^2 - c_{XY} (\sigma_Y^2)^{-1} c_{XY}$$

$$\eta_{X|Y} = \eta_X + \rho_{XY} \frac{\sigma_X}{\sigma_Y} (Y - \eta_Y) = \eta_X + c_{XY} (\sigma_Y^2)^{-1} (Y - \eta_Y)$$

$$\sigma_{X|Y}^2 = \sigma_X^2 (1 - \rho_{XY}^2) = \sigma_X^2 - c_{XY} (\sigma_Y^2)^{-1} c_{YX}$$

$$\begin{aligned}
 E_{Y,X} \left\{ \left(X - \hat{X}(Y) \right)^2 \right\} &\geq BCRB(X) = - \frac{1}{E_{Y,X} \left\{ \frac{\partial^2 \ln f_{Y,X}(y,x)}{\partial x^2} \right\}} \\
 &= - \frac{1}{2(1-\rho_{XY}^2)} E_{Y,X} \left\{ \frac{\partial^2}{\partial x^2} \left[\left(\frac{x-\eta_X}{\sigma_X} \right)^2 - 2\rho_{XY} \left(\frac{x-\eta_X}{\sigma_X} \right) \left(\frac{y-\eta_Y}{\sigma_Y} \right) + \left(\frac{y-\eta_Y}{\sigma_Y} \right)^2 \right] \right\} \\
 &= - \frac{1}{2(1-\rho_{XY}^2)} E_{Y,X} \left\{ \frac{\partial^2}{\partial x^2} \left(\frac{x-\eta_X}{\sigma_X} \right)^2 \right\} = - \frac{1}{\sigma_X^2 (1-\rho_{XY}^2)}
 \end{aligned}$$

$$BCRB(X) = \sigma_X^2 (1 - \rho_{XY}^2) = MSE \left\{ \hat{X}_{MMSE} \right\}$$

È efficiente e non polarizzato

■ **Example:** Observed data = Signal + Noise $\rightarrow Y=S+W$

$S \in \mathcal{N}(\eta_s, \sigma_s^2)$, $W \in \mathcal{N}(0, \sigma_w^2)$, S and W independent

$$SNR = \frac{E\{S^2\}}{E\{W^2\}} = \frac{\eta_s^2 + \sigma_s^2}{\sigma_w^2} = \frac{\sigma_s^2}{\sigma_w^2} \left(1 + \frac{\eta_s^2}{\sigma_s^2} \right) = \gamma (1 + \kappa)$$

MAP se è tutto Gaussiano.

- Find the **MMSE estimator** of the random signal S given the observed data Y : we should derive the pdf of S given Y (the so-called *a-posteriori* pdf).
- Since S and Y are jointly Gaussian, we already know that the pdf of S given Y is a Gaussian pdf, so we can derive directly the conditional mean and variance by using the previous results.

$$\hat{S}_{MMSE} = E\{S|Y\} = \eta_S + \rho_{YS} \frac{\sigma_S}{\sigma_Y} (Y - \eta_Y)$$

$$\eta_Y = E\{Y\} = E\{S + W\} = \eta_S$$

$$\sigma_Y^2 = \text{var}\{Y\} = \text{var}\{S + W\} = \sigma_S^2 + \sigma_W^2 = \sigma_W^2 (1 + \gamma)$$

$$\rho_{YS} = \frac{\text{cov}\{Y, S\}}{\sqrt{\sigma_S^2 (\sigma_S^2 + \sigma_W^2)}} = \frac{E\{(Y - \eta_Y)(S - \eta_S)\}}{\sqrt{\sigma_S^2 (\sigma_S^2 + \sigma_W^2)}} = \frac{E\{(S + W - \eta_S)(S - \eta_S)\}}{\sqrt{\sigma_S^2 (\sigma_S^2 + \sigma_W^2)}}$$

$$\begin{aligned} & \frac{(S - \eta_S)^2 + W(S - \eta_S) \text{cov} = 0}{\sigma_S^2} \\ & = \frac{\sigma_S^2}{\sqrt{\sigma_S^2 (\sigma_S^2 + \sigma_W^2)}} = \sqrt{\frac{\sigma_S^2}{\sigma_S^2 + \sigma_W^2}} = \sqrt{\frac{\gamma}{\gamma + 1}} \end{aligned}$$

$$\gamma \triangleq \frac{\sigma_S^2}{\sigma_W^2}$$

$$\hat{S}_{MMSE} = E\{S|Y\} = \eta_S + \rho_{YS} \frac{\sigma_S}{\sigma_Y} (Y - \eta_Y)$$

$$\eta_Y = \eta_S, \quad \sigma_Y^2 = \sigma_W^2 (1 + \gamma), \quad \rho_{YS} = \sqrt{\frac{\gamma}{\gamma + 1}}$$

$$\hat{S}_{MMSE} = \eta_S + \sqrt{\frac{\gamma}{\gamma + 1}} \sqrt{\frac{\sigma_S^2}{\sigma_Y^2}} (Y - \eta_S) = \eta_S + \frac{\gamma}{\gamma + 1} (Y - \eta_S)$$

$$\hat{S}_{MMSE} = \alpha Y + (1 - \alpha) \eta_S$$

Y migliore stima che posso fare avendo n valori.

$$\text{where } \alpha \triangleq \frac{\gamma}{\gamma + 1} \quad \text{and} \quad \gamma \triangleq \frac{\sigma_S^2}{\sigma_W^2}$$

The Gaussian case: single observation

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■ Hence:

$\alpha \rightarrow 0$, uso meno dati ma più info. a priori
 $\alpha \rightarrow \infty$, Aumenta SNR uso meno info. a priori
 e più ai dati

$$\hat{S}_{MMSE} = \alpha Y + (1 - \alpha)\eta_S, \quad \text{where } \alpha \triangleq \frac{\gamma}{\gamma + 1} \quad \text{and} \quad \gamma \triangleq \frac{\sigma_S^2}{\sigma_W^2}$$

↓
 quanto mi fido dei dati e info a priori (colpo che lineare)

$$MSE\{\hat{S}_{MMSE}\} = \sigma_S^2(1 - \rho_{YS}^2) = \sigma_S^2\left(1 - \frac{\gamma}{\gamma + 1}\right) = (1 - \alpha)\sigma_S^2$$

$$MSE\{\hat{S}_{MMSE}\} = (1 - \alpha) \cdot MSE\{\hat{S}_{ap}\} \leq MSE\{\hat{S}_{ap}\}$$

$$MSE\{\hat{S}_{MMSE}\} = MSE\{\hat{S}_{ap}\} \Leftrightarrow \gamma \rightarrow 0 \quad (\sigma_S^2 \rightarrow 0 \quad \text{or} \quad \sigma_W^2 \rightarrow +\infty)$$

The Gaussian case: single observation

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- Let us compare the **MMSE** estimator with the **ML** estimator:

$$\hat{S}_{ML} = \arg \max_s \ln f_{Y|S}(y|s) = Y, \quad \text{since } Y|S \in \mathcal{N}(S, \sigma_W^2)$$

$$MSE\{\hat{S}_{ML}\} = E\left\{\left(S - \hat{S}_{ML}\right)^2\right\} = E\left\{(S - Y)^2\right\} = E\{W^2\} = \sigma_W^2$$

$$MSE\{\hat{S}_{MMSE}\} = \sigma_S^2 \left(1 - \frac{\gamma}{\gamma + 1}\right) = \sigma_S^2 \left(1 - \frac{\sigma_S^2}{\sigma_S^2 + \sigma_W^2}\right) = \frac{\sigma_W^2 \sigma_S^2}{\sigma_W^2 + \sigma_S^2} = \frac{\gamma}{\gamma + 1} \sigma_W^2 = \alpha \sigma_W^2$$

$$MSE\{\hat{S}_{MMSE}\} = \alpha \cdot MSE\{\hat{S}_{ML}\} \leq MSE\{\hat{S}_{ML}\}$$

info. a priori com
big variance tendem
a zero

$$MSE\{\hat{S}_{MMSE}\} = MSE\{\hat{S}_{ML}\} \Leftrightarrow \gamma \rightarrow +\infty \quad (\sigma_S^2 \rightarrow +\infty \text{ or } \sigma_W^2 \rightarrow 0)$$

The Gaussian case: single observation

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■ In summary:

$$\hat{S}_{MMSE} = \alpha Y + (1 - \alpha)\eta_S = \alpha \hat{S}_{ML} + (1 - \alpha)\hat{S}_{ap}, \quad \text{where } \alpha \triangleq \frac{\gamma}{\gamma + 1} \quad \text{and} \quad \gamma \triangleq \frac{\sigma_S^2}{\sigma_W^2}$$
$$MSE\left\{\hat{S}_{MMSE}\right\} = (1 - \alpha)\sigma_S^2 = (1 - \alpha)MSE\left\{\hat{S}_{ap}\right\} = \alpha\sigma_W^2 = \alpha MSE\left\{\hat{S}_{ML}\right\}$$

$$\sigma_S^2 = \text{cost.} \left\{ \begin{array}{l} \sigma_W^2 \rightarrow +\infty \Rightarrow \gamma = \frac{\sigma_S^2}{\sigma_W^2} \rightarrow 0 \Rightarrow \alpha \rightarrow 0 \Rightarrow \begin{cases} \hat{S}_{MMSE} \rightarrow \eta_S \\ MSE\left\{\hat{S}_{MMSE}\right\} \rightarrow \sigma_S^2 \end{cases} \\ \sigma_W^2 \rightarrow 0 \Rightarrow \gamma = \frac{\sigma_S^2}{\sigma_W^2} \rightarrow +\infty \Rightarrow \alpha \rightarrow 1 \Rightarrow \begin{cases} \hat{S}_{MMSE} \rightarrow Y = S \\ MSE\left\{\hat{S}_{MMSE}\right\} \rightarrow 0 \end{cases} \end{array} \right.$$

- Let us now consider the system formed by 2 vectors $\mathbf{X}$ and $\mathbf{Y}$, jointly Gaussian and mutually correlated. The joint pdf of $\mathbf{X}$ and $\mathbf{Y}$ is derived as follows:

$$\underset{k \times 1}{\mathbf{X}} \in \mathcal{N}(\underset{l \times 1}{\boldsymbol{\eta}}_X, \mathbf{C}_X), \quad \underset{l \times 1}{\mathbf{Y}} \in \mathcal{N}(\underset{l \times 1}{\boldsymbol{\eta}}_Y, \mathbf{C}_Y), \quad \underset{k \times l}{\mathbf{C}}_{XY} = E\left\{(\mathbf{X} - \boldsymbol{\eta}_X)(\mathbf{Y} - \boldsymbol{\eta}_Y)^T\right\}$$

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$$\underset{(k+l) \times 1}{\mathbf{Z}} = \begin{bmatrix} \mathbf{X} \\ \mathbf{Y} \end{bmatrix} \rightarrow \mathbf{Z} \in \mathcal{N}(\boldsymbol{\eta}_Z, \mathbf{C}_Z)$$

$$\underset{(k+l) \times 1}{\boldsymbol{\eta}}_Z = E\{\mathbf{Z}\} = E\left\{\begin{bmatrix} \mathbf{X} \\ \mathbf{Y} \end{bmatrix}\right\} = \begin{bmatrix} \boldsymbol{\eta}_X \\ \boldsymbol{\eta}_Y \end{bmatrix}$$

$$\underset{(k+l) \times (k+l)}{\mathbf{C}}_Z = E\left\{(\mathbf{Z} - \boldsymbol{\eta}_Z)(\mathbf{Z} - \boldsymbol{\eta}_Z)^T\right\} = E\left\{\left(\begin{bmatrix} \mathbf{X} \\ \mathbf{Y} \end{bmatrix} - \begin{bmatrix} \boldsymbol{\eta}_X \\ \boldsymbol{\eta}_Y \end{bmatrix}\right)\left(\begin{bmatrix} \mathbf{X} \\ \mathbf{Y} \end{bmatrix} - \begin{bmatrix} \boldsymbol{\eta}_X \\ \boldsymbol{\eta}_Y \end{bmatrix}\right)^T\right\}$$

$$\begin{aligned} \mathbf{C}_Z &= E \left\{ \begin{bmatrix} \mathbf{X} - \boldsymbol{\eta}_X \\ \mathbf{Y} - \boldsymbol{\eta}_Y \end{bmatrix} \begin{bmatrix} \mathbf{X} - \boldsymbol{\eta}_X \\ \mathbf{Y} - \boldsymbol{\eta}_Y \end{bmatrix}^T \right\} = E \left\{ \begin{bmatrix} \mathbf{X} - \boldsymbol{\eta}_X \\ \mathbf{Y} - \boldsymbol{\eta}_Y \end{bmatrix} \begin{bmatrix} (\mathbf{X} - \boldsymbol{\eta}_X)^T & (\mathbf{Y} - \boldsymbol{\eta}_Y)^T \end{bmatrix} \right\} \\ &= E \left\{ \begin{bmatrix} (\mathbf{X} - \boldsymbol{\eta}_X)(\mathbf{X} - \boldsymbol{\eta}_X)^T & (\mathbf{X} - \boldsymbol{\eta}_X)(\mathbf{Y} - \boldsymbol{\eta}_Y)^T \\ (\mathbf{Y} - \boldsymbol{\eta}_Y)(\mathbf{X} - \boldsymbol{\eta}_X)^T & (\mathbf{Y} - \boldsymbol{\eta}_Y)(\mathbf{Y} - \boldsymbol{\eta}_Y)^T \end{bmatrix} \right\} = \begin{bmatrix} \mathbf{C}_X & \mathbf{C}_{XY} \\ \mathbf{C}_{YX} & \mathbf{C}_Y \end{bmatrix}, \end{aligned}$$

$$\text{where } \mathbf{C}_{XY} \triangleq E \left\{ (\mathbf{X} - \boldsymbol{\eta}_X)(\mathbf{Y} - \boldsymbol{\eta}_Y)^T \right\} = \left(E \left\{ (\mathbf{Y} - \boldsymbol{\eta}_Y)(\mathbf{X} - \boldsymbol{\eta}_X)^T \right\} \right)^T = \left(\mathbf{C}_{YX} \right)^T$$

$$f_Z(\mathbf{z}) \equiv f_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y}) = \frac{1}{(2\pi)^{\frac{k+l}{2}} \sqrt{|\mathbf{C}_Z|}} \exp \left[-\frac{1}{2} \begin{bmatrix} \mathbf{x} - \boldsymbol{\eta}_X \\ \mathbf{y} - \boldsymbol{\eta}_Y \end{bmatrix}^T \begin{bmatrix} \mathbf{C}_X & \mathbf{C}_{XY} \\ \mathbf{C}_{YX} & \mathbf{C}_Y \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{x} - \boldsymbol{\eta}_X \\ \mathbf{y} - \boldsymbol{\eta}_Y \end{bmatrix} \right]$$

- It can be proved that the pdf of $\mathbf{X}$ given $\mathbf{Y}$ is a multidimensional **Gaussian pdf**:

$$f_{\mathbf{X}|\mathbf{Y}}(\mathbf{x}|\mathbf{y}) = \frac{f_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y})}{f_{\mathbf{Y}}(\mathbf{y})} = \frac{1}{(2\pi)^{k/2} \sqrt{|\mathbf{C}_{\mathbf{X}|\mathbf{Y}}|}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\eta}_{\mathbf{X}|\mathbf{Y}})^T \mathbf{C}_{\mathbf{X}|\mathbf{Y}}^{-1} (\mathbf{x} - \boldsymbol{\eta}_{\mathbf{X}|\mathbf{Y}}) \right]$$

- with **mean vector** and **covariance matrix** given by:

$$\boldsymbol{\eta}_{\mathbf{X}|\mathbf{Y}} = E\{\mathbf{X}|\mathbf{Y}\} = \boldsymbol{\eta}_{\mathbf{X}} + \mathbf{C}_{\mathbf{XY}} \mathbf{C}_{\mathbf{Y}}^{-1} (\mathbf{Y} - \boldsymbol{\eta}_{\mathbf{Y}}) \quad [\text{It is a **linear** function of } \mathbf{Y}]$$

$$\mathbf{C}_{\mathbf{X}|\mathbf{Y}} = E\left\{(\mathbf{X} - \boldsymbol{\eta}_{\mathbf{X}|\mathbf{Y}})(\mathbf{X} - \boldsymbol{\eta}_{\mathbf{X}|\mathbf{Y}})^T \middle| \mathbf{Y}\right\} = \mathbf{C}_{\mathbf{X}} - \mathbf{C}_{\mathbf{XY}} \mathbf{C}_{\mathbf{Y}}^{-1} \mathbf{C}_{\mathbf{YX}} = \mathbf{C}_{\mathbf{X}} - \mathbf{C}_{\mathbf{YX}}^T \mathbf{C}_{\mathbf{Y}}^{-1} \mathbf{C}_{\mathbf{YX}}$$

- that are very similar to those derived in the scalar case (see at pag. 27).

- The Gaussian vector case is a straightforward generalization of the Gaussian scalar case:

$$\left\{ \begin{array}{l} \eta_{X|Y} = \eta_X + c_{XY} \left(\sigma_Y^2 \right)^{-1} (Y - \eta_Y) \\ \sigma_{X|Y}^2 = \sigma_X^2 - c_{XY} \left(\sigma_Y^2 \right)^{-1} c_{YX} \end{array} \right.$$



$$\left\{ \begin{array}{l} \boldsymbol{\eta}_{X|Y} = \boldsymbol{\eta}_X + \mathbf{C}_{XY} \mathbf{C}_Y^{-1} (\mathbf{Y} - \boldsymbol{\eta}_Y) \\ \mathbf{C}_{X|Y} = \mathbf{C}_X - \mathbf{C}_{XY} \mathbf{C}_Y^{-1} \mathbf{C}_{YX} = \mathbf{C}_X - \mathbf{C}_{YX}^T \mathbf{C}_Y^{-1} \mathbf{C}_{YX} \end{array} \right.$$

- This means that the **MMSE estimate** of vector **X** given the observed data vector **Y**, when vectors **X** and **Y** are **jointly Gaussian** distributed, is given by:

$$\hat{\mathbf{X}}_{MMSE} = \boldsymbol{\eta}_{X|Y} = E\{\mathbf{X}|\mathbf{Y}\} = \boldsymbol{\eta}_X + \mathbf{C}_{XY}\mathbf{C}_Y^{-1}(\mathbf{Y} - \boldsymbol{\eta}_Y)$$

- The MMSE estimator is **linear**, hence it is **Gaussian** distributed.
- The **MSE matrix** is obtained by averaging with respect to the observed data vector the conditional covariance matrix:

$$MSE\{\hat{\mathbf{X}}_{MMSE}\} = E_{\mathbf{Y}}\{\mathbf{C}_{X|Y}\} = E_{\mathbf{Y}}\{\mathbf{C}_X - \mathbf{C}_{YX}^T\mathbf{C}_Y^{-1}\mathbf{C}_{YX}\} = \mathbf{C}_X - \mathbf{C}_{YX}^T\mathbf{C}_Y^{-1}\mathbf{C}_{YX}$$

- It can be proved that in the Gaussian case, it coincides with the **Bayes CRB**.

- We have introduced the **Deterministic Linear Model**, which applies when the useful signal $\mathbf{S}$ can be expressed as the linear combination of p vectors $\{\mathbf{h}_i\}$, i.e. $\mathbf{S}$ belongs to the subspace generated by the p vectors $\{\mathbf{h}_i\}$, with $p < N$:

$$\mathbf{Y}_{N \times 1} = \mathbf{S} + \mathbf{W} = \mathbf{H}\boldsymbol{\theta} + \mathbf{W} = \sum_{i=1}^p \theta_i \mathbf{h}_i + \mathbf{W}$$

$$\mathbf{H}_{N \times p} = [\mathbf{h}_1 \quad \mathbf{h}_2 \quad \cdots \quad \mathbf{h}_p], \quad \boldsymbol{\theta}_{p \times 1} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_p \end{bmatrix}$$

$$\mathbf{W} \in \mathcal{N}(\mathbf{0}, \mathbf{C}_W), \quad \mathbf{C}_W = E\{\mathbf{W}\mathbf{W}^T\}$$

■ **Problem:** estimate θ given the observed data vector $\mathbf{Y}$, assuming that $\mathbf{H}$ is perfectly known.

$$\mathbf{Y}_{N \times 1} = \mathbf{H}\boldsymbol{\theta} + \mathbf{W} = \sum_{i=1}^p \theta_i \mathbf{h}_i + \mathbf{W}$$

■ We have seen that when θ is modelled as a vector of **deterministic unknown parameters** and the additive noise is Gaussian, we can resort to the **ML method** to estimate θ :

$$\hat{\boldsymbol{\theta}}_{ML} = \arg \max_{\boldsymbol{\theta}} \ln f_{\mathbf{Y}}(\mathbf{y}; \boldsymbol{\theta}) = \left(\mathbf{H}^T \mathbf{C}_W^{-1} \mathbf{H} \right)^{-1} \mathbf{H}^T \mathbf{C}_W^{-1} \mathbf{Y}$$

■ Note that the estimator is **linear**, it is also called **BLUE**.

- The **ML estimator** of θ is **linear, unbiased, Gaussian, efficient, and consistent**:

$$\left\{ \begin{array}{l} \hat{\theta}_{ML} = \left(\mathbf{H}^T \mathbf{C}_W^{-1} \mathbf{H} \right)^{-1} \mathbf{H}^T \mathbf{C}_W^{-1} \mathbf{Y} \\ E \left\{ \hat{\theta}_{ML} \right\} = \theta \\ MSE \left\{ \hat{\theta}_{ML} \right\} = CRB(\theta) = \left(\mathbf{H}^T \mathbf{C}_W^{-1} \mathbf{H} \right)^{-1} \end{array} \right.$$

$$\hat{\theta}_{ML} \in \mathcal{N}(\theta, CRB(\theta))$$

- **Assumption**: vector θ can be modelled as a random vector with known mean vector and covariance matrix $\rightarrow$ the **Bayesian Linear Model**.

- If we assume that it is Gaussian distributed: $\theta \in \mathcal{N}(\boldsymbol{\eta}_\theta, \mathbf{C}_\theta)$

$\rightarrow$ the **Gaussian Linear Model**.

■ **Problem:** find the **MMSE** estimator of θ given the observed data vector $\mathbf{Y}$, assuming that the matrix $\mathbf{H}$ is a-priori perfectly known and that θ can be modelled as a **Gaussian** random vector with known pdf: $\theta \in \mathcal{N}(\boldsymbol{\eta}_\theta, \mathbf{C}_\theta)$

$$\mathbf{Y}_{N \times 1} = \mathbf{H} \boldsymbol{\theta} + \mathbf{W} = \sum_{i=1}^p \theta_i \mathbf{h}_i + \mathbf{W}$$

$$\mathbf{H}_{N \times p} = [\mathbf{h}_1 \quad \mathbf{h}_2 \quad \cdots \quad \mathbf{h}_p], \quad \boldsymbol{\theta}_{p \times 1} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_p \end{bmatrix}$$

$$\mathbf{W} \in \mathcal{N}(\mathbf{0}, \mathbf{C}_W), \quad \mathbf{C}_W = E\{\mathbf{W}\mathbf{W}^T\}$$

$$\mathbf{S}_{N \times 1} = \mathbf{H} \boldsymbol{\theta}_{p \times 1} \quad \text{Signal of interest (SOI)} \equiv \text{Gaussian random vector} \rightarrow \hat{\mathbf{S}}_{MMSE} = \mathbf{H} \hat{\boldsymbol{\theta}}_{MMSE}$$

$$\boldsymbol{\theta} \in \mathcal{N}(\boldsymbol{\eta}_\theta, \mathbf{C}_\theta), \quad \mathbf{Y} | \boldsymbol{\theta} \in \mathcal{N}(\mathbf{H}\boldsymbol{\theta}, \mathbf{C}_w)$$

$$f_{\mathbf{Y}|\boldsymbol{\theta}}(\mathbf{y}|\boldsymbol{\theta}) = \frac{1}{\sqrt{(2\pi)^N |\mathbf{C}_w|}} \exp\left(-\frac{1}{2}(\mathbf{y} - \mathbf{H}\boldsymbol{\theta})^T \mathbf{C}_w^{-1}(\mathbf{y} - \mathbf{H}\boldsymbol{\theta})\right)$$

$$\mathbf{Y} \in \mathcal{N}(\boldsymbol{\eta}_Y, \mathbf{C}_Y)$$

$$\boldsymbol{\eta}_Y = E\{\mathbf{Y}\} = E\{\mathbf{S} + \mathbf{W}\} = E\{\mathbf{H}\boldsymbol{\theta}\} = \mathbf{H}E\{\boldsymbol{\theta}\} = \mathbf{H}\boldsymbol{\eta}_\theta$$

$$\begin{aligned} \mathbf{C}_Y &= E\{(\mathbf{Y} - \boldsymbol{\eta}_Y)(\mathbf{Y} - \boldsymbol{\eta}_Y)^T\} = E\{(\mathbf{H}\boldsymbol{\theta} + \mathbf{W} - \mathbf{H}\boldsymbol{\eta}_\theta)(\mathbf{H}\boldsymbol{\theta} + \mathbf{W} - \mathbf{H}\boldsymbol{\eta}_\theta)^T\} \\ &= E\{(\mathbf{H}(\boldsymbol{\theta} - \boldsymbol{\eta}_\theta) + \mathbf{W})(\mathbf{H}(\boldsymbol{\theta} - \boldsymbol{\eta}_\theta) + \mathbf{W})^T\} \\ &= E\{(\mathbf{H}(\boldsymbol{\theta} - \boldsymbol{\eta}_\theta))(\mathbf{H}(\boldsymbol{\theta} - \boldsymbol{\eta}_\theta))^T\} + E\{\mathbf{W}\mathbf{W}^T\} \\ &= E\{\mathbf{H}(\boldsymbol{\theta} - \boldsymbol{\eta}_\theta)(\boldsymbol{\theta} - \boldsymbol{\eta}_\theta)^T \mathbf{H}^T\} + \mathbf{C}_w \\ &= \mathbf{H}\mathbf{C}_\theta \mathbf{H}^T + \mathbf{C}_w \end{aligned}$$

$$f_{\boldsymbol{\theta}|\mathbf{Y}}(\boldsymbol{\theta}|\mathbf{y}) = \frac{f_{\mathbf{Y}|\boldsymbol{\theta}}(\mathbf{y}|\boldsymbol{\theta})f_{\boldsymbol{\theta}}(\boldsymbol{\theta})}{f_{\mathbf{Y}}(\mathbf{y})} = \frac{1}{\sqrt{(2\pi)^p |\mathbf{C}_{\boldsymbol{\theta}|\mathbf{Y}}|}} \exp\left(-\frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\eta}_{\boldsymbol{\theta}|\mathbf{Y}})^T \mathbf{C}_{\boldsymbol{\theta}|\mathbf{Y}}^{-1} (\boldsymbol{\theta} - \boldsymbol{\eta}_{\boldsymbol{\theta}|\mathbf{Y}})\right)$$

$$\boldsymbol{\eta}_{\boldsymbol{\theta}|\mathbf{Y}} = E\{\boldsymbol{\theta}|\mathbf{Y}\} = \boldsymbol{\eta}_{\boldsymbol{\theta}} + \mathbf{C}_{\boldsymbol{\theta}\mathbf{Y}} \mathbf{C}_{\mathbf{Y}}^{-1} (\mathbf{Y} - \boldsymbol{\eta}_{\mathbf{Y}})$$

$$\mathbf{C}_{\boldsymbol{\theta}|\mathbf{Y}} = E\{(\boldsymbol{\theta} - \boldsymbol{\eta}_{\boldsymbol{\theta}|\mathbf{Y}})(\boldsymbol{\theta} - \boldsymbol{\eta}_{\boldsymbol{\theta}|\mathbf{Y}})^T | \mathbf{Y}\} = \mathbf{C}_{\boldsymbol{\theta}} - \mathbf{C}_{\boldsymbol{\theta}\mathbf{Y}} \mathbf{C}_{\mathbf{Y}}^{-1} \mathbf{C}_{\mathbf{Y}\boldsymbol{\theta}}$$

where: $\boldsymbol{\eta}_{\mathbf{Y}} = \mathbf{H}\boldsymbol{\eta}_{\boldsymbol{\theta}}, \quad \mathbf{C}_{\mathbf{Y}} = \mathbf{H}\mathbf{C}_{\boldsymbol{\theta}}\mathbf{H}^T + \mathbf{C}_{\mathbf{W}}$

$$\begin{aligned} \mathbf{C}_{\boldsymbol{\theta}\mathbf{Y}} &= (\mathbf{C}_{\mathbf{Y}\boldsymbol{\theta}})^T = E\{(\boldsymbol{\theta} - \boldsymbol{\eta}_{\boldsymbol{\theta}})(\mathbf{Y} - \boldsymbol{\eta}_{\mathbf{Y}})^T\} = E\{(\boldsymbol{\theta} - \boldsymbol{\eta}_{\boldsymbol{\theta}})(\mathbf{H}(\boldsymbol{\theta} - \boldsymbol{\eta}_{\boldsymbol{\theta}}) + \mathbf{W})^T\} \\ &= E\{(\boldsymbol{\theta} - \boldsymbol{\eta}_{\boldsymbol{\theta}})(\boldsymbol{\theta} - \boldsymbol{\eta}_{\boldsymbol{\theta}})^T \mathbf{H}^T + (\boldsymbol{\theta} - \boldsymbol{\eta}_{\boldsymbol{\theta}})\mathbf{W}^T\} \\ &= E\{(\boldsymbol{\theta} - \boldsymbol{\eta}_{\boldsymbol{\theta}})(\boldsymbol{\theta} - \boldsymbol{\eta}_{\boldsymbol{\theta}})^T \mathbf{H}^T\} + E\{(\boldsymbol{\theta} - \boldsymbol{\eta}_{\boldsymbol{\theta}})\mathbf{W}^T\} \\ &= E\{(\boldsymbol{\theta} - \boldsymbol{\eta}_{\boldsymbol{\theta}})(\boldsymbol{\theta} - \boldsymbol{\eta}_{\boldsymbol{\theta}})^T\} \mathbf{H}^T = \mathbf{C}_{\boldsymbol{\theta}} \mathbf{H}^T \end{aligned}$$

■ Hence, we find:

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$$\boldsymbol{\eta}_{\theta|Y} = \boldsymbol{\eta}_{\theta} + \mathbf{C}_{\theta} \mathbf{H}^T \left(\mathbf{H} \mathbf{C}_{\theta} \mathbf{H}^T + \overset{N \times N}{\mathbf{C}_W} \right)^{-1} (\mathbf{Y} - \mathbf{H} \boldsymbol{\eta}_{\theta})$$

$$\mathbf{C}_{\theta|Y} = \mathbf{C}_{\theta} - \mathbf{C}_{\theta} \mathbf{H}^T \left(\mathbf{H} \mathbf{C}_{\theta} \mathbf{H}^T + \mathbf{C}_W \right)^{-1} \mathbf{H} \mathbf{C}_{\theta}$$

e $N \gg P$

■ Using the **Matrix Inversion Lemma**, they can be expressed as:

$$\boldsymbol{\eta}_{\theta|Y} = \boldsymbol{\eta}_{\theta} + \left(\overset{P \times P}{\mathbf{C}_{\theta}^{-1}} + \mathbf{H}^T \overset{\text{Grosso calcolo computazionale}}{\mathbf{C}_W^{-1}} \mathbf{H} \right)^{-1} \mathbf{H}^T \mathbf{C}_W^{-1} (\mathbf{Y} - \mathbf{H} \boldsymbol{\eta}_{\theta})$$

$$\mathbf{C}_{\theta|Y} = \left(\mathbf{C}_{\theta}^{-1} + \mathbf{H}^T \mathbf{C}_W^{-1} \mathbf{H} \right)^{-1}$$



■ Matrix Inversion Lemma (a.k.a. Woodbury Identity):

$$\left(\mathbf{A} + \mathbf{B}\mathbf{D}^{-1}\mathbf{C}\right)^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{B}\left(\mathbf{D} + \mathbf{C}\mathbf{A}^{-1}\mathbf{B}\right)^{-1}\mathbf{C}\mathbf{A}^{-1}$$

where $\mathbf{A}$ and $\mathbf{D}$ are square matrices of full rank and $\mathbf{B}$ and $\mathbf{C}$ are rectangular matrices of proper dimensions.

$$\mathbf{A} = \mathbf{C}_{\theta}^{-1}, \quad \mathbf{B} = \mathbf{H}^T, \quad \mathbf{C} = \mathbf{H}, \quad \mathbf{D} = \mathbf{C}_w$$

$\Downarrow$

$$\left(\mathbf{C}_{\theta}^{-1} + \mathbf{H}^T\mathbf{C}_w^{-1}\mathbf{H}\right)^{-1} = \mathbf{C}_{\theta} - \mathbf{C}_{\theta}\mathbf{H}^T\left(\mathbf{C}_w + \mathbf{H}\mathbf{C}_{\theta}\mathbf{H}^T\right)^{-1}\mathbf{H}\mathbf{C}_{\theta}$$



■ Woodbury Identity (it comes in many variants):

$$\left(\mathbf{P}^{-1} + \mathbf{B}^T \mathbf{R}^{-1} \mathbf{B}\right)^{-1} \mathbf{B}^T \mathbf{R}^{-1} = \mathbf{P} \mathbf{B}^T \left(\mathbf{B} \mathbf{P} \mathbf{B}^T + \mathbf{R}\right)^{-1}$$

where $\mathbf{P}$ and $\mathbf{R}$ are square matrices of full rank and $\mathbf{B}$ a rectangular matrix of proper dimensions.

$$\mathbf{P} = \mathbf{C}_\theta, \quad \mathbf{B} = \mathbf{H}, \quad \mathbf{R} = \mathbf{C}_w$$

$\Downarrow$

$$\mathbf{C}_\theta \mathbf{H}^T \left(\mathbf{H} \mathbf{C}_\theta \mathbf{H}^T + \mathbf{C}_w\right)^{-1} = \left(\mathbf{C}_\theta^{-1} + \mathbf{H}^T \mathbf{C}_w^{-1} \mathbf{H}\right)^{-1} \mathbf{H}^T \mathbf{C}_w^{-1}$$

- In summary, the **MMSE** estimator of the Gaussian random vector θ and the covariance matrix of the estimation error (which coincides with the BCRB) are:

$$\hat{\theta}_{MMSE} = \eta_{\theta|Y} = \eta_{\theta} + \left(\mathbf{C}_{\theta}^{-1} + \mathbf{H}^T \mathbf{C}_w^{-1} \mathbf{H} \right)^{-1} \mathbf{H}^T \mathbf{C}_w^{-1} (\mathbf{Y} - \mathbf{H} \eta_{\theta})$$

$$MSE \left\{ \hat{\theta}_{MMSE} \right\} = \mathbf{C}_{\theta|Y} = \left(\mathbf{C}_{\theta}^{-1} + \mathbf{H}^T \mathbf{C}_w^{-1} \mathbf{H} \right)^{-1} = BCRB(\theta)$$

- The **MMSE estimator** of the parameters of the **Gaussian linear model** is a **linear** function of the data vector $\mathbf{Y}$, it is **efficient** and **unbiased**:

$$\begin{aligned} E \left\{ \theta - \hat{\theta}_{MMSE} \right\} &= \eta_{\theta} - E \left\{ \eta_{\theta|Y} \right\} = \eta_{\theta} - \eta_{\theta} - \left(\mathbf{C}_{\theta}^{-1} + \mathbf{H}^T \mathbf{C}_w^{-1} \mathbf{H} \right)^{-1} \mathbf{H}^T \mathbf{C}_w^{-1} E \left\{ (\mathbf{Y} - \mathbf{H} \eta_{\theta}) \right\} \\ &= - \left(\mathbf{C}_{\theta}^{-1} + \mathbf{H}^T \mathbf{C}_w^{-1} \mathbf{H} \right)^{-1} \mathbf{H}^T \mathbf{C}_w^{-1} \left(E \left\{ \mathbf{Y} \right\} - \mathbf{H} \eta_{\theta} \right) \\ &= - \left(\mathbf{C}_{\theta}^{-1} + \mathbf{H}^T \mathbf{C}_w^{-1} \mathbf{H} \right)^{-1} \mathbf{H}^T \mathbf{C}_w^{-1} \left(\mathbf{H} E \left\{ \theta \right\} - \mathbf{H} \eta_{\theta} \right) = \mathbf{0} \end{aligned}$$

■ **Example:** Observed signal $Y(t)$ = constant Gaussian signal S + additive white Gaussian noise (AWGN) $W(t)$.

$$Y(t) = S + W(t), \quad t \in [0, T) \quad \text{where } S \in \mathcal{N}(\eta_s, \sigma_s^2) \quad \text{and } S_w(f) = \frac{N_0}{2}$$

■ The observed signal is filtered with a low-pass anti-aliasing filter of bandwidth B (such that $2BT \gg 1$) and then sampled at intervals $T_s = 1/(2B)$:

$$Y_B(t) = Y(t) \otimes h_{LP}(t) \cong S + W(t) \otimes h_{LP}(t) = S + W_B(t), \quad t \in [0, T)$$

■ If $2BT \gg 1$, the spectrum of the useful signal is not affected by the filter:

$$S(f)H_{LP}(f) = S \cdot T \text{sinc}(fT) \cdot \text{rect}\left(\frac{f}{2B}\right) \cong S \cdot T \text{sinc}(fT) \quad \text{if } 2B \gg 1/T$$

- Power Spectral Density (PSD) and power of the filtered (bandlimited) noise:

$$S_{W_B}(f) = S_W(f) |H_{LP}(f)|^2 = \frac{N_0}{2} \text{rect}\left(\frac{f}{2B}\right) \rightarrow \sigma_{W_B}^2 = \int_{-\infty}^{+\infty} S_{W_B}(f) df = N_0 B$$

- The Autocorrelation Function (ACF) and the Power Spectral Density (PSD) of a process are related by the Fourier transform (**Wiener-Khintchine theorem**):

$$R_{W_B}(\tau) \stackrel{FT}{\Leftrightarrow} S_{W_B}(f)$$

$$S_{W_B}(f) = \frac{N_0}{2} \text{rect}\left(\frac{f}{2B}\right) \rightarrow R_{W_B}(\tau) = N_0 B \cdot \text{sinc}(2B\tau)$$

- We sample at intervals $T_s = 1/(2B)$.

■ The image of the observed (noisy) signal is:

$$\mathbf{Y} \triangleq \left[\frac{1}{\sqrt{2B}} Y_B(0) \quad \frac{1}{\sqrt{2B}} Y_B\left(\frac{1}{2B}\right) \quad \cdots \quad \frac{1}{\sqrt{2B}} Y_B\left(\frac{N-1}{2B}\right) \right]^T$$

$$Y_i \triangleq \frac{1}{\sqrt{2B}} Y_B\left(\frac{i}{2B}\right) = \frac{S}{\sqrt{2B}} + \frac{1}{\sqrt{2B}} W_B\left(\frac{i}{2B}\right) = A + W_i, \quad i = 0, 1, \dots, N-1$$

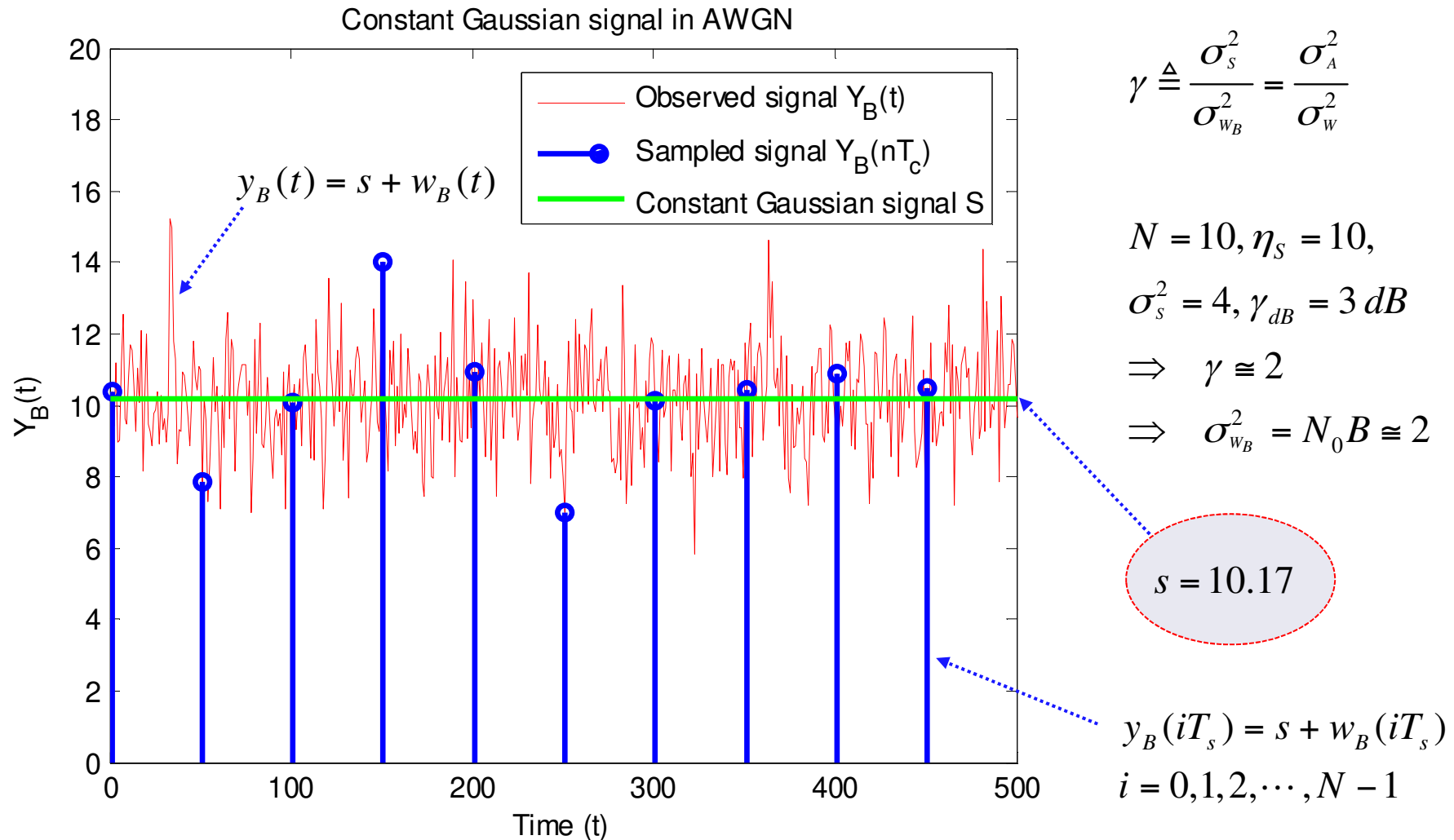
$$\text{where } A \triangleq \frac{S}{\sqrt{2B}}, \quad W_i \triangleq \frac{1}{\sqrt{2B}} W_B\left(\frac{i}{2B}\right), \quad \text{and } N \triangleq T/T_s = 2BT$$

$$A \in \mathcal{N}(\eta_A, \sigma_A^2) \quad \text{where } \eta_A = \frac{1}{\sqrt{2B}} \eta_s \quad \text{and} \quad \sigma_A^2 = \frac{1}{2B} \sigma_s^2$$

$$W_i \in \mathcal{N}(0, \sigma_w^2) \quad \text{where } \sigma_w^2 = \frac{1}{2B} \cdot N_0 B = \frac{N_0}{2}, \quad \{W_i\}_{i=0}^{N-1} \quad \text{IID}$$

Estimation of a constant Gaussian signal in AWGN

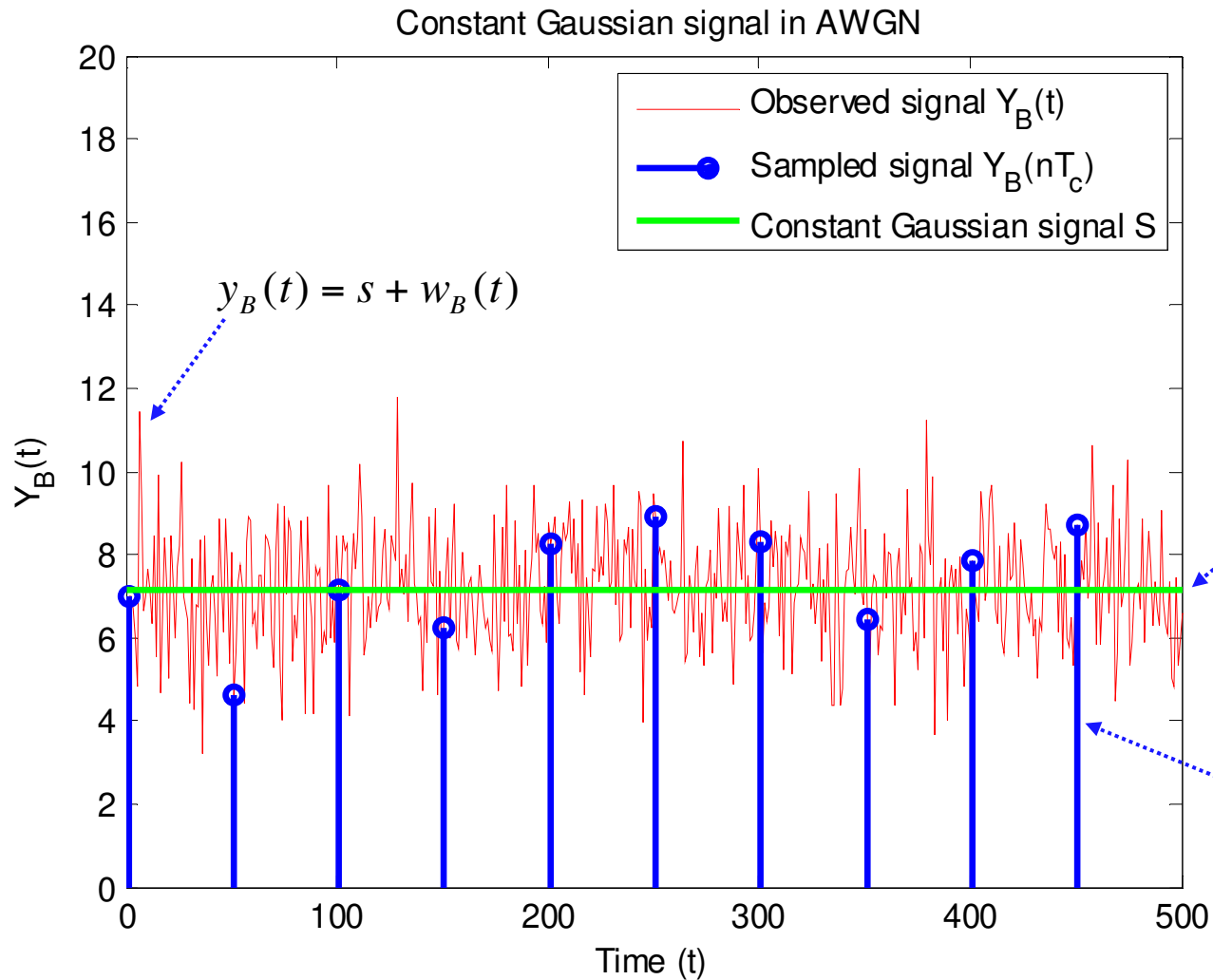
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■ Note that we do not want to estimate mean and variance of S , but the value assumed by S in the specific realization we observe.

Estimation of a constant Gaussian signal in AWGN

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$$N = 10, \eta_s = 10,$$

$$\sigma_s^2 = 4, \gamma_{dB} = 3 \text{ dB}$$

$$\Rightarrow \gamma \cong 2$$

$$\Rightarrow \sigma_{w_B}^2 = N_0 B \cong 2$$

■ In the previous realization $s=10.17$ and in this one $s=7.15$.

Observed data

Noise samples

Constant random signal

$$Y_i = A + W_i, \quad i = 1, 2, \dots, N \quad W_i \in \mathcal{N}(0, \sigma_w^2), \quad \text{IID} \quad A \in \mathcal{N}(\eta_A, \sigma_A^2)$$

Observed data vector

$$\mathbf{Y} \in \mathcal{N}(\boldsymbol{\eta}_Y, \mathbf{C}_Y)$$

$$\mathbf{Y} = A\mathbf{1} + \mathbf{W}$$

$$\mathbf{Y} = [Y_0 \ Y_1 \ \dots \ Y_{N-1}]^T$$

$$\mathbf{1} = [1 \ 1 \ \dots \ 1]^T$$

$$\mathbf{W} = [W_0 \ W_1 \ \dots \ W_{N-1}]^T$$

■ Let us concentrate on the estimate of A (once we have the estimate of A , the estimate of S is immediately obtained: $\hat{S}_{MMSE} = \sqrt{2B} \cdot \hat{A}_{MMSE}$)

■ The signal vector $\mathbf{S}$ is a particular case of the Gaussian linear model for $p=1$:

$$\mathbf{S} = \mathbf{H}\boldsymbol{\theta} = A\mathbf{1} \quad \rightarrow \quad \mathbf{H} = \mathbf{1}, \quad \boldsymbol{\theta} = A$$

■ Let us derive the **ML** and **MMSE** estimators by using previous results for the deterministic and Gaussian linear model, respectively:

$$\boldsymbol{\theta} = A, \quad \boldsymbol{\eta}_{\theta} = \eta_A, \quad \mathbf{C}_{\theta} = \sigma_A^2, \quad \mathbf{H} = \mathbf{1}, \quad \mathbf{C}_W = \sigma_W^2 \mathbf{I}$$

■ The **ML** estimator of S , i.e. the estimator that we derived under the assumption that A is deterministic (so which does not use the *a-priori* pdf, but only the observed data vector $\mathbf{Y}$) is given by the **Sample Mean**, in fact:

$$\hat{A}_{ML} = \left(\mathbf{H}^T \mathbf{C}_W^{-1} \mathbf{H} \right)^{-1} \mathbf{H}^T \mathbf{C}_W^{-1} \mathbf{Y} = \left(\frac{1}{\sigma_W^2} \mathbf{1}^T \mathbf{1} \right)^{-1} \frac{1}{\sigma_W^2} \mathbf{1}^T \mathbf{Y} = \frac{1}{N} \mathbf{1}^T \mathbf{Y} = \frac{1}{N} \sum_{i=0}^{N-1} Y_i$$

$$E \left\{ A - \hat{A}_{ML} \right\} = \eta_A - \frac{1}{N} \mathbf{1}^T E \left\{ \mathbf{Y} \right\} = \eta_A - \frac{\eta_A}{N} \mathbf{1}^T \mathbf{1} = \eta_A - \eta_A = 0$$

$$MSE \left\{ \hat{A}_{ML} \right\} = E \left\{ (A - \hat{A}_{ML})^2 \right\} = \left(\mathbf{H}^T \mathbf{C}_W^{-1} \mathbf{H} \right)^{-1} = \left(\frac{1}{\sigma_W^2} \mathbf{1}^T \mathbf{1} \right)^{-1} = \left(\frac{N}{\sigma_W^2} \right)^{-1} = \frac{\sigma_W^2}{N}$$

■ The ML estimator is linear, Gaussian distributed, unbiased, efficient, and consistent.

- The MSE when A is random can be derived as follows: $\mathbf{Y} = A\mathbf{1} + \mathbf{W}$

$$\begin{aligned} MSE\{\hat{A}_{ML}\} &= E\{(A - \hat{A}_{ML})^2\} = E\left\{\left(A - \frac{1}{N}\mathbf{1}^T \mathbf{Y}\right)^2\right\} = E\left\{\left(A - \frac{A}{N}\mathbf{1}^T \mathbf{1} - \frac{1}{N}\mathbf{1}^T \mathbf{W}\right)^2\right\} \\ &= E\left\{\left(-\frac{1}{N}\mathbf{1}^T \mathbf{W}\right)^2\right\} = \frac{1}{N^2} E\left\{(\mathbf{1}^T \mathbf{W})(\mathbf{1}^T \mathbf{W})^T\right\} = \frac{1}{N^2} \mathbf{1}^T E\{\mathbf{W}\mathbf{W}^T\} \mathbf{1} \\ &= \frac{\sigma_W^2}{N^2} \mathbf{1}^T \mathbf{1} = \frac{\sigma_W^2}{N} \end{aligned}$$

- which is the same as that we derived under the assumption that A is a deterministic parameter:

$$MSE\{\hat{A}_{ML}\}\Big|_{A \text{ det}} = CRB(A)\Big|_{A \text{ det}} = (\mathbf{H}^T \mathbf{C}_W^{-1} \mathbf{H})^{-1} = \frac{\sigma_W^2}{N}$$

■ Now let us derive the **MMSE estimator** of A , i.e. the optimal estimator which makes use of both the *a-priori* information (the *a-priori* pdf) and the N observed data $\mathbf{Y}$, by exploiting previous results for the **Gaussian linear model**:

$$\boldsymbol{\theta} = A, \quad \boldsymbol{\eta}_\theta = \eta_A, \quad \mathbf{C}_\theta = \sigma_A^2, \quad \mathbf{H} = \mathbf{1}, \quad \mathbf{C}_W = \sigma_W^2 \mathbf{I}$$

$$\hat{A}_{MMSE} = \boldsymbol{\eta}_\theta + \left(\mathbf{C}_\theta^{-1} + \mathbf{H}^T \mathbf{C}_W^{-1} \mathbf{H} \right)^{-1} \mathbf{H}^T \mathbf{C}_W^{-1} (\mathbf{Y} - \mathbf{H} \boldsymbol{\eta}_\theta)$$

$$= \eta_A + \left(\frac{1}{\sigma_A^2} + \frac{N}{\sigma_W^2} \right)^{-1} \frac{1}{\sigma_W^2} \mathbf{1}^T (\mathbf{Y} - \mathbf{1} \eta_A) = \eta_A + \left(\frac{\sigma_W^2 + N \sigma_A^2}{\sigma_A^2 \sigma_W^2} \right)^{-1} \frac{1}{\sigma_W^2} \mathbf{1}^T (\mathbf{Y} - \mathbf{1} \eta_A)$$

$$= \eta_A + \frac{\sigma_A^2}{\sigma_W^2 + N \sigma_A^2} (\mathbf{1}^T \mathbf{Y} - N \eta_A) = \frac{\sigma_A^2}{\sigma_W^2 + N \sigma_A^2} \mathbf{1}^T \mathbf{Y} + \left(1 - \frac{N \sigma_A^2}{\sigma_W^2 + N \sigma_A^2} \right) \eta_A$$

$$\begin{aligned}
 \hat{A}_{MMSE} &= \frac{\sigma_A^2}{\sigma_W^2 + N\sigma_A^2} \mathbf{1}^T \mathbf{Y} + \left(1 - \frac{N\sigma_A^2}{\sigma_W^2 + N\sigma_A^2} \right) \eta_A \\
 &= \frac{N\sigma_A^2}{\sigma_W^2 + N\sigma_A^2} \cdot \frac{1}{N} \mathbf{1}^T \mathbf{Y} + \frac{\sigma_W^2}{\sigma_W^2 + N\sigma_A^2} \eta_A \\
 &= \alpha \cdot \frac{1}{N} \mathbf{1}^T \mathbf{Y} + (1 - \alpha) \eta_A = \alpha \hat{A}_{ML} + (1 - \alpha) \hat{A}_{ap}
 \end{aligned}$$

■ where we defined:

$$\alpha \triangleq \frac{N\sigma_A^2}{\sigma_W^2 + N\sigma_A^2} = \frac{N\gamma}{1 + N\gamma} = \frac{PG \cdot \gamma}{1 + PG \cdot \gamma} \in [0, 1), \quad \gamma \triangleq \frac{\sigma_A^2}{\sigma_W^2} \in [0, +\infty)$$

$$MSE\left\{\hat{A}_{MMSE}\right\} = \left(\mathbf{C}_{\theta}^{-1} + \mathbf{H}^T \mathbf{C}_W^{-1} \mathbf{H}\right)^{-1}$$

$$= \left(\frac{1}{\sigma_A^2} + \frac{N}{\sigma_W^2}\right)^{-1} = \left(\frac{\sigma_W^2 + N\sigma_A^2}{\sigma_A^2 \sigma_W^2}\right)^{-1}$$

$$= \frac{\sigma_A^2 \sigma_W^2}{\sigma_W^2 + N\sigma_A^2} = \left(1 - \frac{N\sigma_A^2}{\sigma_W^2 + N\sigma_A^2}\right) \sigma_A^2 = (1 - \alpha) \sigma_A^2 = (1 - \alpha) \cdot MSE\left\{\hat{A}_{ap}\right\}$$

$$= \frac{\sigma_A^2 \sigma_W^2}{\sigma_W^2 + N\sigma_A^2} = \frac{N\sigma_A^2}{\sigma_W^2 + N\sigma_A^2} \cdot \frac{\sigma_W^2}{N} = \alpha \cdot \frac{\sigma_W^2}{N} = \alpha \cdot MSE\left\{\hat{A}_{ML}\right\}$$

$$\alpha \triangleq \frac{N\sigma_A^2}{\sigma_W^2 + N\sigma_A^2} = \frac{N\gamma}{1 + N\gamma} = \frac{PG \cdot \gamma}{1 + PG \cdot \gamma} \in [0, 1), \quad \gamma \triangleq \frac{\sigma_A^2}{\sigma_W^2} \in [0, +\infty)$$

■ In summary:

$$\left\{ \begin{array}{l} \hat{A}_{ap} = \eta_A \\ \hat{A}_{ML} = \frac{1}{N} \mathbf{1}^T \mathbf{Y} \\ \hat{A}_{MMSE} = \alpha \hat{A}_{ML} + (1 - \alpha) \hat{A}_{ap} \end{array} \right.$$

$$MSE\{\hat{A}_{ap}\} = \sigma_A^2, \quad MSE\{\hat{A}_{ML}\} = \frac{\sigma_W^2}{N}, \quad MSE\{\hat{A}_{MMSE}\} = \alpha \frac{\sigma_W^2}{N} = (1 - \alpha) \sigma_A^2$$

$$\alpha = \frac{N\gamma}{1 + N\gamma} \in [0, 1) \Rightarrow \left\{ \begin{array}{l} MSE\{\hat{A}_{MMSE}\} = (1 - \alpha) \cdot MSE\{\hat{A}_{ap}\} \leq MSE\{\hat{A}_{ap}\} \\ MSE\{\hat{A}_{MMSE}\} = \alpha \cdot MSE\{\hat{A}_{ML}\} \leq MSE\{\hat{A}_{ML}\} \end{array} \right.$$

- All the three estimators are **linear**, **Gaussian distributed**, and **unbiased**:

$$\varepsilon_{ap} = A - \hat{A}_{ap} \in \mathcal{N}(0, \sigma_A^2),$$

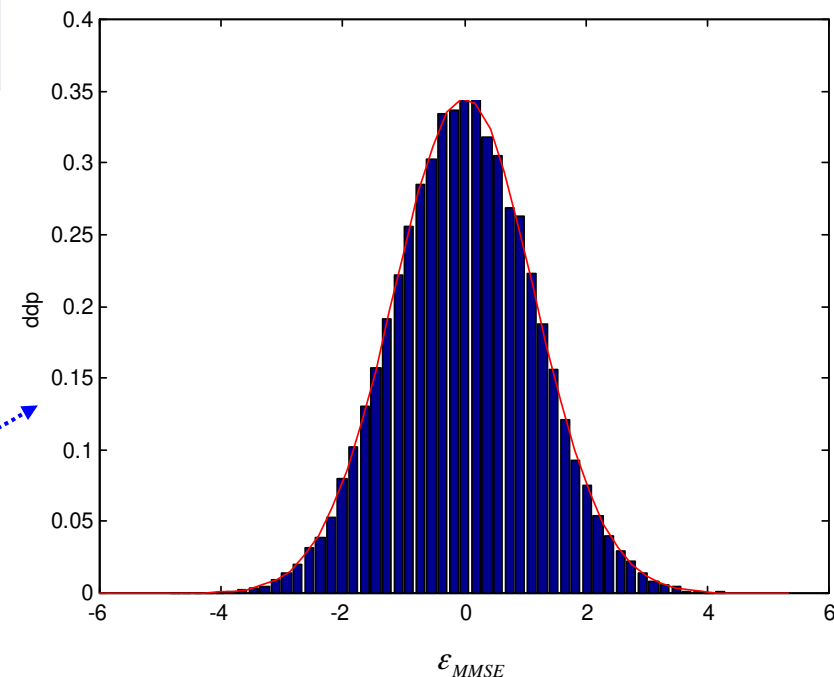
$$\varepsilon_{ML} = A - \hat{A}_{ML} \in \mathcal{N}\left(0, \frac{\sigma_w^2}{N}\right),$$

$$\varepsilon_{MMSE} = A - \hat{A}_{MMSE} \in \mathcal{N}\left(0, \alpha \frac{\sigma_w^2}{N}\right)$$

$$\eta_A = 10, \quad \sigma_A^2 = \sigma_w^2 = 4 \quad (\gamma = 0 \text{ dB})$$

$$N = 2 \quad M = 5 \cdot 10^4 \text{ Monte Carlo runs}$$

$$\varepsilon_{MMSE} = A - \hat{A}_{MMSE} \in \mathcal{N}\left(0, \alpha \frac{\sigma_w^2}{N}\right)$$



- The MSE's of the three estimators do not depend on the mean value of A but only on its variance and/or the noise variance.
- Coefficient α represents the improvement in terms of MSE of the MMSE estimator w.r.t. the ML estimator, achieved by properly exploiting the knowledge of the *a-priori* pdf of A . The ML estimator uses only the observed data, the MMSE uses the observed data and the *a-priori* information.
- The ML and MMSE estimators coincide when α tends to 1, i.e. when N and/or γ tend to infinity.
- Coefficient $1-\alpha$ represents the improvement in terms of MSE of the MMSE estimator w.r.t. the *A-Priori* estimator, achieved by properly exploiting the observed data, in addition to the *a-priori* information. The *A-Priori* estimator uses only the knowledge of the *a-priori* pdf of A , but not observed data. The MMSE uses observed data and *a-priori* information.
- The *A-Priori* and MMSE estimators coincide when α tends to 0, i.e. when N and/or γ tend to 0.

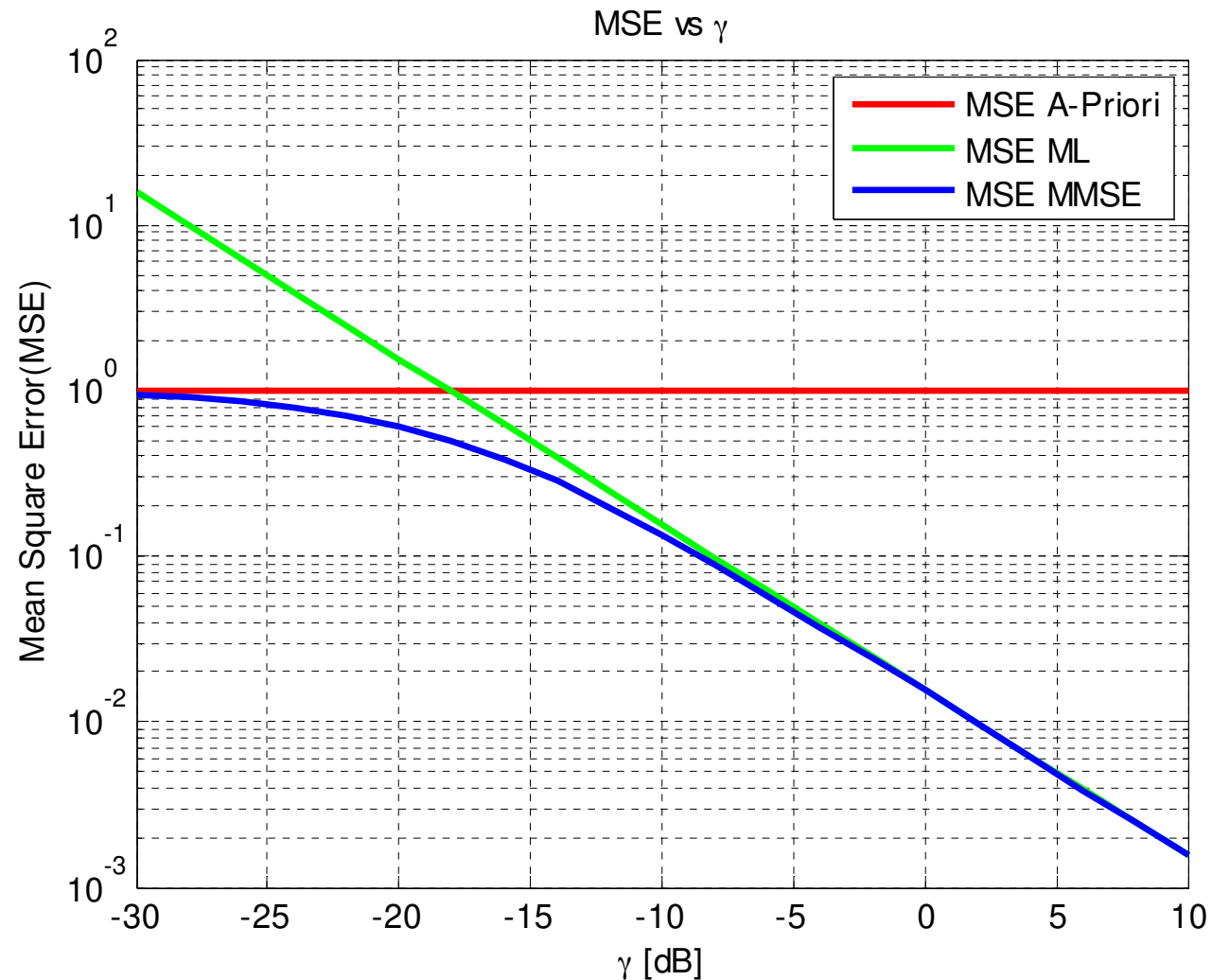


Figure 1 - MSE vs. γ [curve for $\sigma_A^2 = 1$ constant, for varying σ_W^2 , $N=64$]

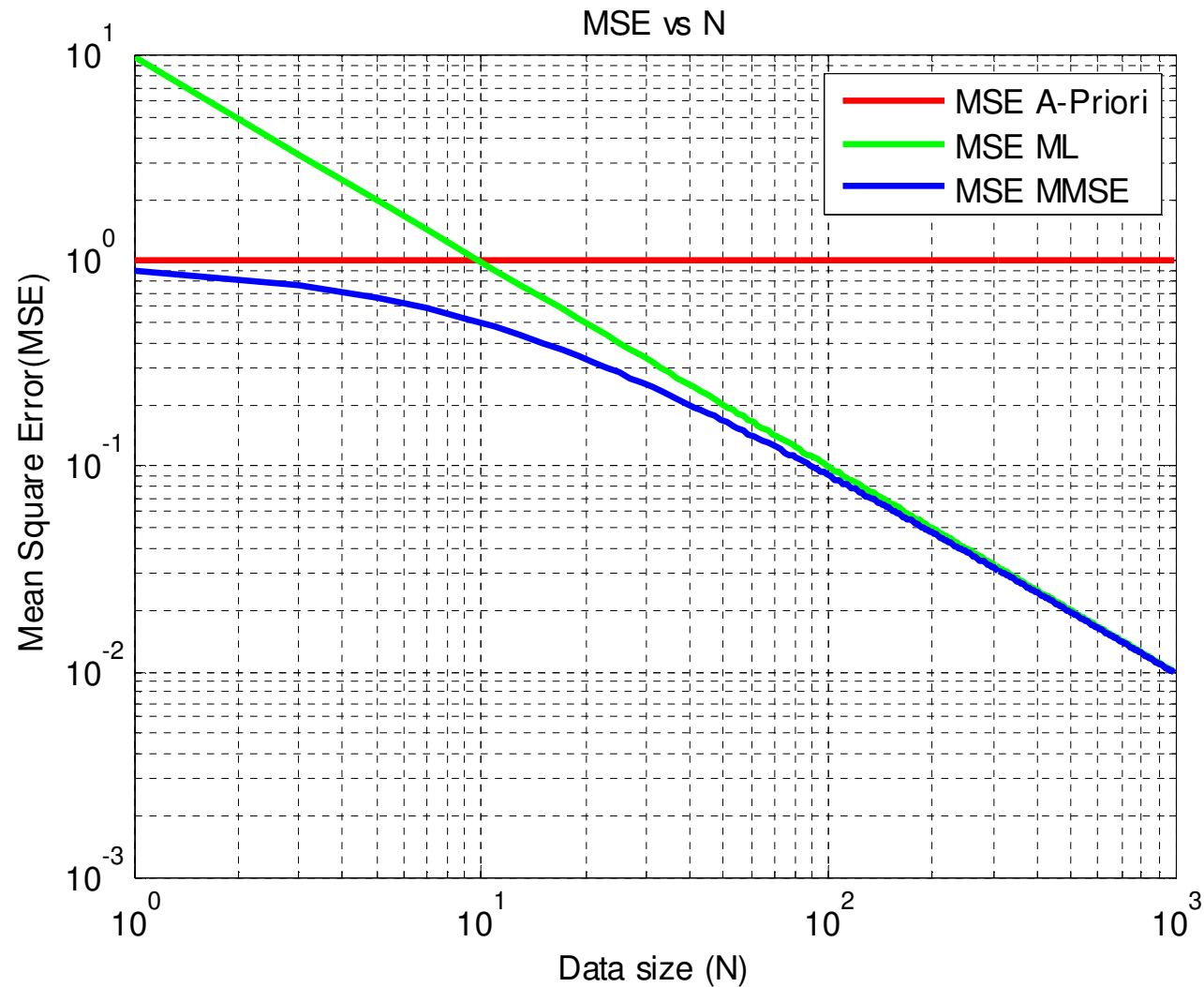


Figure 2 - MSE vs. N [curve for $\sigma_A^2 = 1$, $\sigma_W^2 = 10$, so $\gamma = -10$ dB]

■ **Example:** Let us consider again the **problem of estimating amplitude and phase of a sinusoid in AWGN.**

$$X_k = \frac{1}{\sqrt{2B}} X_B \left(\frac{k}{2B} \right) = \frac{A}{\sqrt{2B}} \cos \left(2\pi f_0 \frac{k}{2B} + \theta \right) + \frac{1}{\sqrt{2B}} W_B \left(\frac{k}{2B} \right), \quad 0 \leq k \leq N-1$$

$$= \frac{A}{\sqrt{2B}} \cos(\theta) \cos(2\pi F_0 k) - \frac{A}{\sqrt{2B}} \sin(\theta) \sin(2\pi F_0 k) + W_k, \quad \text{where } F_0 \triangleq \frac{f_0}{2B}$$

$$\begin{cases} \theta_1 \triangleq \frac{A}{\sqrt{2B}} \cos(\theta) \\ \theta_2 \triangleq -\frac{A}{\sqrt{2B}} \sin(\theta) \end{cases} \quad \begin{cases} \mathbf{h}_1 \triangleq [1 \quad \cos(2\pi F_0) \quad \cdots \quad \cos(2\pi F_0(N-1))]^T \\ \mathbf{h}_2 \triangleq [0 \quad \sin(2\pi F_0) \quad \cdots \quad \sin(2\pi F_0(N-1))]^T \end{cases}$$

⇓

$$\mathbf{X} = A\mathbf{p}(\theta) = \theta_1 \mathbf{h}_1 + \theta_2 \mathbf{h}_2 + \mathbf{W} = \mathbf{H}\boldsymbol{\theta} + \mathbf{W}$$

- Thanks to a proper re-parametrization of the SOI, we obtained the so-called **Deterministic Linear Model**:

$$\underset{2 \times 1}{\boldsymbol{\theta}} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}, \quad \underset{N \times 2}{\mathbf{H}} = [\mathbf{h}_1 \quad \mathbf{h}_2] \Rightarrow \mathbf{X} = \mathbf{H}\boldsymbol{\theta} + \mathbf{W}$$

- If we assume that $W(t)$ is AWGN with Power spectral Density (PSD) equal to $N_0/2$, we get:

$$\mathbf{W} \in \mathcal{N}(\mathbf{0}, \sigma_w^2 \mathbf{I}), \quad \text{where } \sigma_w^2 = \frac{N_0}{2}$$

- Under these assumptions, we found that the **ML estimator** coincides with the **Linear Least Squares (LS) estimator**.

■ **ML estimator = Linear LS estimator** → it is **linear, unbiased, Gaussian distributed, efficient, and consistent**:

$$\hat{\boldsymbol{\theta}}_{ML} = \hat{\boldsymbol{\theta}}_{LS} = \mathbf{H}^\# \mathbf{X} = \left(\mathbf{H}^H \mathbf{H} \right)^{-1} \mathbf{H}^H \mathbf{X}$$

$$E \left\{ \hat{\boldsymbol{\theta}}_{ML} \right\} = \boldsymbol{\theta}, \quad MSE \left\{ \hat{\boldsymbol{\theta}}_{ML} \right\} = CRB(\boldsymbol{\theta}) = \sigma_w^2 \left(\mathbf{H}^H \mathbf{H} \right)^{-1}$$

$$\hat{\boldsymbol{\theta}}_{ML} \in \mathcal{N}(\boldsymbol{\theta}, CRB(\boldsymbol{\theta}))$$

■ Taking into account that sine and cosine at the same frequency are orthogonal and that the energy of the two vectors $\mathbf{h}_i$ is equal to $N/2$, we found that:

$$\mathbf{H}^T \mathbf{H} \cong \frac{N}{2} \mathbf{I} \quad \text{if } N = 2BT \gg 1$$

- The ML estimates of θ_1 e θ_2 and their MSE's were derived to be:

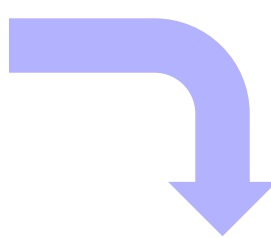
$$\hat{\boldsymbol{\theta}}_{ML} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{X} \cong \left(\frac{N}{2} \mathbf{I} \right)^{-1} \begin{bmatrix} \mathbf{h}_1^T \mathbf{X} \\ \mathbf{h}_2^T \mathbf{X} \end{bmatrix} = \frac{2}{N} \begin{bmatrix} \mathbf{h}_1^T \mathbf{X} \\ \mathbf{h}_2^T \mathbf{X} \end{bmatrix}$$

$$\hat{\theta}_{1,ML} = \frac{2}{N} \sum_{n=0}^{N-1} X_n \cos(2\pi F_0 n)$$

$$\hat{\theta}_{2,ML} = \frac{2}{N} \sum_{n=0}^{N-1} X_n \sin(2\pi F_0 n)$$

$$MSE\{\hat{\theta}_{k,ML}\} = CRB(\theta_k) = \frac{2\sigma_w^2}{N}, \quad k = 1, 2$$

- The ML estimate of A and θ is immediately derived once we have the ML estimate of vector θ , thanks to the **invariance property** of the ML estimate:

$$\hat{\boldsymbol{\theta}}_{ML} = \begin{bmatrix} \hat{\theta}_{1,ML} \\ \hat{\theta}_{2,ML} \end{bmatrix} = \mathbf{H}^{\#} \mathbf{X}$$

$$\begin{cases} A = \sqrt{2B} \sqrt{\theta_1^2 + \theta_2^2} \\ \theta = -\arctg\left(\frac{\theta_2}{\theta_1}\right) \end{cases} \Rightarrow \begin{cases} \hat{A}_{ML} = \sqrt{2B} \sqrt{\hat{\theta}_{1,ML}^2 + \hat{\theta}_{2,ML}^2} \\ \hat{\theta}_{ML} = -\arctg\left(\frac{\hat{\theta}_{2,ML}}{\hat{\theta}_{1,ML}}\right) \end{cases}$$

■ **Assumption:** The amplitude A is random, e.g. due to propagation effects on a random channel, and can be modelled as a Rayleigh r.v. (in this case we talk about “Rayleigh channel”). Whereas, the phase θ is typically modelled as a r.v. uniformly distributed in $[-\pi, \pi)$. Amplitude and phase can be assumed to be independent.

$$A \in \mathcal{R}(\sigma_A^2) \quad \Theta \in \mathcal{U}(-\pi, \pi)$$


$$f_A(a) = \frac{a}{\sigma_A^2} e^{-\frac{a^2}{2\sigma_A^2}} u(a), \quad \text{with } E\{A^2\} = 2\sigma_A^2$$

Data model: $\mathbf{X} = A\mathbf{p}(\theta) = \theta_1 \mathbf{h}_1 + \theta_2 \mathbf{h}_2 + \mathbf{W} = \mathbf{H}\boldsymbol{\theta} + \mathbf{W}$


$$\boldsymbol{\theta} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} \triangleq \begin{bmatrix} \frac{A}{\sqrt{2B}} \cos(\theta) \\ -\frac{A}{\sqrt{2B}} \sin(\theta) \end{bmatrix}$$

- It is basically a transformation from Polar coordinates to Cartesian coordinates.
- To find the joint pdf of θ_1 and θ_2 we should apply **the Fundamental Theorem** for the transformation of random vectors.
- Since $A > 0$, the system of equations has a unique solution:

$$\begin{cases} A = \sqrt{2B} \sqrt{\theta_1^2 + \theta_2^2} \\ \theta = -\arctg\left(\frac{\theta_2}{\theta_1}\right) \end{cases} \leftarrow \text{atan2}(-\theta_2, \theta_1)$$

$$f_{\theta_1, \theta_2}(\theta_1, \theta_2) = \frac{f_{A, \Theta}(a, \theta)}{|\mathbf{J}|} \bigg|_{\substack{a = \sqrt{2B} \sqrt{\theta_1^2 + \theta_2^2} \\ \theta = -\arctg(\theta_2/\theta_1)}} = \frac{1}{|\mathbf{J}|} \cdot \frac{a}{\sigma_A^2} e^{-\frac{a^2}{2\sigma_A^2}} u(a) \cdot \frac{1}{2\pi} \text{rect}\left(\frac{\theta}{2\pi}\right) \bigg|_{\substack{a = \sqrt{2B} \sqrt{\theta_1^2 + \theta_2^2} \\ \theta = -\arctg(\theta_2/\theta_1)}}$$

■ **J** is the Jacobian of the transformation:



$$\mathbf{J} = \begin{vmatrix} \frac{\partial \theta_1}{\partial a} & \frac{\partial \theta_1}{\partial \theta} \\ \frac{\partial \theta_2}{\partial a} & \frac{\partial \theta_2}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \frac{1}{\sqrt{2B}} \cos \theta & -\frac{a}{\sqrt{2B}} \sin \theta \\ -\frac{1}{\sqrt{2B}} \sin \theta & -\frac{a}{\sqrt{2B}} \cos \theta \end{vmatrix} = -\frac{a}{2B}$$

$$f_{\theta_1, \theta_2}(\theta_1, \theta_2) = \frac{1}{|\mathbf{J}|} \cdot \frac{a}{\sigma_A^2} e^{-\frac{a^2}{2\sigma_A^2}} u(a) \cdot \frac{1}{2\pi} \text{rect}\left(\frac{\theta}{2\pi}\right) \bigg|_{\substack{a=\sqrt{2B}\sqrt{\theta_1^2+\theta_2^2} \\ \theta=-\arctg(\theta_2/\theta_1)}}$$

$$= \frac{2B}{2\pi\sigma_A^2} e^{-\frac{a^2}{2\sigma_A^2}} u(a) \cdot \text{rect}\left(\frac{\theta}{2\pi}\right) \bigg|_{\substack{a=\sqrt{2B}\sqrt{\theta_1^2+\theta_2^2} \\ \theta=-\arctg(\theta_2/\theta_1)}}$$

- The joint pdf of θ_1 and θ_2 is finally obtained:

$$f_{\theta_1, \theta_2}(\theta_1, \theta_2) = \frac{2B}{2\pi\sigma_A^2} e^{-\frac{a^2}{2\sigma_A^2}} u(a) \text{rect}\left(\frac{\theta}{2\pi}\right) \bigg|_{\substack{a=\sqrt{2B}\sqrt{\theta_1^2+\theta_2^2} \\ \theta=-\arctg(\theta_2/\theta_1)}} = \frac{2B}{2\pi\sigma_A^2} e^{-\frac{2B(\theta_1^2+\theta_2^2)}{2\sigma_A^2}}$$

$$= \frac{1}{\sqrt{2\pi\frac{\sigma_A^2}{2B}}} e^{-\frac{\theta_1^2}{2\cdot\frac{\sigma_A^2}{2B}}} \cdot \frac{1}{\sqrt{2\pi\frac{\sigma_A^2}{2B}}} e^{-\frac{\theta_2^2}{2\cdot\frac{\sigma_A^2}{2B}}} = f_{\theta_1}(\theta_1) f_{\theta_2}(\theta_2)$$

$$\theta_1 \in \mathcal{N}\left(0, \frac{\sigma_A^2}{2B}\right), \quad \theta_2 \in \mathcal{N}\left(0, \frac{\sigma_A^2}{2B}\right), \quad \text{IID}$$

■ Hence, the SOI can be modelled by the Gaussian Linear Model:

$$\mathbf{X} = \mathbf{H}\boldsymbol{\theta} + \mathbf{W}$$

$$\boldsymbol{\theta} \in \mathcal{N}(\boldsymbol{\eta}_\theta, \mathbf{C}_\theta), \quad \boldsymbol{\eta}_\theta = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{C}_\theta = \sigma_\theta^2 \mathbf{I} = \frac{\sigma_A^2}{2B} \mathbf{I}$$

$$\mathbf{W} \in \mathcal{N}(\mathbf{0}, \mathbf{C}_W), \quad \text{where } \mathbf{C}_W = \sigma_W^2 \mathbf{I} \quad \text{and} \quad \sigma_W^2 = \frac{N_0}{2}$$

$$\mathbf{H} = [\mathbf{h}_1 \quad \mathbf{h}_2], \quad \begin{cases} \mathbf{h}_1 = [1 \quad \cos(2\pi F_0) \quad \cdots \quad \cos(2\pi F_0(N-1))]^T \\ \mathbf{h}_2 = [0 \quad \sin(2\pi F_0) \quad \cdots \quad \sin(2\pi F_0(N-1))]^T \end{cases}$$

■ **Problem:** find the MMSE estimator of the Gaussian random vector $\boldsymbol{\theta}$ given the observed data vector $\mathbf{X}$ (under the assumption that F_0 is known).

- The general expressions of the **MMSE estimator** of the Gaussian random vector θ and of the covariance matrix of the estimation errors (i.e. the MSE matrix) have already been derived:

$$\hat{\theta}_{MMSE} = \eta_{\theta|X} = \eta_{\theta} + \left(\mathbf{C}_{\theta}^{-1} + \mathbf{H}^T \mathbf{C}_W^{-1} \mathbf{H} \right)^{-1} \mathbf{H}^T \mathbf{C}_W^{-1} (\mathbf{X} - \mathbf{H} \eta_{\theta})$$

$$MSE \left\{ \hat{\theta}_{MMSE} \right\} = \mathbf{C}_{\theta|X} = \left(\mathbf{C}_{\theta}^{-1} + \mathbf{H}^T \mathbf{C}_W^{-1} \mathbf{H} \right)^{-1} = BCRB(\theta)$$

linear function of $\mathbf{X}$

$$\hat{\theta}_{MMSE} = \frac{1}{\sigma_W^2} \left(\frac{1}{\sigma_{\theta}^2} \mathbf{I} + \frac{1}{\sigma_W^2} \mathbf{H}^T \mathbf{H} \right)^{-1} \mathbf{H}^T \mathbf{X}$$

$$MSE \left\{ \hat{\theta}_{MMSE} \right\} = \left(\frac{1}{\sigma_{\theta}^2} \mathbf{I} + \frac{1}{\sigma_W^2} \mathbf{H}^T \mathbf{H} \right)^{-1}, \quad \text{where } \sigma_{\theta}^2 = \frac{\sigma_A^2}{2B}, \quad \sigma_W^2 = \frac{N_0}{2}$$

$$\mathbf{H} = [\mathbf{h}_1 \quad \mathbf{h}_2], \quad \begin{cases} \mathbf{h}_1 = [1 \quad \cos(2\pi F_0) \quad \cdots \quad \cos(2\pi F_0(N-1))]^T \\ \mathbf{h}_2 = [0 \quad \sin(2\pi F_0) \quad \cdots \quad \sin(2\pi F_0(N-1))]^T \end{cases}$$

■ Recalling that: $\mathbf{H}^T \mathbf{H} \cong \frac{N}{2} \mathbf{I}$ if $N = 2BT \gg 1$

■ we find:

$$\hat{\boldsymbol{\theta}}_{MMSE} = \frac{1}{\sigma_w^2} \left(\frac{1}{\sigma_\theta^2} \mathbf{I} + \frac{N}{2\sigma_w^2} \mathbf{I} \right)^{-1} \begin{bmatrix} \mathbf{h}_1^T \mathbf{X} \\ \mathbf{h}_2^T \mathbf{X} \end{bmatrix} = \frac{2\sigma_\theta^2}{2\sigma_w^2 + N\sigma_\theta^2} \begin{bmatrix} \mathbf{h}_1^T \mathbf{X} \\ \mathbf{h}_2^T \mathbf{X} \end{bmatrix}$$

$$MSE\{\hat{\boldsymbol{\theta}}_{MMSE}\} = \frac{2\sigma_w^2\sigma_\theta^2}{2\sigma_w^2 + N\sigma_\theta^2} \mathbf{I}$$

■ In summary, the MMSE estimates of θ_1 and θ_2 and their MSE are given by:

$$\hat{\theta}_{1,MMSE} = \frac{2\sigma_{\theta}^2}{2\sigma_w^2 + N\sigma_{\theta}^2} \mathbf{h}_1^T \mathbf{X} = \frac{N\sigma_{\theta}^2}{2\sigma_w^2 + N\sigma_{\theta}^2} \cdot \frac{2}{N} \sum_{n=0}^{N-1} X_n \cos(2\pi F_0 n)$$

$$\hat{\theta}_{2,MMSE} = \frac{2\sigma_{\theta}^2}{2\sigma_w^2 + N\sigma_{\theta}^2} \mathbf{h}_2^T \mathbf{X} = \frac{N\sigma_{\theta}^2}{2\sigma_w^2 + N\sigma_{\theta}^2} \cdot \frac{2}{N} \sum_{n=0}^{N-1} X_n \sin(2\pi F_0 n)$$

$$MSE\left\{\hat{\theta}_{k,MMSE}\right\} = \frac{2\sigma_w^2\sigma_{\theta}^2}{2\sigma_w^2 + N\sigma_{\theta}^2} = \frac{N\sigma_{\theta}^2}{2\sigma_w^2 + N\sigma_{\theta}^2} \cdot \frac{2\sigma_w^2}{N}, \quad k = 1, 2$$

■ The **MMSE estimator** for the **Gaussian linear model** in **AWGN** is **linear**, **Gaussian distributed**, **unbiased**, **efficient**, and **consistent**.

- The MMSE estimates of θ_1 and θ_2 and their MSE can be expressed immediately in terms of the ML estimators and their MSE:

$$\hat{\theta}_{1,MMSE} = \frac{N\sigma_{\theta}^2}{2\sigma_w^2 + N\sigma_{\theta}^2} \cdot \frac{2}{N} \sum_{n=0}^{N-1} X_n \cos(2\pi F_0 n) = \alpha \hat{\theta}_{1,ML}$$

$$\hat{\theta}_{2,MMSE} = \frac{N\sigma_{\theta}^2}{2\sigma_w^2 + N\sigma_{\theta}^2} \cdot \frac{2}{N} \sum_{n=0}^{N-1} X_n \sin(2\pi F_0 n) = \alpha \hat{\theta}_{2,ML}$$

$$MSE\left\{\hat{\theta}_{k,MMSE}\right\} = \frac{N\sigma_{\theta}^2}{2\sigma_w^2 + N\sigma_{\theta}^2} \cdot \frac{2\sigma_w^2}{N} = \alpha MSE\left(\hat{\theta}_{k,ML}\right), \quad k = 1, 2$$

$$\text{where } \alpha \triangleq \frac{N\sigma_{\theta}^2}{2\sigma_w^2 + N\sigma_{\theta}^2} = \frac{\frac{N}{2}\sigma_{\theta}^2}{\sigma_w^2 + \frac{N}{2}\sigma_{\theta}^2} = \frac{PG \cdot \gamma}{1 + PG \cdot \gamma} \in [0, 1), \quad PG = \frac{N}{2}$$

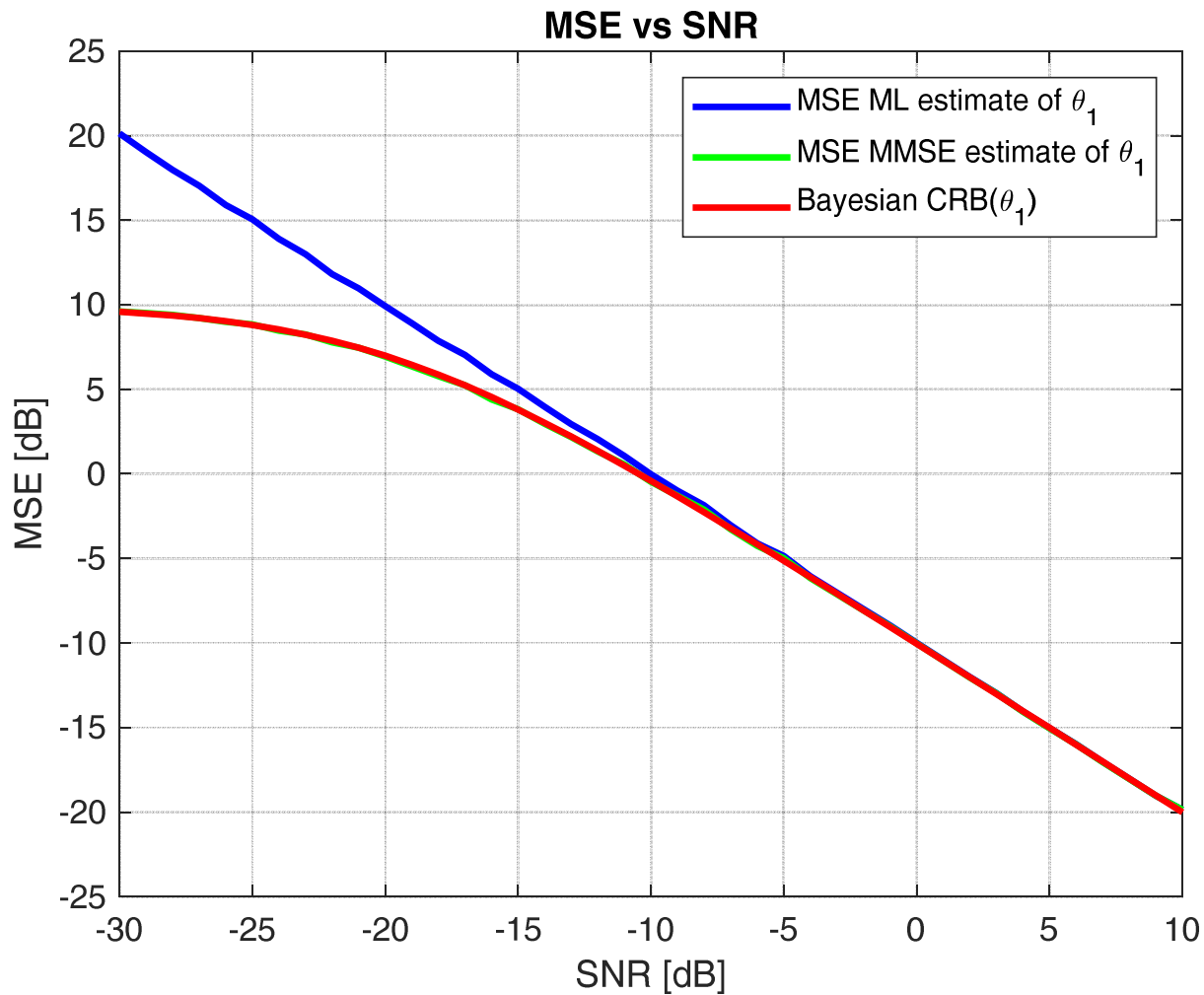
- Parameter α is related to the input Signal-to-Noise power Ratio (SNR) γ :

$$\mathbf{X} = \mathbf{H}\boldsymbol{\theta} + \mathbf{W}$$

$$\begin{aligned}\gamma &\triangleq \frac{E\{\|\mathbf{H}\boldsymbol{\theta}\|^2\}}{E\{\|\mathbf{W}\|^2\}} = \frac{E\{(\mathbf{H}\boldsymbol{\theta})^T \mathbf{H}\boldsymbol{\theta}\}}{E\{\mathbf{W}^T \mathbf{W}\}} = \frac{E\{\boldsymbol{\theta}^T \mathbf{H}^T \mathbf{H}\boldsymbol{\theta}\}}{E\left\{\sum_{n=1}^N W_n^2\right\}} = \frac{\frac{N}{2} E\{\boldsymbol{\theta}^T \boldsymbol{\theta}\}}{E\left\{\sum_{n=1}^N W_n^2\right\}} \\ &= \frac{N\sigma_{\theta}^2}{N\sigma_W^2} = \frac{\sigma_{\theta}^2}{\sigma_W^2} = \frac{\sigma_A^2}{N_0 B}\end{aligned}$$

$$\alpha = \frac{\frac{N}{2} \sigma_{\theta}^2}{\sigma_W^2 + \frac{N}{2} \sigma_{\theta}^2} = \frac{\frac{N}{2} \gamma}{1 + \frac{N}{2} \gamma} = \frac{PG \cdot \gamma}{1 + PG \cdot \gamma}$$

- When $\gamma \rightarrow \infty$ or $N \rightarrow \infty$, $\alpha \rightarrow 1$, and so the MMSE estimator tends to coincide with the ML estimator.



$$f_0 = 1 \text{ KHz}$$

$$\sigma_\theta^2 = 10$$

$$B = 5 \text{ KHz}$$

$$T = 20 \text{ msec}$$

$$MC \text{ runs} = 10^4$$

- Once we have the MMSE estimate of θ_1 and θ_2 we can get an estimate of A and θ as follows:

$$\begin{cases} \hat{A} = \sqrt{2B} \sqrt{\hat{\theta}_{1,MMSE}^2 + \hat{\theta}_{2,MMSE}^2} \\ \hat{\theta} = -\arctg\left(\hat{\theta}_{2,MMSE} / \hat{\theta}_{1,MMSE}\right) \end{cases}$$

- However, we cannot claim that these are the MMSE estimators of A and θ , since for the MMSE and MAP estimators the **invariance property** holds only for **affine transformations** (i.e. of the kind: $\mathbf{Y}=\mathbf{A}\cdot\mathbf{X}+\mathbf{b}$), but not for nonlinear transformations.

$$\hat{A}_{ML} = \sqrt{2B} \sqrt{\hat{\theta}_{1,ML}^2 + \hat{\theta}_{2,ML}^2}, \quad \text{but} \quad \hat{A}_{MMSE} \neq \sqrt{2B} \sqrt{\hat{\theta}_{1,MMSE}^2 + \hat{\theta}_{2,MMSE}^2}$$

- We may refer to the above estimators as the **pseudo-MMSE**.

■ The **pseudo-MMSE** estimator of A can be expressed in terms of the **ML estimator** of A :

■ Recalling that:

$$\hat{A}_{ML} = \sqrt{2B} \sqrt{\hat{\theta}_{1,ML}^2 + \hat{\theta}_{2,ML}^2}, \quad \hat{\theta}_{1,MMSE} = \alpha \hat{\theta}_{1,ML}, \quad \hat{\theta}_{2,MMSE} = \alpha \hat{\theta}_{2,ML}$$

■ we get:

$$\hat{A}_{pseudo-MMSE} = \sqrt{2B} \sqrt{\hat{\theta}_{1,MMSE}^2 + \hat{\theta}_{2,MMSE}^2} = \sqrt{2B} \sqrt{\alpha^2 \hat{\theta}_{1,ML}^2 + \alpha^2 \hat{\theta}_{2,ML}^2} = \alpha \hat{A}_{ML}$$

$$\begin{aligned} \hat{\theta}_{pseudo-MMSE} &= -\arctg\left(\hat{\theta}_{2,MMSE} / \hat{\theta}_{1,MMSE}\right) = -\arctg\left(\alpha \hat{\theta}_{2,ML} / \alpha \hat{\theta}_{1,ML}\right) \\ &= -\arctg\left(\hat{\theta}_{2,ML} / \hat{\theta}_{1,ML}\right) = \hat{\theta}_{ML} \end{aligned}$$

$$BCRB(A) = - \frac{1}{E_{A,\mathbf{X}} \left\{ \frac{\partial^2 \ln f_{A,\mathbf{X}}(a, \mathbf{x})}{\partial a^2} \right\}}$$

■ The performance of the **pseudo-MMSE** can be assessed by calculating the BCRB.

$$E_{A,\mathbf{X}} \left\{ \frac{\partial^2 \ln f_{A,\mathbf{X}}(a, \mathbf{x})}{\partial a^2} \right\} = E_{A,\mathbf{X}} \left\{ \frac{\partial^2 \ln \left(f_{\mathbf{X}|A}(\mathbf{x}|a) f_A(a) \right)}{\partial a^2} \right\}$$

$$= E_{A,\mathbf{X}} \left\{ \frac{\partial^2 \ln f_{\mathbf{X}|A}(\mathbf{x}|a)}{\partial a^2} + \frac{\partial^2 \ln f_A(a)}{\partial a^2} \right\}$$

$$= E_{A,\mathbf{X}} \left\{ \frac{\partial^2 \ln f_{\mathbf{X}|A}(\mathbf{x}|a)}{\partial a^2} \right\} + E_{A,\mathbf{X}} \left\{ \frac{\partial^2 \ln f_A(a)}{\partial a^2} \right\}$$

$$= E_A \left\{ E_{\mathbf{X}|A} \left\{ \frac{\partial^2 \ln f_{\mathbf{X}|A}(\mathbf{x}|a)}{\partial a^2} \right\} \right\} + E_A \left\{ \frac{\partial^2 \ln f_A(a)}{\partial a^2} \right\}$$

- The first term has already been derived when we calculated the Fisher Information Matrix (FIM) for the estimation of amplitude, phase and frequency of a sinusoid in AWGN (Ch. 4, p. 188):

$$\mathbf{I}(\boldsymbol{\theta}) = \begin{bmatrix} \frac{T}{N_0} & 0 & 0 \\ 0 & \frac{A^2 T}{N_0} & \frac{\pi A^2 T (N-1)}{N_0} \\ 0 & \frac{\pi A^2 T (N-1)}{N_0} & \frac{2\pi^2 A^2 T (N-1)(2N-1)}{3N_0} \end{bmatrix}$$

- Hence, we have that:

$$-E_{\mathbf{x}|A} \left\{ \frac{\partial^2 \ln f_{\mathbf{x}|A}(\mathbf{x}|a)}{\partial a^2} \right\} = \frac{T}{N_0}$$

■ So, we have:

$$\begin{aligned} -E_{A,\mathbf{X}} \left\{ \frac{\partial^2 \ln f_{A,\mathbf{X}}(a, \mathbf{X})}{\partial a^2} \right\} &= E_A \left\{ -E_{\mathbf{X}|A} \left\{ \frac{\partial^2 \ln f_{\mathbf{X}|A}(\mathbf{X}|a)}{\partial a^2} \right\} \right\} - E_A \left\{ \frac{\partial^2 \ln f_A(a)}{\partial a^2} \right\} \\ &= E_A \left\{ \frac{T}{N_0} \right\} - E_A \left\{ \frac{\partial^2 \ln f_A(a)}{\partial a^2} \right\} \end{aligned}$$

■ Recalling that:

$$f_A(a) = \frac{a}{\sigma_A^2} e^{-\frac{a^2}{2\sigma_A^2}} u(a), \quad \text{where } E\{A^2\} = 2\sigma_A^2$$

$$\text{and } \gamma = \frac{\sigma_\theta^2}{\sigma_w^2} = \frac{\sigma_A^2}{N_0 B} \quad \text{is the input SNR}$$

■ we get:

$$\begin{aligned}
 -E_{A,\mathbf{X}} \left\{ \frac{\partial^2 \ln f_{A,\mathbf{X}}(a, \mathbf{X})}{\partial a^2} \right\} &= \frac{T}{N_0} - E_A \left\{ \frac{\partial^2}{\partial a^2} \left(\ln \left(\frac{1}{\sigma_A^2} \right) + \ln(a) - \frac{a^2}{2\sigma_A^2} \right) \right\} \\
 &= \frac{T}{N_0} + E_A \left\{ \frac{1}{a^2} + \frac{1}{\sigma_A^2} \right\} = \frac{T}{N_0} + E_A \left\{ \frac{1}{a^2} \right\} + \frac{1}{\sigma_A^2} \\
 &\cong \frac{T}{N_0} + \frac{1}{2\sigma_A^2} + \frac{1}{\sigma_A^2}
 \end{aligned}$$

we approximate the term $E\{1/a^2\}$, that cannot be calculated in closed-form, with $1/E\{a^2\}$.

$$BCRB(A) = - \frac{1}{E_{A,\mathbf{X}} \left\{ \frac{\partial^2 \ln f_{A,\mathbf{X}}(a, \mathbf{X})}{\partial a^2} \right\}} \cong \frac{1}{\frac{T}{N_0} + \frac{3}{2\sigma_A^2}} = \frac{\sigma_A^2}{\frac{T\sigma_A^2}{N_0} + \frac{3}{2}} = \frac{\sigma_A^2}{\frac{N}{2}\gamma + \frac{3}{2}}$$

where: $\gamma = \frac{\sigma_A^2}{N_0 B} = \frac{2T\sigma_A^2}{N_0 N}$ is the input SNR

Jensen's Inequality:

$$E_A \left\{ \frac{1}{a^2} \right\} \geq \frac{1}{E_A \{a^2\}}$$

■ MAP estimators of A and θ

■ The log-likelihood function (log-LF) has already been derived:

$$\ln L(A, \theta) = \kappa - \frac{A^2}{N_0 2B} \cdot \frac{N}{2} + \frac{2A}{N_0 \sqrt{2B}} \sum_{i=0}^{N-1} x_i \cos(2\pi F_0 i + \theta)$$

■ The amplitude A is Rayleigh distributed, the phase θ is uniformly distributed in $[-\pi, \pi)$, A and θ are independent.

■ The **joint MAP estimators** of A and θ are obtained by solving the **MAP equation**:

$$\begin{cases} \frac{\partial \ln L(A, \theta) + \ln f_A(A) + \ln f_\theta(\theta)}{\partial A} = \frac{\partial \ln L(A, \theta) + \ln f_A(A)}{\partial A} = 0 \\ \frac{\partial \ln L(A, \theta) + \ln f_A(A) + \ln f_\theta(\theta)}{\partial \theta} = \frac{\partial \ln L(A, \theta)}{\partial \theta} = 0 \end{cases}$$

■ As a consequence of the 2nd equation:

$$\hat{\theta}_{MAP} = \hat{\theta}_{ML} = -\arctan \left(\frac{\sum_{i=0}^{N-1} x_i \sin(2\pi F_0 i)}{\sum_{i=0}^{N-1} x_i \cos(2\pi F_0 i)} \right) = -\arctan \left(\frac{\mathbf{s}^T \mathbf{x}}{\mathbf{c}^T \mathbf{x}} \right)$$

where we defined the two orthonormal (orthogonal and unit norm vectors):

$$\mathbf{c} \triangleq [c_0 \quad c_1 \quad \cdots \quad c_{N-1}], \text{ where } c_i \triangleq \sqrt{\frac{2}{N}} \cos(2\pi F_0 i), \quad \|\mathbf{c}\|^2 = \mathbf{c}^T \mathbf{c} = 1$$

$$\mathbf{s} \triangleq [s_0 \quad s_1 \quad \cdots \quad s_{N-1}], \text{ where } s_i \triangleq \sqrt{\frac{2}{N}} \sin(2\pi F_0 i), \quad \|\mathbf{s}\|^2 = \mathbf{s}^T \mathbf{s} = 1$$

$$\mathbf{s}^T \mathbf{c} = \mathbf{c}^T \mathbf{s} = 0 \quad \rightarrow \quad \mathbf{s} \perp \mathbf{c}$$

■ From the 1st equation:

$$\begin{aligned}
 & \frac{\partial \ln L(A, \theta) + \ln f_A(A)}{\partial A} \\
 &= \frac{\partial}{\partial A} \left[-\frac{A^2}{N_0 2B} \cdot \frac{N}{2} + \frac{2A}{N_0 \sqrt{2B}} \sum_{i=0}^{N-1} x_i \cos(2\pi F_0 i + \theta) + \ln(A) - \frac{A^2}{2\sigma_A^2} \right] \\
 &= -\frac{AN}{N_0 2B} + \frac{2}{N_0 \sqrt{2B}} \sum_{i=0}^{N-1} x_i \cos(2\pi F_0 i + \theta) + \frac{1}{A} - \frac{A}{\sigma_A^2} = 0
 \end{aligned}$$

■ where:

$$\begin{aligned}
 \sum_{i=0}^{N-1} x_i \cos(2\pi F_0 i + \hat{\theta}_{MAP}) &= \sum_{i=0}^{N-1} x_i \cos(2\pi F_0 i + \hat{\theta}_{ML}) \\
 &= \frac{N}{2} \cdot \frac{1}{\sqrt{2B}} \hat{A}_{ML} = \frac{N}{2} \cdot \frac{1}{\sqrt{2B}} \left(\sqrt{\frac{2}{T}} \sqrt{(\mathbf{c}^T \mathbf{x})^2 + (\mathbf{s}^T \mathbf{x})^2} \right)
 \end{aligned}$$

■ The **MAP equation** for the amplitude becomes:

$$\frac{\partial \ln L(A, \theta) + \ln f_A(A)}{\partial A} = -\frac{AN}{N_0 2B} + \frac{2}{N_0 \sqrt{2B}} \left(\frac{N}{2} \frac{1}{\sqrt{2B}} \hat{A}_{ML} \right) + \frac{1}{A} - \frac{A}{\sigma_A^2} = 0$$

$$\frac{A}{\sigma_A^2} + \frac{AN}{N_0 2B} - \frac{N}{N_0 2B} \hat{A}_{ML} - \frac{1}{A} = \frac{1}{A} \left[\left(\frac{1}{\sigma_A^2} + \frac{T}{N_0} \right) A^2 - \frac{T}{N_0} \hat{A}_{ML} A - 1 \right] = 0$$

$$\left(\frac{1}{\sigma_A^2} + \frac{T}{N_0} \right) A^2 - \frac{T}{N_0} \hat{A}_{ML} A - 1 = 0$$

■ The **MAP equation** has two roots: the maximum is obtained from the larger root.

■ Hence, the **MAP estimator** is given by:

$$\hat{A}_{MAP} = \frac{\frac{T}{N_0} \hat{A}_{ML} + \sqrt{\left(\frac{T}{N_0} \hat{A}_{ML}\right)^2 + 4\left(\frac{1}{\sigma_A^2} + \frac{T}{N_0}\right)}}{2\left(\frac{1}{\sigma_A^2} + \frac{T}{N_0}\right)} = \frac{\hat{A}_{ML} + \sqrt{\left(\hat{A}_{ML}\right)^2 + 4\frac{N_0}{T}\left(\frac{N_0}{\sigma_A^2 T} + 1\right)}}{2\left(\frac{N_0}{\sigma_A^2 T} + 1\right)}$$

$$= \frac{\hat{A}_{ML} + \sqrt{\left(\hat{A}_{ML}\right)^2 + 4\sigma_A^2 \cdot \frac{2}{N\gamma}\left(1 + \frac{2}{N\gamma}\right)}}{2\left(1 + \frac{2}{N\gamma}\right)} = \frac{\hat{A}_{ML} + \sqrt{\left(\hat{A}_{ML}\right)^2 + \frac{4\sigma_A^2}{\alpha}\left(\frac{1}{\alpha} - 1\right)}}{\frac{2}{\alpha}}$$

$$= \frac{\alpha \hat{A}_{ML} + \sqrt{\left(\alpha \hat{A}_{ML}\right)^2 + 4\sigma_A^2(1 - \alpha)}}{2}, \quad \text{where } \gamma = \frac{\sigma_A^2}{N_0 B} = \frac{2T\sigma_A^2}{N_0 N} \quad \text{and} \quad \alpha \triangleq \frac{\frac{N}{2}\gamma}{1 + \frac{N}{2}\gamma}$$

$$\hat{A}_{MAP} = \frac{\alpha \hat{A}_{ML} + \sqrt{\left(\alpha \hat{A}_{ML}\right)^2 + 4\sigma_A^2(1-\alpha)}}{2},$$

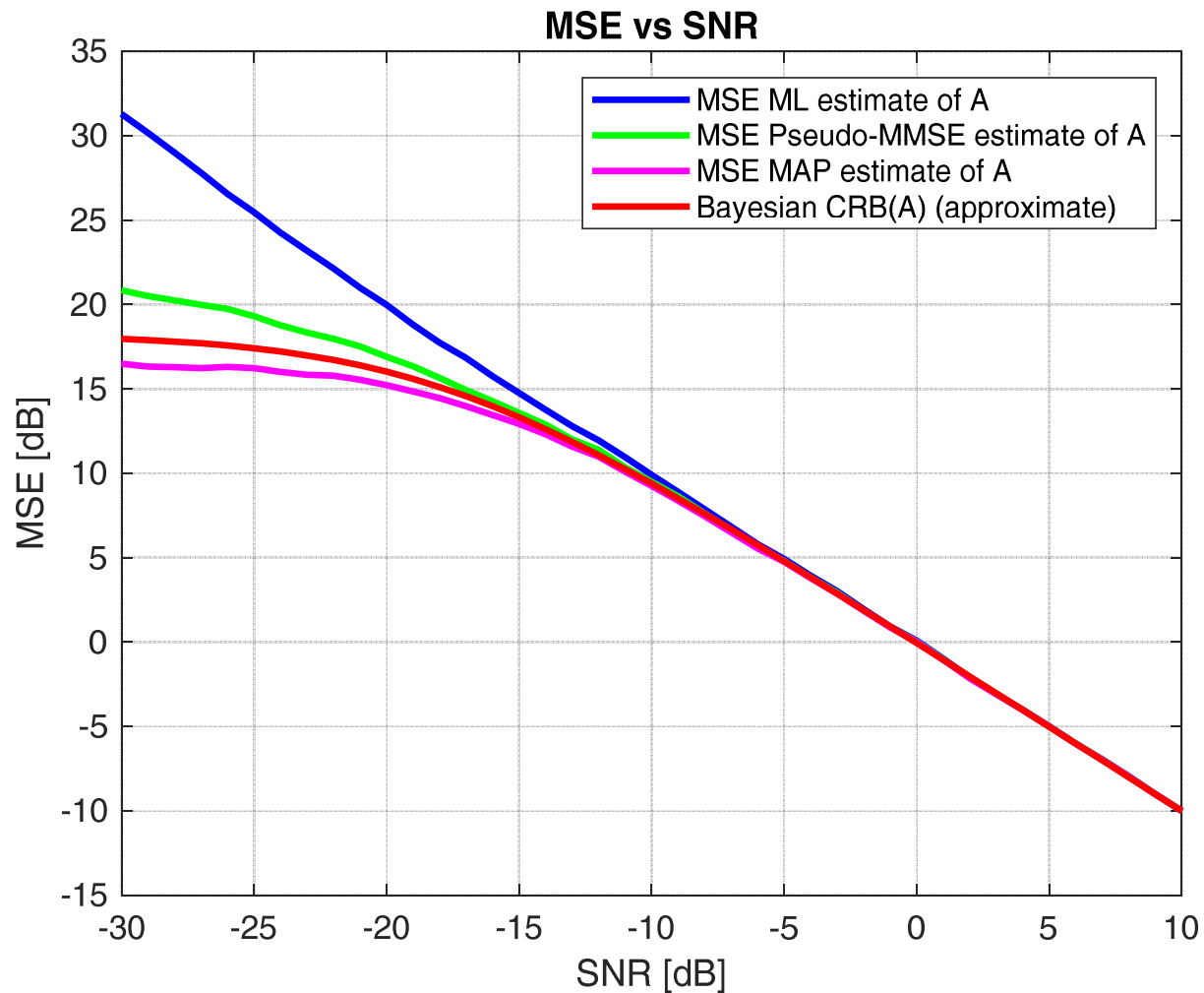
$$\text{where } \gamma \triangleq \frac{\sigma_A^2}{N_0 B} \text{ and } \alpha \triangleq \frac{\frac{N}{2}\gamma}{1 + \frac{N}{2}\gamma}$$

■ Note that:

$$\text{if } \alpha \rightarrow 1 \text{ (i.e. } N \rightarrow \infty \text{ or } \gamma \rightarrow \infty): \hat{A}_{MAP} \rightarrow \hat{A}_{ML}$$

$$\text{if } \alpha \rightarrow 0 \text{ (i.e. } N \rightarrow 0 \text{ or } \gamma \rightarrow 0): \hat{A}_{MAP} \rightarrow \sigma_A \text{ [mode of } f_A(A)]$$

$$\text{if } \left(\alpha \hat{A}_{ML}\right)^2 \gg 4\sigma_A^2(1-\alpha): \hat{A}_{MAP} \cong \alpha \hat{A}_{ML} = \hat{A}_{pseudo-MMSE}$$



$$f_0 = 1 \text{ KHz}$$

$$\sigma_\theta^2 = 10$$

$$B = 5 \text{ KHz}$$

$$T = 20 \text{ msec}$$

$$MC \text{ runs} = 10^4$$

MAP Estimation of a constant Laplacian signal in AWGN⁵

Problem: find the MAP estimator of a constant random signal S in AWGN.

Model of the observed data

Random constant signal

Noise samples

$$Y_i = S + W_i, \quad i = 0, 1, 2, \dots, N-1; \quad S \in \mathcal{L}(\lambda), \quad W_i \in \mathcal{N}(0, \sigma_w^2), \quad \{W_i\}_{i=0}^{N-1} \text{ IID}$$

■ The difference with respect to the previous case, already analyzed, is that the constant random signal S is not Gaussian distributed but it follows the **Laplace distribution**, so the *a-posteriori* pdf is no more Gaussian.

■ In the Gaussian case, the MMSE and MAP estimators coincide and are given by:

$$\hat{S}_{MMSE} = \hat{S}_{MAP} = \alpha \hat{S}_{ML} + (1 - \alpha) \eta_S \quad \text{where: } \hat{S}_{ML} \triangleq \underbrace{\frac{1}{N} \mathbf{1}^T \mathbf{Y}}_{\text{Sample Mean}}, \quad \alpha \triangleq \frac{N\gamma}{1 + N\gamma}, \quad \gamma \triangleq \frac{\sigma_S^2}{\sigma_w^2}$$

■ In this case, the MMSE and MAP estimators are different, and it is not possible to find the MMSE estimator in closed form. Let us derive the **MAP estimator**.

MAP Estimation of a constant Laplacian signal in AWGN⁶

- The **Maximum A Posteriori (MAP)** estimator is given by the **mode** of the *a-posteriori* pdf:

$$\hat{S}_{MAP} = \arg \max_s f_{S|Y}(s|\mathbf{y})$$

- To maximize the *a-posteriori* pdf w.r.t. S is equivalent to maximize the joint pdf of the random parameter S and of the data vector $\mathbf{Y}$:


$$\begin{aligned}\hat{S}_{MAP} &= \arg \max_s f_{S|Y}(s|\mathbf{y}) = \arg \max_s \frac{f_{S,Y}(s,\mathbf{y})}{f_Y(\mathbf{y})} = \arg \max_s f_{S,Y}(s,\mathbf{y}) \\ &= \arg \max_s f_{Y|S}(\mathbf{y}|s) f_S(s)\end{aligned}$$

MAP equation: $\left. \frac{\partial}{\partial s} \left[\ln f_{Y|S}(\mathbf{y}|s) + \ln f_S(s) \right] \right|_{s=\hat{S}_{MAP}} = 0$

MAP Estimation of a constant Laplacian signal in AWGN⁷

Model of the observed data

Random constant signal

Noise samples

$$Y_i = S + W_i, \quad i = 0, 1, 2, \dots, N-1; \quad S \in \mathcal{L}(\lambda), \quad W_i \in \mathcal{N}(0, \sigma_W^2), \quad \{W_i\}_{i=0}^{N-1} \text{ IID}$$

$$f_S(s) = \frac{1}{2\lambda} e^{-\frac{|s|}{\lambda}} \rightarrow E\{S^2\} = 2\lambda^2 \quad \text{Laplace distribution}$$

$$f_{\mathbf{Y}|S}(\mathbf{y}|s) = \prod_{i=0}^{N-1} f_{Y_i|S}(y_i|s) = \prod_{i=0}^{N-1} \frac{1}{\sqrt{2\pi\sigma_W^2}} e^{-\frac{(y_i-s)^2}{2\sigma_W^2}} = (2\pi\sigma_W^2)^{-\frac{N}{2}} e^{-\frac{1}{2\sigma_W^2} \sum_{i=0}^{N-1} (y_i-s)^2}$$

$$\ln f_{\mathbf{Y}|S}(\mathbf{y}|s) + \ln f_S(s) = \kappa_1 - \frac{1}{2\sigma_W^2} \sum_{i=0}^{N-1} (y_i - s)^2 - \frac{|s|}{\lambda}$$

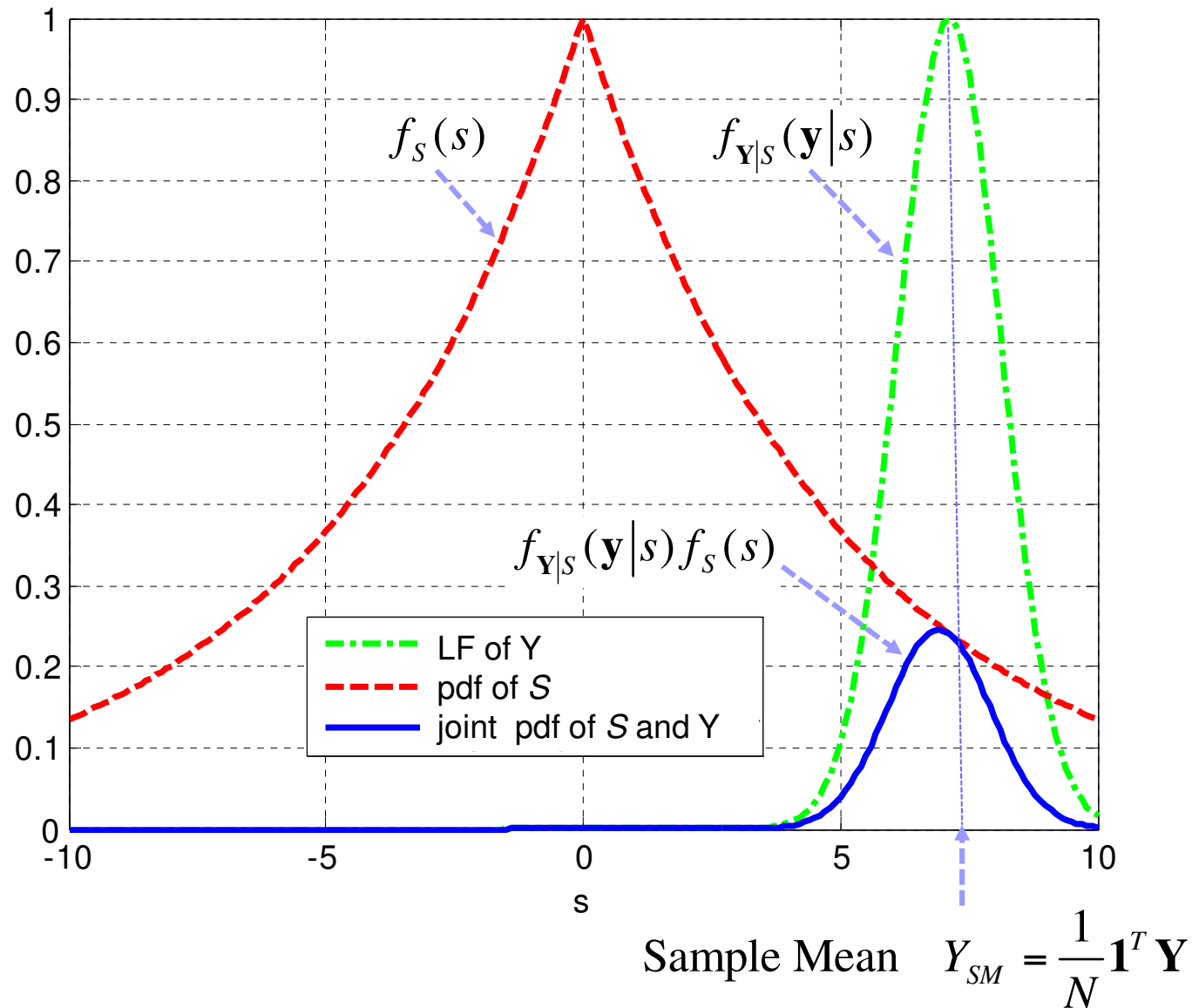
MAP Estimation of a constant Laplacian signal in AWGN⁸

$$\begin{aligned}\ln f_{\mathbf{Y}|s}(\mathbf{y}|s) &= \kappa_1 - \frac{1}{2\sigma_W^2} \sum_{i=0}^{N-1} (y_i - s)^2 \\ &= \kappa_1 - \frac{1}{2\sigma_W^2} \sum_{i=0}^{N-1} (y_i^2 - 2y_i s + s^2) \\ &= \kappa_1 - \frac{1}{2\sigma_W^2} \sum_{i=0}^{N-1} y_i^2 + \frac{s}{\sigma_W^2} \sum_{i=0}^{N-1} y_i - \frac{N}{2\sigma_W^2} s^2 \\ &= -\frac{N}{2\sigma_W^2} (s^2 - 2Y_{SM}s + \kappa_2) \\ &= -\frac{N}{2\sigma_W^2} (s - Y_{SM})^2 + \kappa_3\end{aligned}$$

$$\text{where } Y_{SM} \triangleq \frac{1}{N} \sum_{i=0}^{N-1} Y_i = \frac{1}{N} \mathbf{1}^T \mathbf{Y} \quad \text{Sample Mean}$$

The maximum of $\ln f_{\mathbf{Y}|s}(\mathbf{y}|s)$ is for $s = Y_{SM} \equiv ML$ estimate of s

MAP Estimation of a constant Laplacian signal in AWGN⁹



MAP Estimation of a constant Laplacian signal in AWGN

$$\begin{aligned}\ln f_{\mathbf{Y}|s}(\mathbf{y}|s) + \ln f_s(s) &= -\frac{N}{2\sigma_w^2} (s - Y_{SM})^2 + \kappa_3 - \frac{|s|}{\lambda} \\ &= -\frac{N}{2\sigma_w^2} \left(s^2 + \frac{2\sigma_w^2}{N\lambda} |s| - 2Y_{SM}s + (Y_{SM})^2 \right) + \kappa_3 \\ &= -\frac{N}{2\sigma_w^2} \left(s^2 + \frac{2\sigma_w^2}{N\lambda} |s| - 2Y_{SM}s + \kappa_4 \right)\end{aligned}$$

$$\text{where } Y_{SM} \triangleq \frac{1}{N} \sum_{i=0}^{N-1} Y_i = \frac{1}{N} \mathbf{1}^T \mathbf{Y} \quad \text{Sample Mean}$$

$$\text{MAP equation: } \frac{\partial \left[\ln f_{\mathbf{Y}|s}(\mathbf{y}|s) + \ln f_s(s) \right]}{\partial s} = \frac{\partial}{\partial s} \left(s^2 + \frac{2\sigma_w^2}{N\lambda} |s| - 2Y_{SM}s + \kappa_4 \right) = 0$$

MAP Estimation of a constant Laplacian signal in AWGN¹

■ We can find the solution by differentiating w.r.t. s only if the maximum is at a value where the joint pdf is continuous, i.e. in this case only if the solution is different from $s=0$ (there is corner point at $s=0$):

$$\frac{\partial}{\partial s} \left(s^2 + \frac{2\sigma_w^2}{N\lambda} |s| - 2Y_{SM}s + \kappa_4 \right) = 2s + \frac{2\sigma_w^2}{N\lambda} \text{sign}\{s\} - 2Y_{SM} = 0$$

$$s = Y_{SM} - \frac{\sigma_w^2}{N\lambda} \text{sign}\{s\}$$

$$\begin{cases} \hat{S}_{MAP} = Y_{SM} - \frac{\sigma_w^2}{N\lambda} \text{sign}\{\hat{S}_{MAP}\} = Y_{SM} - \frac{\sigma_w^2}{N\lambda} & \text{if } Y_{SM} - \frac{\sigma_w^2}{N\lambda} > 0 \\ \hat{S}_{MAP} = Y_{SM} - \frac{\sigma_w^2}{N\lambda} \text{sign}\{\hat{S}_{MAP}\} = Y_{SM} + \frac{\sigma_w^2}{N\lambda} & \text{if } Y_{SM} + \frac{\sigma_w^2}{N\lambda} < 0 \end{cases}$$

MAP Estimation of a constant Laplacian signal in AWGN²

$$\begin{cases} \hat{S}_{MAP} = Y_{SM} - \frac{\sigma_w^2}{N\lambda} \text{sign}\{\hat{S}_{MAP}\} = Y_{SM} - \frac{\sigma_w^2}{N\lambda} & \text{if } Y_{SM} > \frac{\sigma_w^2}{N\lambda} \\ \hat{S}_{MAP} = Y_{SM} - \frac{\sigma_w^2}{N\lambda} \text{sign}\{\hat{S}_{MAP}\} = Y_{SM} + \frac{\sigma_w^2}{N\lambda} & \text{if } Y_{SM} < -\frac{\sigma_w^2}{N\lambda} \end{cases}$$

■ In compact form, taking into account that $\text{sign}\{\hat{S}_{MAP}\} = \text{sign}\{Y_{SM}\}$, we have:

$$\hat{S}_{MAP} = Y_{SM} - \frac{\sigma_w^2}{N\lambda} \text{sign}\{Y_{SM}\}, \quad \text{if } |Y_{SM}| > \frac{\sigma_w^2}{N\lambda}$$

■ The MAP estimate is 0 otherwise:

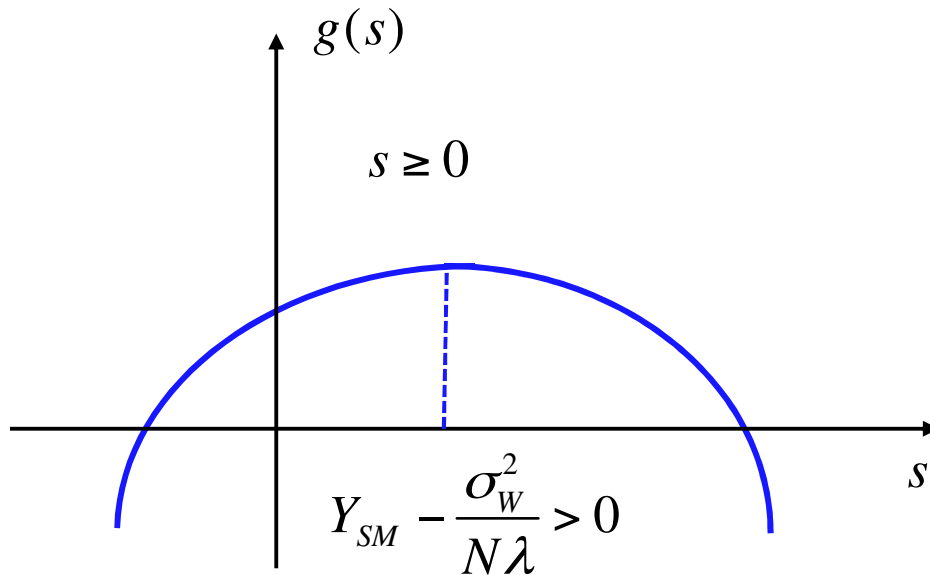
$$\hat{S}_{MAP} = 0 \quad \text{if } |Y_{SM}| \leq \frac{\sigma_w^2}{N\lambda}$$

$$\begin{aligned} \text{sign}\{\hat{S}_{MAP}\} &= \text{sign}\{Y_{SM}\} \\ &\text{when } \hat{S}_{MAP} \neq 0 \end{aligned}$$

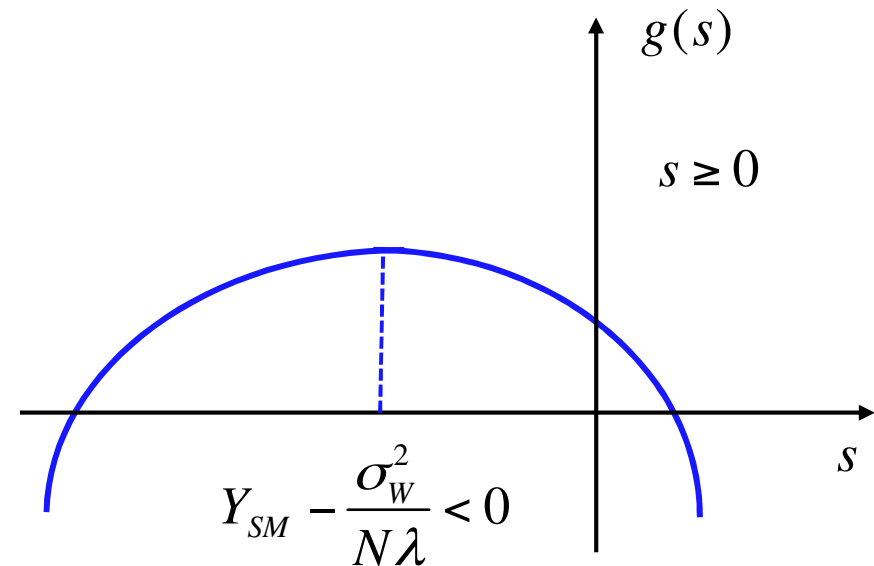
MAP Estimation of a constant Laplacian signal in AWGN³

$$g(s) = \ln f_{\mathbf{Y}|s}(\mathbf{y}|s) + \ln f_s(s) = -\frac{N}{2\sigma_w^2} \left(s^2 + \frac{2\sigma_w^2}{N\lambda} |s| - 2Y_{SM}s + \kappa_4 \right)$$

$$\text{If } s \geq 0 \rightarrow g(s) = -\frac{N}{2\sigma_w^2} \left(s^2 - 2 \left(Y_{SM} - \frac{\sigma_w^2}{N\lambda} \right) s + \kappa_4 \right)$$



$$\hat{S}_{MAP} = Y_{SM} - \frac{\sigma_w^2}{N\lambda}$$

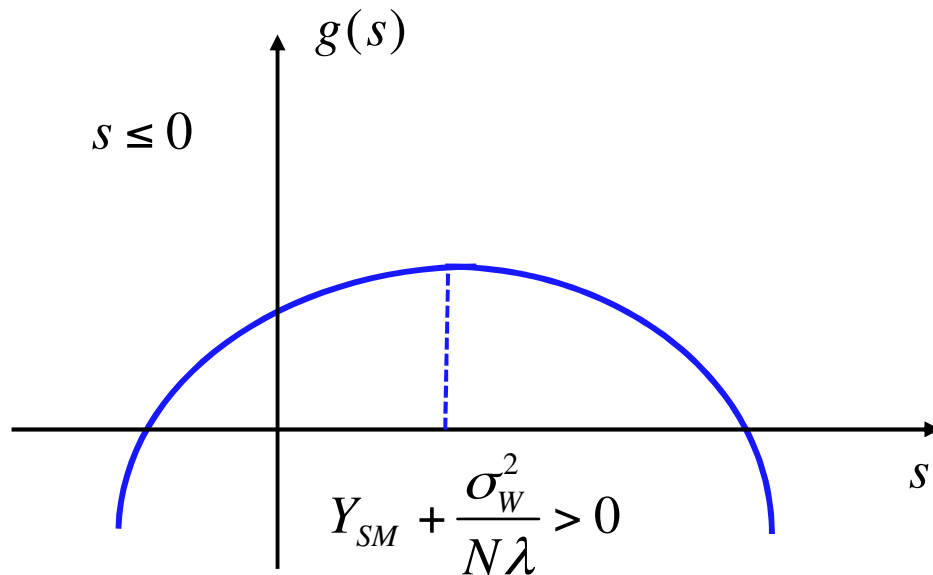


$$\hat{S}_{MAP} = 0$$

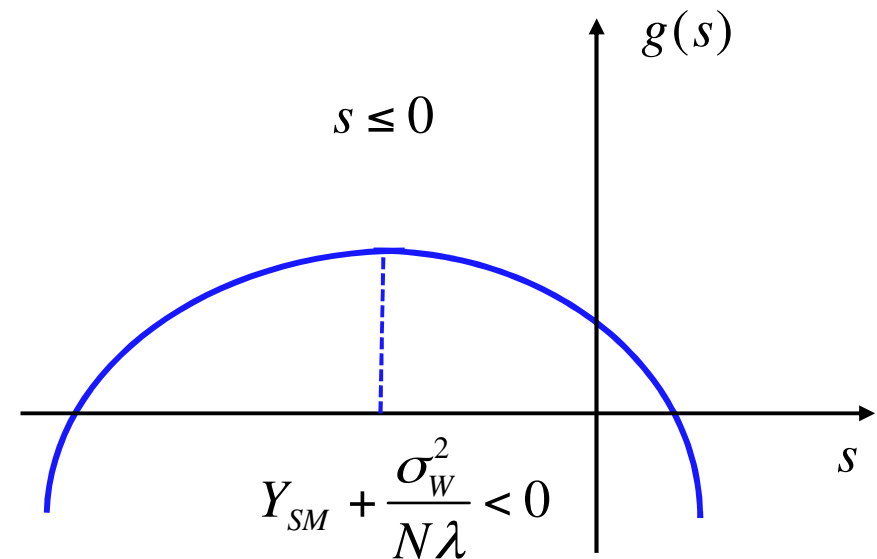
MAP Estimation of a constant Laplacian signal in AWGN⁴

$$g(s) = \ln f_{\mathbf{Y}|s}(\mathbf{y}|s) + \ln f_s(s) = -\frac{N}{2\sigma_w^2} \left(s^2 + \frac{2\sigma_w^2}{N\lambda} |s| - 2Y_{SM}s + \kappa_4 \right)$$

$$\text{If } s \leq 0 \rightarrow g(s) = -\frac{N}{2\sigma_w^2} \left(s^2 - 2 \left(Y_{SM} + \frac{\sigma_w^2}{N\lambda} \right) s + \kappa_4 \right)$$



$$\hat{S}_{MAP} = 0$$

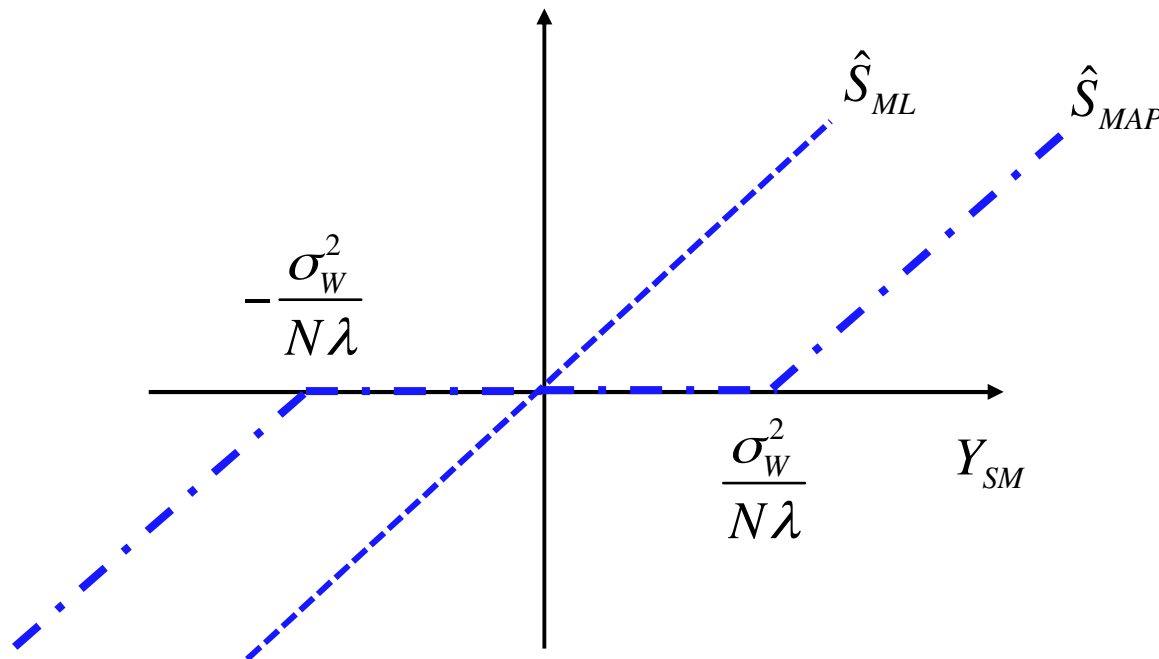


$$\hat{S}_{MAP} = Y_{SM} + \frac{\sigma_w^2}{N\lambda}$$

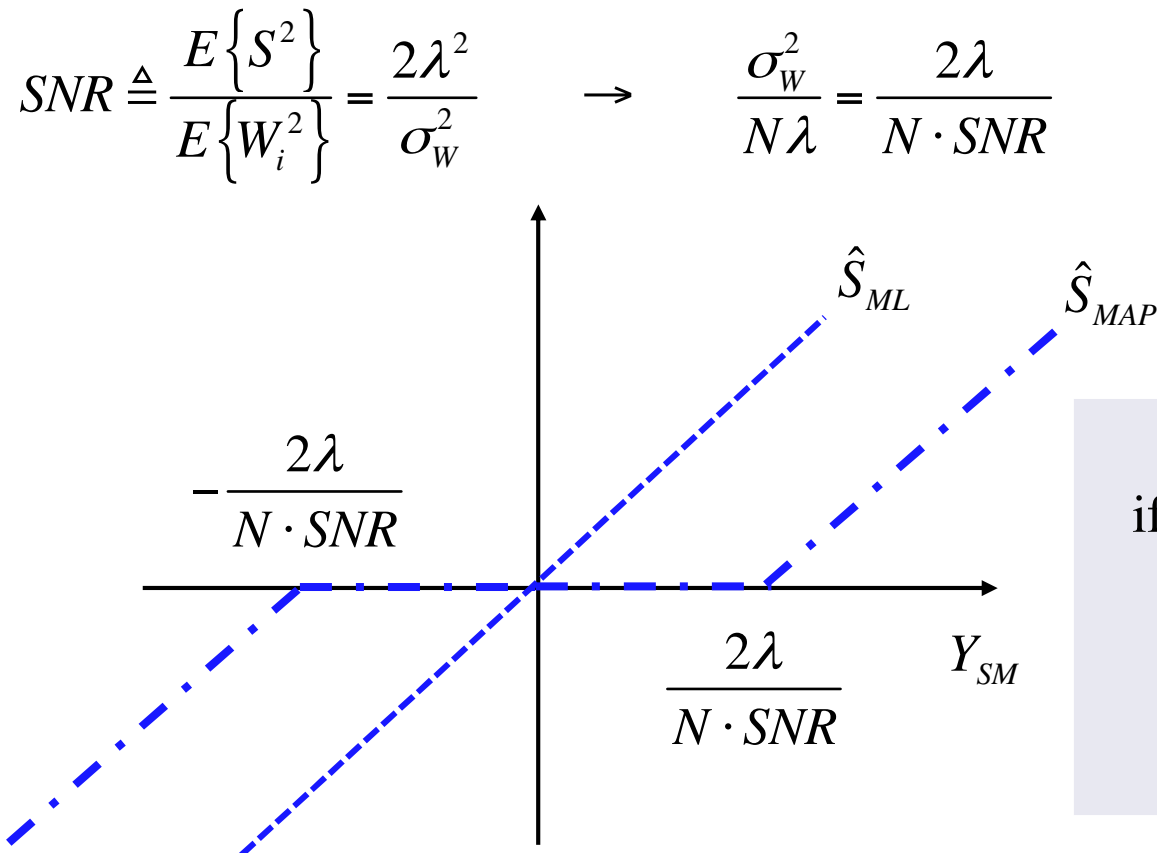
MAP Estimation of a constant Laplacian signal in AWGN⁵

■ MAP estimate: $\hat{S}_{MAP} = \begin{cases} 0, & |Y_{SM}| \leq \frac{\sigma_w^2}{N\lambda} \\ Y_{SM} - \frac{\sigma_w^2}{N\lambda} \text{sign}\{Y_{SM}\}, & |Y_{SM}| > \frac{\sigma_w^2}{N\lambda} \end{cases}$

■ ML estimate: $\hat{S}_{ML} = Y_{SM}$



MAP Estimation of a constant Laplacian signal in AWGN⁶



if $N \rightarrow \infty$ or $SNR \rightarrow \infty$

$\Downarrow$

$$\hat{S}_{MAP} \rightarrow \hat{S}_{ML}$$

■ **MAP estimator** → **Thresholding & Denoising effect:** it uses the observed data, actually the sample mean of the data, only if its absolute value is greater than a given threshold, that is a function of the SNR and of the data size N . Otherwise, the estimate is given by the mode of the *a-priori* pdf, which is 0.

MAP Estimation of a constant Laplacian signal in AWGN⁷

$$\begin{aligned} BCRB(S) &= - \frac{1}{E_{S,Y} \left\{ \frac{\partial^2 \ln f_{S,Y}(s, \mathbf{y})}{\partial s^2} \right\}} \\ &\quad \curvearrowright E_{S,Y} \left\{ \frac{\partial^2 \ln f_{S,Y}(s, \mathbf{y})}{\partial s^2} \right\} = E_{S,Y} \left\{ \frac{\partial^2 \ln \left(f_{Y|S}(\mathbf{y}|s) f_A(s) \right)}{\partial s^2} \right\} \\ &= E_{S,Y} \left\{ \frac{\partial^2 \ln f_{Y|S}(\mathbf{y}|s)}{\partial s^2} + \frac{\partial^2 \ln f_S(s)}{\partial s^2} \right\} \\ &= E_{S,Y} \left\{ \frac{\partial^2 \ln f_{Y|S}(\mathbf{y}|s)}{\partial s^2} \right\} + E_{S,Y} \left\{ \frac{\partial^2 \ln f_S(s)}{\partial s^2} \right\} \\ &= E_S \left\{ E_{Y|S} \left\{ \frac{\partial^2 \ln f_{Y|S}(\mathbf{y}|s)}{\partial s^2} \right\} \right\} + E_S \left\{ \frac{\partial^2 \ln f_S(s)}{\partial s^2} \right\} \end{aligned}$$

MAP Estimation of a constant Laplacian signal in AWGN⁸

$$\ln f_{\mathbf{Y}|s}(\mathbf{y}|s) = \ln \left(\frac{1}{(2\sigma_w^2)^{N/2}} \right) - \frac{1}{2\sigma_w^2} \sum_{i=0}^{N-1} (y_i - s)^2, \quad \ln f_s(s) = \ln \left(\frac{1}{2\lambda} \right) - \frac{|s|}{\lambda}$$

$$\left\{ \begin{array}{l} \frac{\partial \ln f_{\mathbf{Y}|s}(\mathbf{y}|s)}{\partial s} = \frac{1}{\sigma_w^2} \sum_{i=0}^{N-1} (y_i - s) = \frac{1}{\sigma_w^2} \sum_{i=0}^{N-1} y_i - \frac{N}{\sigma_w^2} s, \quad \frac{\partial^2 \ln f_{\mathbf{Y}|s}(\mathbf{y}|s)}{\partial s^2} = -\frac{N}{\sigma_w^2} \\ \frac{\partial \ln f_s(s)}{\partial s} = -\frac{1}{\lambda} \text{sign}(s), \quad \frac{\partial^2 \ln f_s(s)}{\partial s^2} = -\frac{2}{\lambda} \delta(s) \end{array} \right.$$

$$E_s \left\{ E_{\mathbf{Y}|s} \left\{ \frac{\partial^2 \ln f_{\mathbf{Y}|s}(\mathbf{y}|s)}{\partial s^2} \right\} \right\} = -\frac{N}{\sigma_w^2}$$

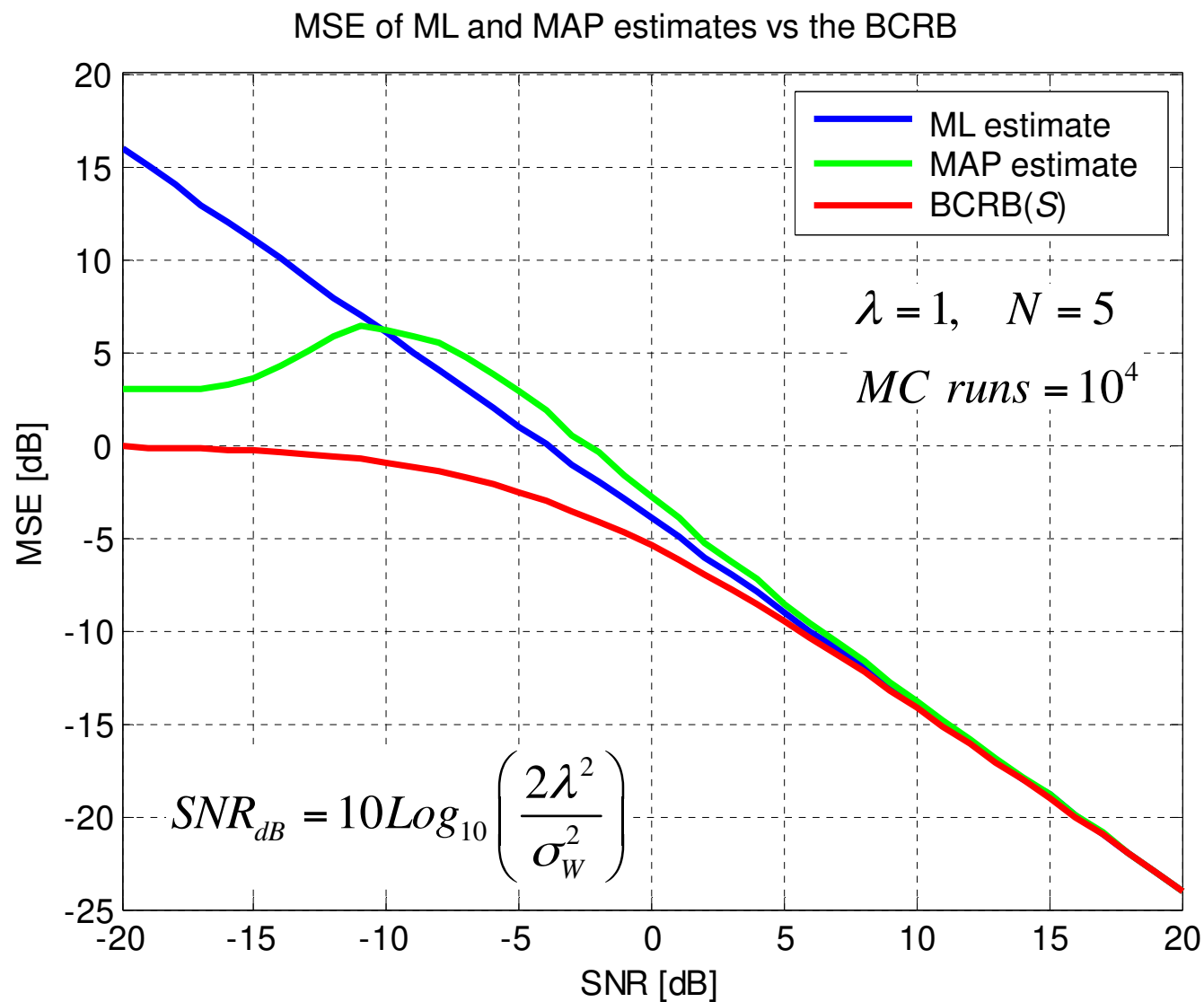
MAP Estimation of a constant Laplacian signal in AWGN⁹

$$E_s \left\{ \frac{\partial^2 \ln f_s(s)}{\partial s^2} \right\} = \int_{-\infty}^{+\infty} -\frac{2}{\lambda} \delta(s) f_s(s) ds = -\frac{2}{\lambda} f_s(0) = -\frac{2}{\lambda} \cdot \frac{1}{2\lambda} = -\frac{1}{\lambda^2}$$

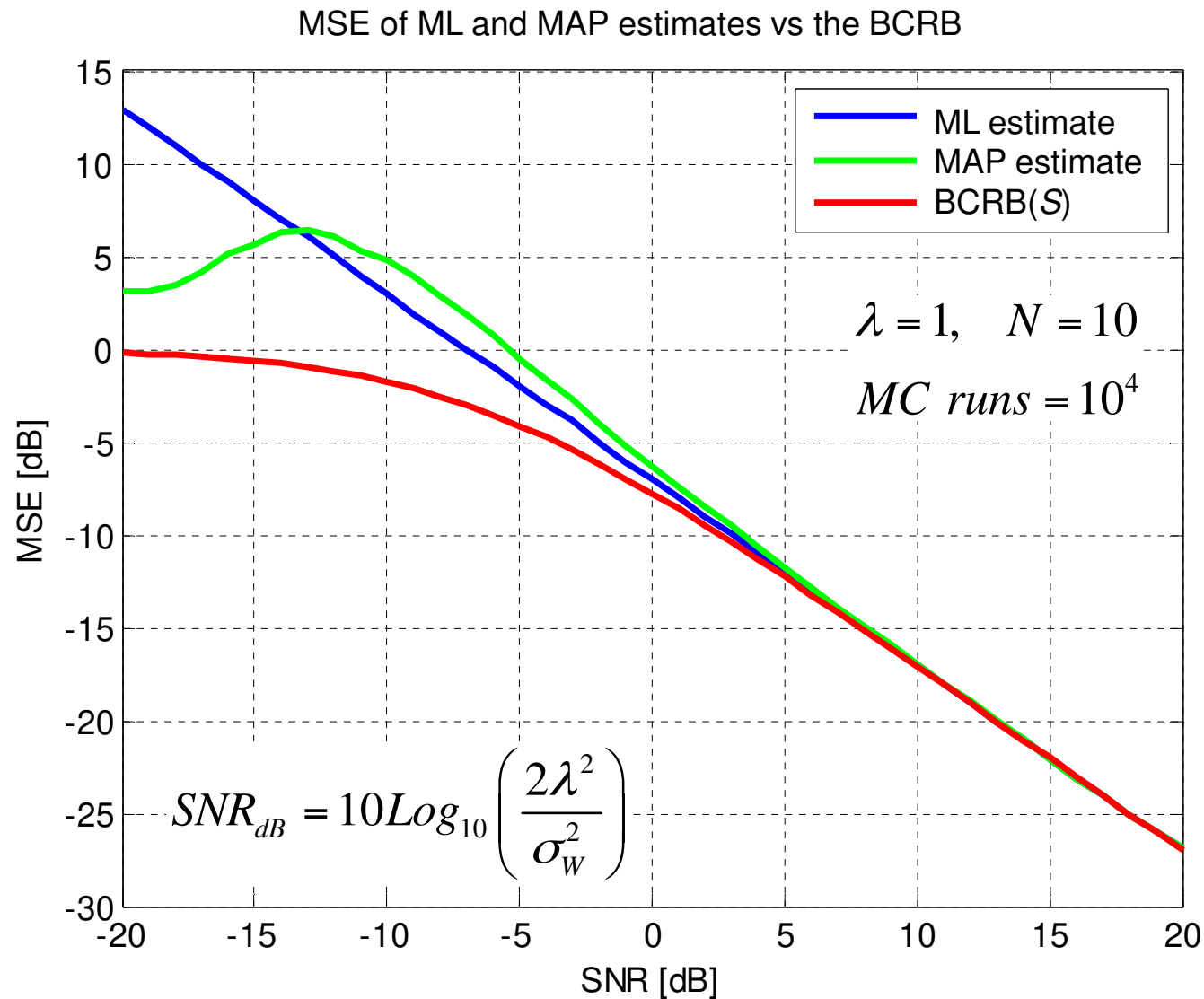
$$\begin{aligned} BCRB(S) &= -\frac{1}{E_{s,Y} \left\{ \frac{\partial^2 \ln f_{s,Y}(s, \mathbf{y})}{\partial s^2} \right\}} = \frac{1}{\frac{N}{\sigma_w^2} + \frac{1}{\lambda^2}} = \frac{\lambda^2}{1 + \frac{N\lambda^2}{\sigma_w^2}} \\ &= \frac{\lambda^2}{1 + N \cdot SNR/2} \end{aligned}$$

$$\text{where: } SNR = \frac{2\lambda^2}{\sigma_w^2}$$

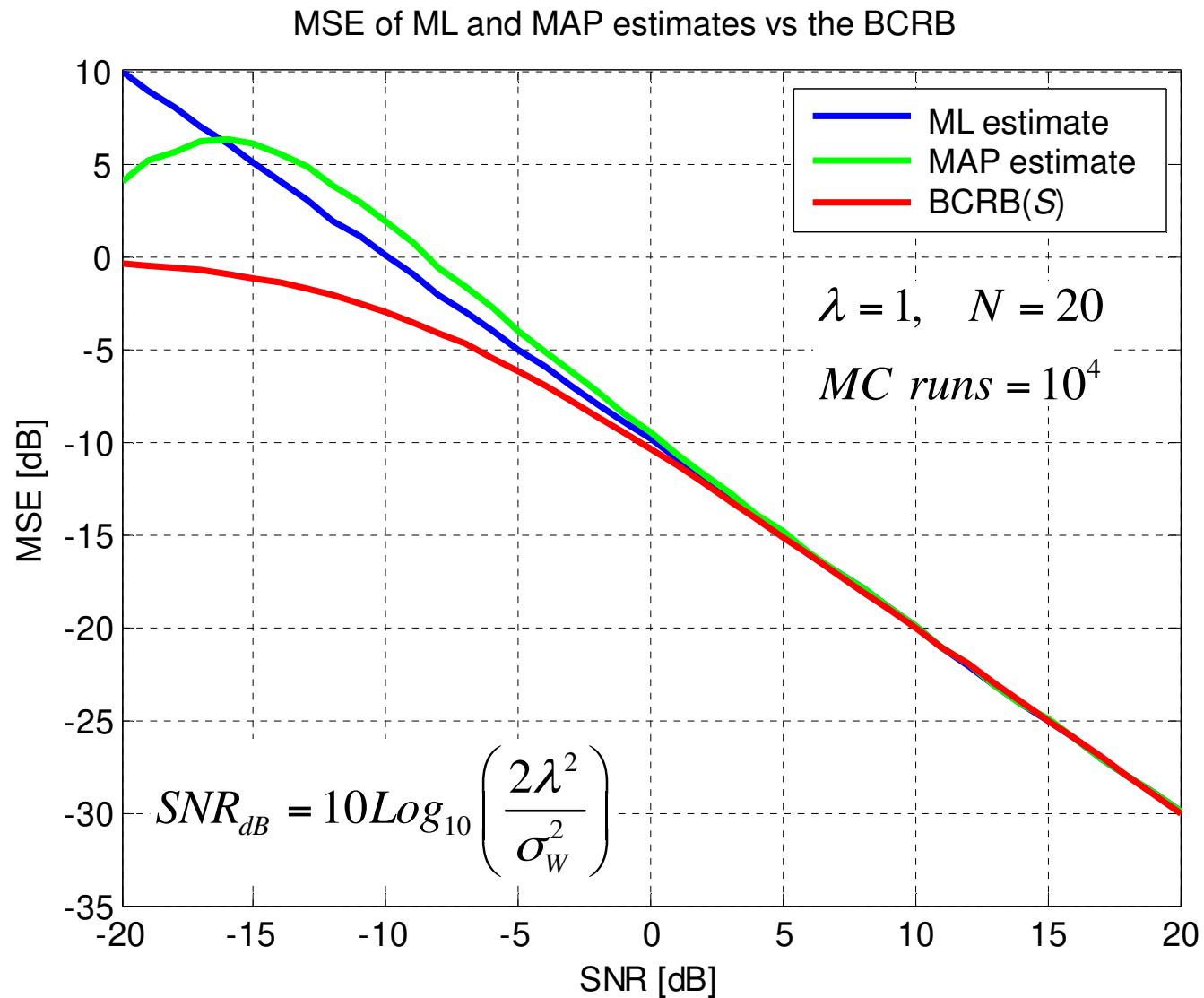
MAP Estimation of a constant Laplacian signal in AWGN⁰



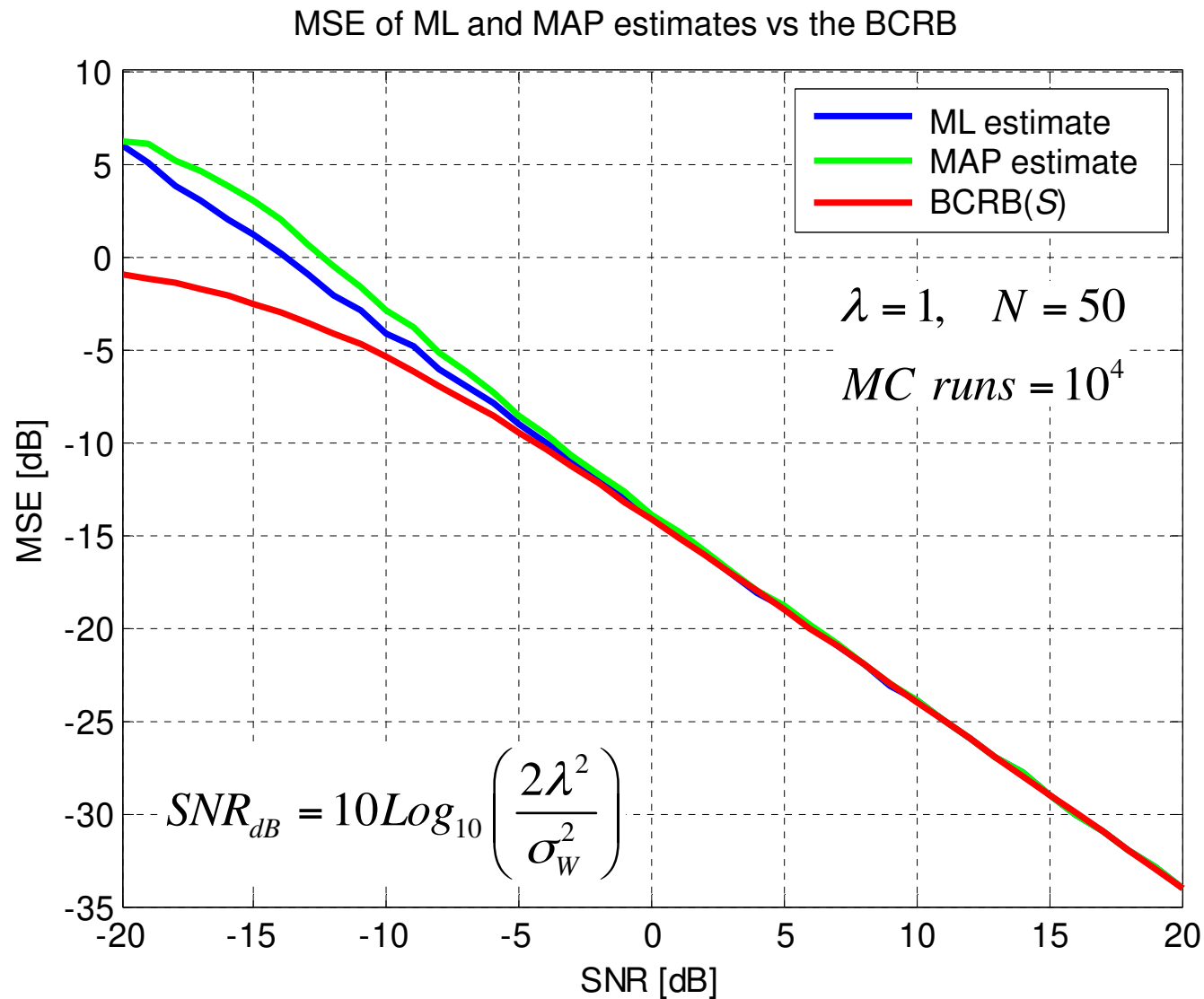
MAP Estimation of a constant Laplacian signal in AWGN¹



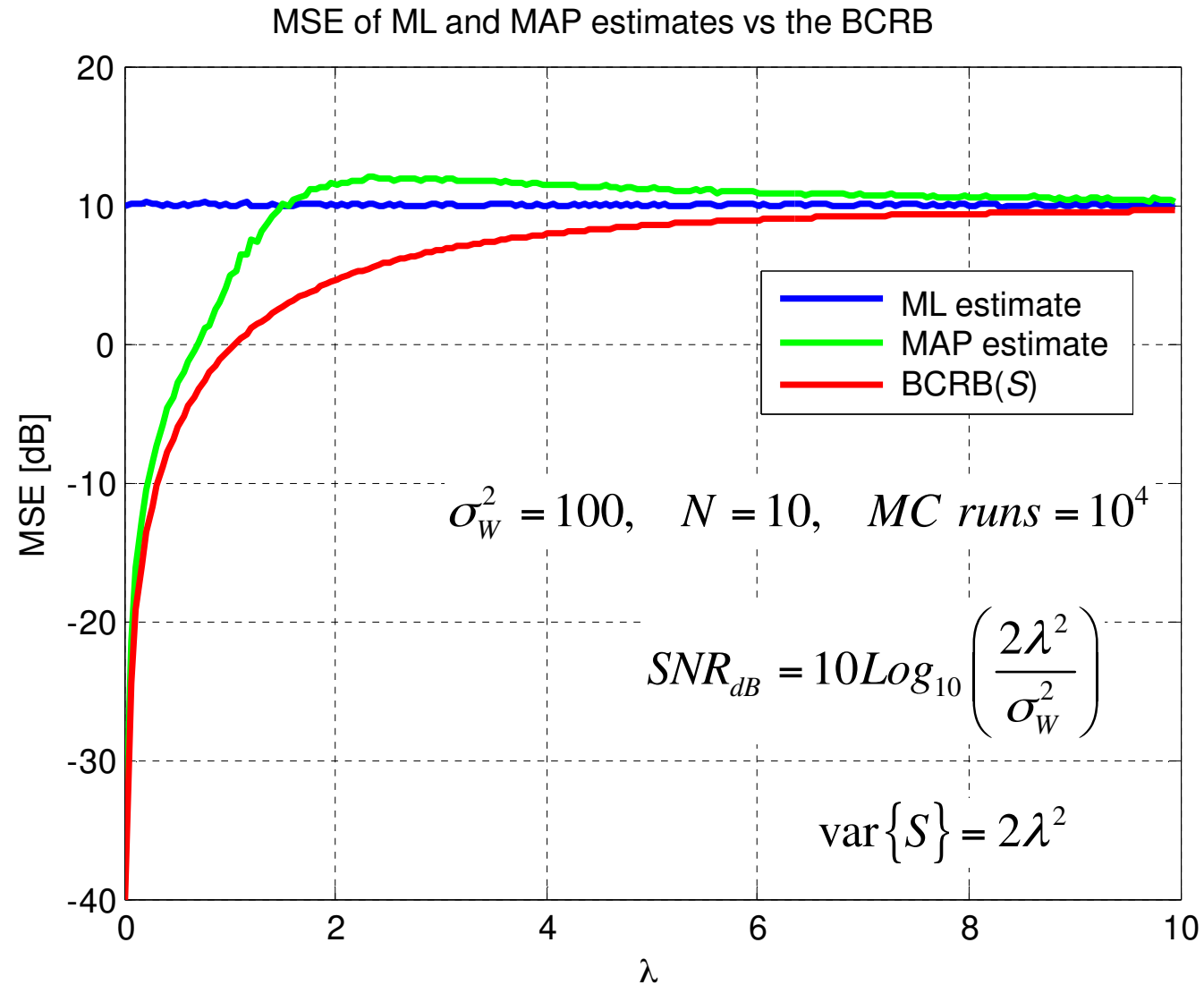
MAP Estimation of a constant Laplacian signal in AWGN²



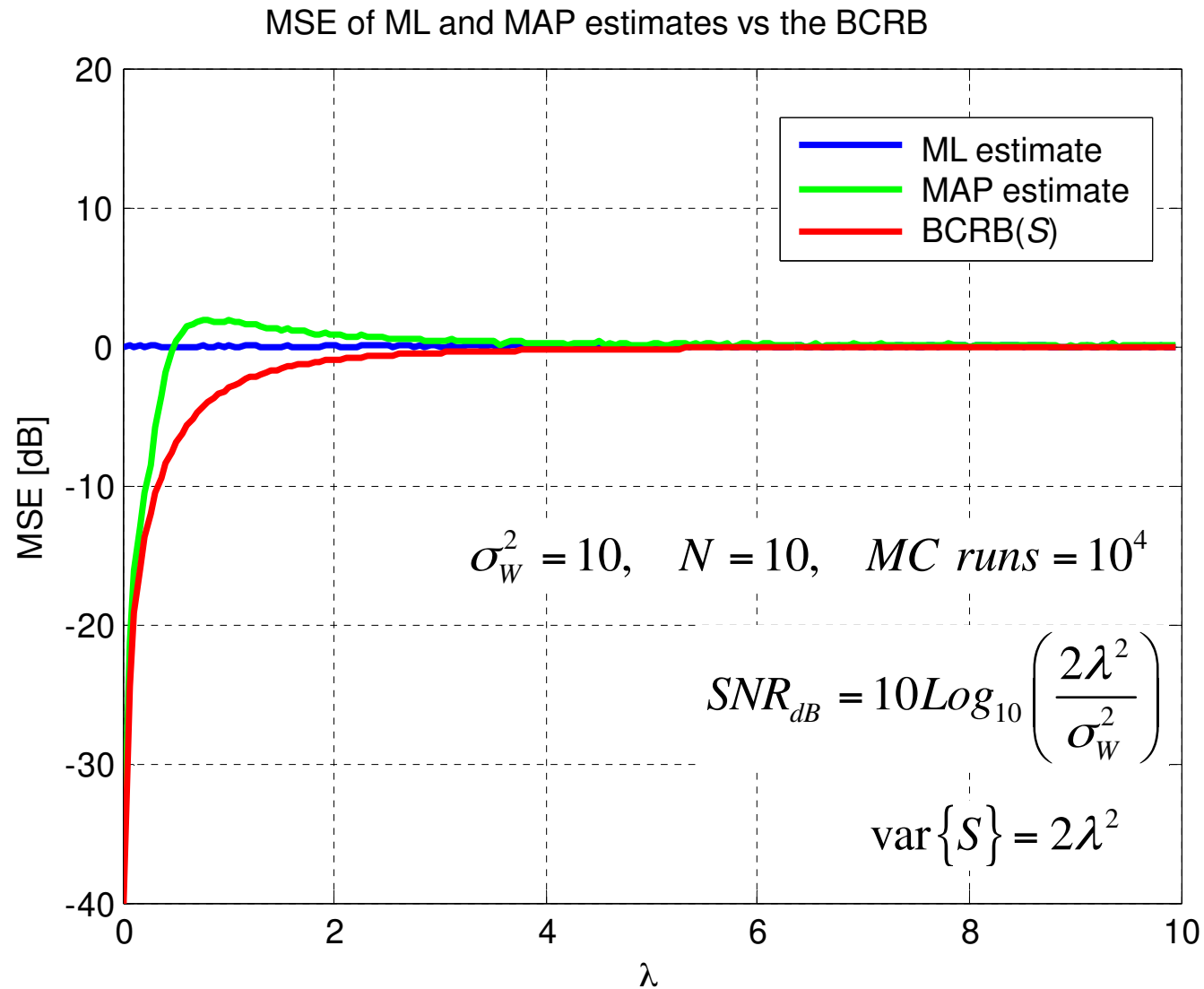
MAP Estimation of a constant Laplacian signal in AWGN³



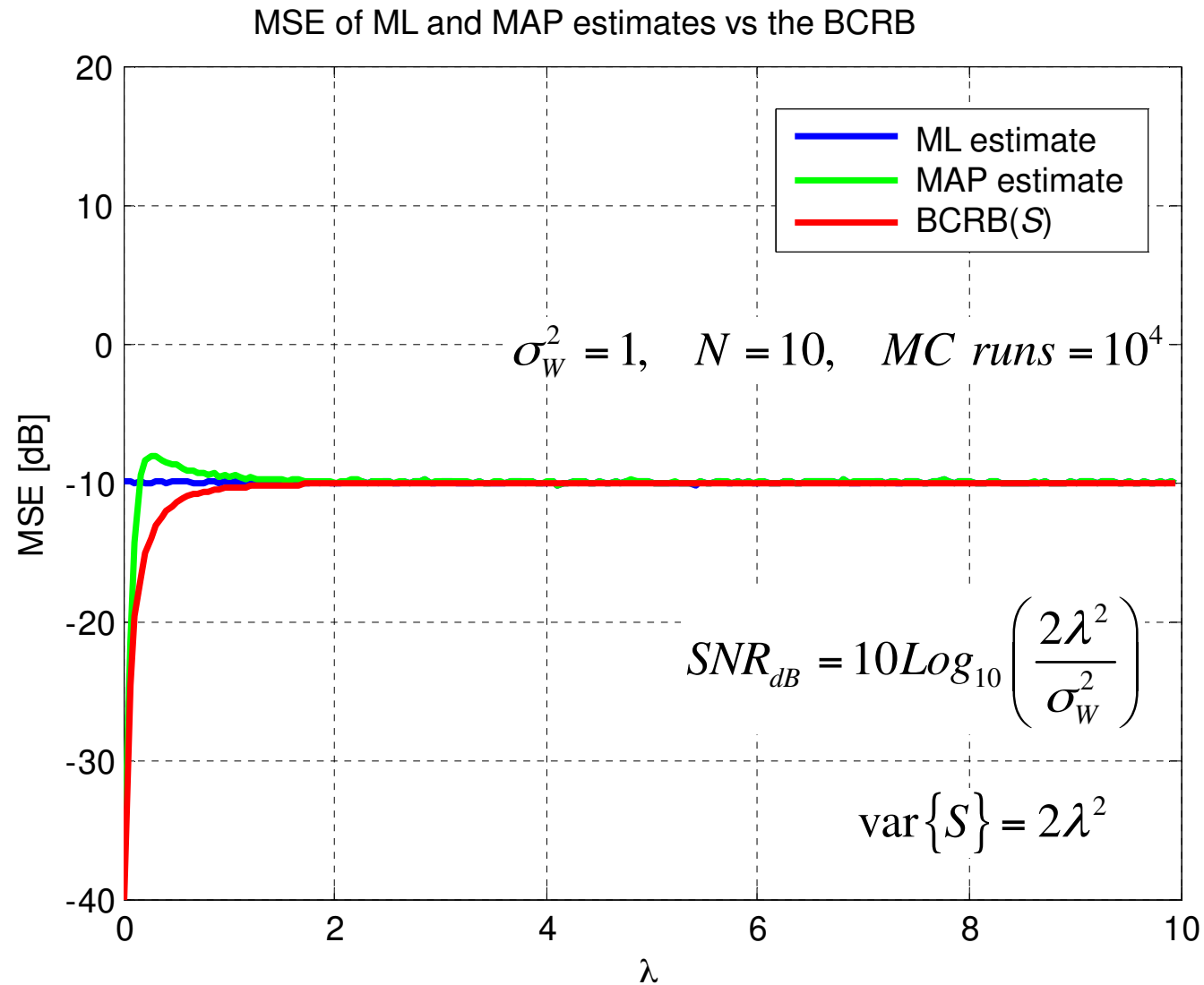
MAP Estimation of a constant Laplacian signal in AWGN⁴



MAP Estimation of a constant Laplacian signal in AWGN⁵



MAP Estimation of a constant Laplacian signal in AWGN⁶



MAP Estimation of the parameter of an Exp distribution¹⁷

Problem: MAP estimation of the parameter η of an Exponential distribution

■ We observe N realizations of an exponentially distributed r.v. Y and we want to estimate the parameter η of the pdf under the assumption that the unknown parameter can be modelled as a r.v. with *a-priori* pdf which is exponential with known parameter λ . Conditionally to a given value of η the observed data are IID.

■ Find the **MAP estimator** and compare it with the **ML estimator** of η .

Observed data: $\{Y_i ; i = 0, 1, 2, \dots, N-1\}$ $Y_i | \eta \in \text{Exp}(\eta)$, IID, $E\{Y_i | \eta\} = \frac{1}{\eta}$

$$f_{\eta}(\eta) = \lambda e^{-\lambda \eta} u(\eta) \quad \rightarrow \quad E\{\eta\} = \frac{1}{\lambda}, \quad \sigma_{\eta}^2 = \text{var}\{\eta\} = \frac{1}{\lambda^2}$$

$$f_{\mathbf{Y}|\eta}(\mathbf{y}|\eta) = \prod_{i=0}^{N-1} f_{Y_i|\eta}(y_i|\eta) = \prod_{i=0}^{N-1} \eta e^{-\eta y_i} u(y_i) = \eta^N e^{-\eta \sum_{i=0}^{N-1} y_i} \prod_{i=0}^{N-1} u(y_i)$$

MAP Estimation of the parameter of an Exp distribution¹⁸

■ The **Maximum Likelihood (ML)** estimator is given by the **mode** of the log-LF, i.e. the logarithm of the pdf of the observed data $\mathbf{Y}$ conditioned to a value of the unknown parameter η :

$$\hat{\eta}_{ML} = \arg \max_{\eta} f_{\mathbf{Y}|\eta}(\mathbf{y}|\eta)$$

■ If the log-LF is differentiable w.r.t. η , the ML estimate can be derived by solving the following equation

ML equation: $\left. \frac{\partial \ln f_{\mathbf{Y}|\eta}(\mathbf{y}|\eta)}{\partial \eta} \right|_{\eta=\hat{\eta}_{ML}} = 0$

$\ln f_{\mathbf{Y}|\eta}(\mathbf{y}|\eta) = \ln \left(\eta^N e^{-\eta \sum_{i=0}^{N-1} y_i} \prod_{i=0}^{N-1} u(y_i) \right) = N \ln \eta - \eta \sum_{i=0}^{N-1} y_i$

MAP Estimation of the parameter of an Exp distribution¹⁹

■ Solving the ML equation we get:

$$\frac{\partial}{\partial \eta} \left(N \ln \eta - \eta \sum_{i=0}^{N-1} y_i \right) = \frac{N}{\eta} - \sum_{i=0}^{N-1} y_i = 0 \Rightarrow \hat{\eta}_{ML} = \frac{1}{\frac{1}{N} \sum_{i=0}^{N-1} y_i} = \frac{1}{Y_{SM}}$$

■ The **Maximum A Posteriori (MAP)** estimator is instead given by the **mode** of the a-posteriori pdf or equivalently by the value which maximizes the joint pdf of **Y** and η :

$$\hat{\eta}_{MAP} = \arg \max_{\eta} f_{\eta|Y}(\eta|y) = \arg \max_{\eta} f_{Y|\eta}(y|\eta) f_{\eta}(\eta)$$

MAP equation:
$$\left. \frac{\partial \left[\ln f_{Y|\eta}(y|\eta) + \ln f_{\eta}(\eta) \right]}{\partial \eta} \right|_{\eta=\hat{\eta}_{MAP}} = 0$$

MAP Estimation of the parameter of an Exp distribution¹²⁰

$$\begin{aligned}\frac{\partial \left[\ln f_{\mathbf{Y}|\eta}(\mathbf{y}|\eta) + \ln f_{\eta}(\eta) \right]}{\partial \eta} &= \frac{N}{\eta} - \sum_{i=0}^{N-1} y_i + \frac{\partial}{\partial \eta} \ln \left(\lambda e^{-\lambda \eta} u(\eta) \right) \\ &= \frac{N}{\eta} - \sum_{i=0}^{N-1} y_i - \lambda = 0\end{aligned}$$

$$\hat{\eta}_{MAP} = \frac{1}{\frac{\lambda}{N} + \frac{1}{N} \sum_{i=0}^{N-1} y_i} = \frac{1}{Y_{SM} + \lambda/N}$$


■ It is worth observing that when N tends to infinity (i.e. we have many data) or when λ tends to 0 (i.e. when the variance of the *a-priori* pdf tends to be very large, so that the *a-priori* pdf is *non-informative*), the MAP estimator tends to coincide with the ML estimator:

$$\lim_{N \rightarrow \infty} \hat{\eta}_{MAP} = \hat{\eta}_{ML}, \quad \lim_{\lambda \rightarrow 0} \hat{\eta}_{MAP} = \hat{\eta}_{ML}$$

MAP Estimation of the parameter of an Exp distribution¹²¹

- Let us assume now that the data pdf is parametrized by $\alpha=1/\eta$ instead of η :

$$Y_i | \alpha \in \text{Exp}(\alpha), \quad \text{IID}, \quad E\{Y_i | \alpha\} = \alpha$$

$$f_{\mathbf{Y}|\alpha}(\mathbf{y}|\alpha) = \prod_{i=0}^{N-1} f_{Y_i|\alpha}(y_i|\alpha) = \prod_{i=0}^{N-1} \frac{1}{\alpha} e^{-\frac{y_i}{\alpha}} u(y_i) = \left(\frac{1}{\alpha}\right)^N e^{-\frac{1}{\alpha} \sum_{i=0}^{N-1} y_i} \prod_{i=0}^{N-1} u(y_i)$$

$$f_{\alpha}(\alpha) = ? \quad \rightarrow \quad \alpha = g(\eta) = \frac{1}{\eta} \quad \rightarrow \quad f_{\alpha}(\alpha) = \left. \frac{f_{\eta}(\eta)}{\left| \frac{dg(\eta)}{d\eta} \right|} \right|_{\eta=g^{-1}(\alpha)}$$

MAP Estimation of the parameter of an Exp distribution¹²²

$$f_{\alpha}(\alpha) = \left. \frac{f_{\eta}(\eta)}{\left| \frac{dg(\eta)}{d\eta} \right|} \right|_{\eta=g^{-1}(\alpha)} = \left. \frac{\lambda e^{-\lambda\eta} u(\eta)}{1/\eta^2} \right|_{\eta=1/\alpha} = \frac{\lambda}{\alpha^2} e^{-\frac{\lambda}{\alpha}} u(\alpha)$$


$$\ln f_{\mathbf{Y}|\alpha}(\mathbf{y}|\alpha) = \ln \left[\left(\frac{1}{\alpha} \right)^N e^{-\frac{1}{\alpha} \sum_{i=0}^{N-1} y_i} \prod_{i=0}^{N-1} u(y_i) \right] = -N \ln \alpha - \frac{1}{\alpha} \sum_{i=0}^{N-1} y_i$$

$$\frac{\partial \ln f_{\mathbf{Y}|\alpha}(\mathbf{y}|\alpha)}{\partial \alpha} = -\frac{N}{\alpha} + \frac{1}{\alpha^2} \sum_{i=0}^{N-1} y_i = 0 \Rightarrow \hat{\alpha}_{ML} = \frac{1}{N} \sum_{i=0}^{N-1} y_i = Y_{SM} = \frac{1}{\hat{\eta}_{ML}}$$

MAP Estimation of the parameter of an Exp distribution¹²³

Invariance property of the ML estimate: $\alpha = \frac{1}{\eta} \Rightarrow \hat{\alpha}_{ML} = \frac{1}{\hat{\eta}_{ML}}$

■ For the MAP estimate the invariance property holds only for affine transformations, but not for nonlinear transformations:

$$\alpha = \frac{1}{\eta} \not\Rightarrow \hat{\alpha}_{MAP} = \frac{1}{\hat{\eta}_{MAP}}$$

■ Let us derive the MAP estimator of α :

$$\begin{aligned} \ln f_{Y|\alpha}(\mathbf{y}|\alpha) + \ln f_{\alpha}(\alpha) &= -N \ln \alpha - \frac{1}{\alpha} \sum_{i=0}^{N-1} y_i + \ln \left(\frac{\lambda}{\alpha^2} e^{-\frac{\lambda}{\alpha}} u(\alpha) \right) \\ &= -N \ln \alpha - \frac{N}{\alpha} Y_{SM} + \ln \lambda - 2 \ln \alpha - \frac{\lambda}{\alpha} \end{aligned}$$

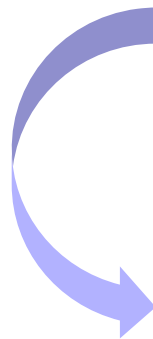
MAP Estimation of the parameter of an Exp distribution¹²⁴

■ Differentiating w.r.t. α and solving the MAP equation we get:

$$\frac{\partial}{\partial \alpha} \left(\ln f_{\mathbf{y}|\alpha}(\mathbf{y}|\alpha) + \ln f_{\alpha}(\alpha) \right) = \frac{\partial}{\partial \alpha} \left(-N \ln \alpha - \frac{N}{\alpha} Y_{SM} + \ln \lambda - 2 \ln \alpha - \frac{\lambda}{\alpha} \right)$$

$$= -\frac{N}{\alpha} + \frac{N}{\alpha^2} Y_{SM} - \frac{2}{\alpha} + \frac{\lambda}{\alpha^2}$$

$$= \frac{NY_{SM} + \lambda - (N+2)\alpha}{\alpha^2} = 0$$



$$\hat{\alpha}_{MAP} = \frac{Y_{SM} + \lambda/N}{1 + 2/N} \neq \frac{1}{\hat{\eta}_{MAP}} = Y_{SM} + \lambda/N$$

MAP Estimation of the parameter of an Exp distribution¹²⁵

■ However, when N tends to infinity, the MAP estimator tends to coincide with the ML estimator, so the *a-priori* information is less and less useful when the data size increases:

$$N \gg 1 \rightarrow \hat{\eta}_{MAP} = \frac{1}{Y_{SM} + \lambda/N} \cong \frac{1}{Y_{SM}} = \hat{\eta}_{ML}$$

$$N \gg 1 \rightarrow \hat{\alpha}_{MAP} = \frac{Y_{SM} + \lambda/N}{1 + 2/N} \cong Y_{SM} = \hat{\alpha}_{ML}$$

■ This result holds true in general: when the sample size N tends to infinity, the MAP estimator tends to coincide with the ML estimator, which does not use the *a-priori* information.

MAP Estimation of the parameter of an Exp distribution²⁶

- Let us now derive the BCRB for the random parameter η :

$$\begin{aligned} BCRB(\eta) &= - \frac{1}{E_{\eta, \mathbf{Y}} \left\{ \frac{\partial^2 \ln f_{\eta, \mathbf{Y}}(\eta, \mathbf{Y})}{\partial \eta^2} \right\}} \\ &\quad \curvearrowright E_{\eta, \mathbf{Y}} \left\{ \frac{\partial^2 \ln f_{\eta, \mathbf{Y}}(\eta, \mathbf{y})}{\partial \eta^2} \right\} = E_{\eta, \mathbf{Y}} \left\{ \frac{\partial^2 \ln \left(f_{\mathbf{Y}|\eta}(\mathbf{y}|\eta) f_{\eta}(\eta) \right)}{\partial \eta^2} \right\} \\ &= E_{\eta, \mathbf{Y}} \left\{ \frac{\partial^2 \ln f_{\mathbf{Y}|\eta}(\mathbf{y}|\eta)}{\partial \eta^2} + \frac{\partial^2 \ln f_{\eta}(\eta)}{\partial \eta^2} \right\} \\ &= E_{\eta, \mathbf{Y}} \left\{ \frac{\partial^2 \ln f_{\mathbf{Y}|\eta}(\mathbf{y}|\eta)}{\partial \eta^2} \right\} + E_{\eta, \mathbf{Y}} \left\{ \frac{\partial^2 \ln f_{\eta}(\eta)}{\partial \eta^2} \right\} \\ &= E_{\eta} \left\{ E_{\mathbf{Y}|\eta} \left\{ \frac{\partial^2 \ln f_{\mathbf{Y}|\eta}(\mathbf{y}|\eta)}{\partial \eta^2} \right\} \right\} + E_{\eta} \left\{ \frac{\partial^2 \ln f_{\eta}(\eta)}{\partial \eta^2} \right\} \end{aligned}$$

MAP Estimation of the parameter of an Exp distribution¹²⁷

$$\frac{\partial \ln f_{\mathbf{Y}|\eta}(\mathbf{y}|\eta)}{\partial \eta} + \frac{\partial \ln f_{\eta}(\eta)}{\partial \eta} = \frac{N}{\eta} - \sum_{i=1}^N y_i - \lambda$$


$$\frac{\partial^2 \ln f_{\mathbf{Y}|\eta}(\mathbf{y}|\eta)}{\partial \eta^2} + \frac{\partial^2 \ln f_{\eta}(\eta)}{\partial \eta^2} = -\frac{N}{\eta^2}$$

$$E_{\eta, \mathbf{Y}} \left\{ \frac{\partial^2 \ln f_{\eta, \mathbf{Y}}(\eta, \mathbf{y})}{\partial \eta^2} \right\} = -N \cdot E_{\eta} \left\{ \frac{1}{\eta^2} \right\}$$

$$MSE\{\hat{\eta}\} \geq BCRB(\eta) = \frac{1}{N \cdot E\{1/\eta^2\}} \leq \frac{1}{N} E\{\eta^2\} = \frac{2}{N\lambda^2} = ACRB(\eta)$$

$$E\{\Phi(\eta^2)\} \geq \Phi(E\{\eta^2\}) \quad \text{due to the Jensen's inequality and the convexity of } \Phi(x) = \frac{1}{x}$$

MAP Estimation of the parameter of an Exp distribution²⁸

If η is deterministic:

$$\frac{\partial^2 \ln f_{\mathbf{Y}|\eta}(\mathbf{y}|\eta)}{\partial \eta^2} = -\frac{N}{\eta^2} \rightarrow CRB(\eta) = -\frac{1}{E_{\mathbf{Y}} \left\{ \frac{\partial^2 \ln f_{\mathbf{Y}|\eta}(\mathbf{y}|\eta)}{\partial \eta^2} \right\}} = \frac{\eta^2}{N}$$

The "average" CRB (ACRB) is given by:

$$ACRB(\eta) = E_{\eta} \{ CRB(\eta) \} = \frac{E_{\eta} \{ \eta^2 \}}{N} = \frac{2}{N\lambda^2}$$

However, we cannot say that $MSE\{\hat{\eta}\} \geq ACRB(\eta)$ is always valid.

[it can be proved that it is true only under some particular conditions]

MAP Estimation of the parameter of an Exp distribution²⁹

